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ON RINGS OF OPERATORS. REDUCTION THEORY

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Note. This paper was written in 1937–38. Various other commitments prevented the author from effecting some changes, which he had intended to carry out before publishing the paper. This delayed the publication until the present time.

The paper is now published in the 1938 form, with only minor modifications:

1) Footnote 17 following Definition 9 and footnote 18 following Theorem IX, both in §22, are new.
2) The second reference 12 is new.
3) Lemma 22 and the derivation of the properties of $E \# F$ based on it differs from the 1938 version. This Lemma was, however, used by the author in his Princeton Lectures on Operator Theory in 1935.
4) The second part of (δ) in §24 differs from the 1938 version.

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Part I: Generalized direct sums

1 Description of unitary spaces

We will consider spaces which fulfill the postulates A, B, C, E of (1, pp. 64–66). They were discussed in (10, Chapter I). Such a space is either a finite-dimensional Euclidean space, or a Hilbert space. Loc. cit. (10) they were called "linear spaces with an inner product", but we now prefer to follow the terminology of (11, pp. 16–18), and call them "unitary spaces". (It would perhaps be more consequent, to require A, B only for "unitary" spaces, and to add the further specifications "separable, complete" for C, E. In the present context, however, we need not pay much attention to these nuances. Omission of C, separability, would leave most of our results unaffected, but we propose to discuss this problem, as well as various other ones connected with the generalization of the results of (10) to the non-separable case, at another occasion.)

Unitary spaces will always be denoted, as in (10), by the letter $\mathfrak{H}$. If several unitary spaces occur simultaneously, we will distinguish them by indices.

The 0-dimensional space (with the sole element 0) will again be explicitly excluded from the considerations of this paper, as it was the case in (10) too. Thus the only possible dimensions for unitary spaces are: $1, 2, \ldots$ (Euclidean spaces), and $+\infty$ (Hilbert space).

The fundamental operations of a unitary space $\mathfrak{H}$ are the "linear operations" of αf, $f + g$ (α a complex number, $f, g \in \mathfrak{H}$, and αf, $f + g \in \mathfrak{H}$) and the "inner product" (f, g) ($f, g \in \mathfrak{H}$ and (f, g) a complex number). From these the "absolute value" $\|f\| = \sqrt{(f, f)}$ and the "distance" $\|f - g\|$ are derived (cf. (1, pp. 64–65)). Even when we consider several unitary spaces $\mathfrak{H}$ simultaneously, we will use these symbols αf, $f + g$, (f, g), $\|f\|$ irrespectively of the $\mathfrak{H}$ in which the f, g are. This will not lead to misunderstandings, because it will never happen that an f belongs to several $\mathfrak{H}$'s with different definitions of these notions.

2 N-functions, Stieltjes measure, and total continuity

Another essential element of our considerations are the functions $\rho(\lambda)$ which satisfy the following conditions:

(i) $\rho(\lambda)$ is real valued, defined for all $\lambda > -\infty$, $< +\infty$;

(ii) $\rho(\lambda)$ is monotonous non-decreasing. ($\lambda_1 \leq \lambda_2$ implies $\rho(\lambda_1) \leq \rho(\lambda_2)$);

(iii) $\rho(\lambda)$ is right semi-continuous. ($\lambda \geq \lambda_0$, $\lambda \to \lambda_0$ imply $\rho(\lambda) \to \rho(\lambda_0)$, or: $\rho(\lambda_0 + 0) = \rho(\lambda_0)$);

(iv) $\rho(\lambda)$ is bounded. (A constant C with $|\rho(\lambda)| \leq C$ for all $\lambda > -\infty$, $< +\infty$ exists.)

Such a $\rho(\lambda)$ will be called an N-function.

This definition differs from the one given in (4, p. 193) for M-functions only by the domain of $\rho(\lambda)$: it was then $0 \leq \lambda \leq a$ ($< +\infty$), now it is $-\infty < \lambda < +\infty$. However these functions too were introduced in (4, p. 199). The restriction of boundedness, (iv), could be omitted without essentially affecting the use which we wish to make of the $\rho(\lambda)$, and either procedure has its specific advantages. Our choice, requiring boundedness, seems however to be technically preferable.

We will need the notion of Lebesgue-Stieltjes integration with respect to N-functions $\rho(\lambda)$. We will use this theory in the form in which it appears in (11, pp. 198–221). For the convenience of the reader we list below the essential notions, definitions and facts which we wish to use, as well as some minor deviations in terminology.

(α) Some subsets K of $-\infty < \lambda < +\infty$ are called $\rho(\lambda)$-*measurable* or *measurable with respect to* $\rho(\lambda)$, cf. (11, pp. 203–205). For them a $\rho(\lambda)$-measure is defined, which we will denote by $\mu_{\rho(\lambda)}(K)$. The situation which we discuss is somewhat simpler than the one loc. cit., because our $\rho(\lambda)$ are real and monotonous, while the $\rho(\lambda)$ loc. cit. are complex and of bounded variation. In the notation loc. cit. we have $\rho(\lambda) = \rho_1(\lambda)$, while $\rho_2(\lambda) = \rho_3(\lambda) = \rho_4(\lambda) = 0$. Consequently we have $\mu_{\rho(\lambda)}(K) = M(K) = M_1(K)$. (We write K for the set E loc. cit., as we wish to use the letter E for projections only.) As discussed loc. cit., the notion of $\rho(\lambda)$-measurability and the $\rho(\lambda)$-measure $\mu_{\rho(\lambda)}(K)$ share all essential properties with Lebesgue-measurability and measure, except its invariance under translations. They are somewhat more convenient insofar as the $\rho(\lambda)$-measure of the entire interval $-\infty < \lambda < +\infty$ is finite: $\rho(+\infty) - \rho(-\infty)$.

(β) Based on the notions (α) the $\rho(\lambda)$-integral or *Lebesgue-Stieltjes integral with respect to* $\rho(\lambda)$ can be defined, cf. (11, pp. 205–213); an alternative procedure is described in (4, pp. 198–202). A complex-valued function $f(\lambda)$ is $\rho(\lambda)$-measurable, if the set

$$K_f\begin{pmatrix} a_2, b_2 \\ a_1, b_1 \end{pmatrix}: \quad a_1 < \Re f(\lambda) \leq a_2, \qquad b_1 < \Im f(\lambda) \leq b_2$$

is $\rho(\lambda)$-measurable for every choice of a_1, a_2, b_1, b_2. If $f(\lambda)$ is $\rho(\lambda)$-measurable and $f(\lambda) \geq 0$, then the integral $\int f(\lambda)\, d\rho(\lambda)$ is defined, its value being ≥ 0, $\leq +\infty$ (it may be $= +\infty$!); if $f(\lambda)$ is $\rho(\lambda)$-measurable and complex-valued, then $\int f(\lambda)\, d\rho(\lambda)$ is only defined if $\int |f(\lambda)|\, d\rho(\lambda)$ is finite (this applies to real-valued $f(\lambda)$'s too, if they are not ≥ 0!). In this case $f(\lambda)$ is $\rho(\lambda)$-summable, and $\int f(\lambda)\, d\rho(\lambda)$ is complex, but necessarily finite.

If K is a $\rho(\lambda)$-measurable set, then

$$1_K(\lambda) = \begin{cases} 1, & \lambda \in K, \\ 0, & \lambda \notin K \end{cases}$$

is a $\rho(\lambda)$-measurable function. The notions of $\rho(\lambda)$-measurability and $\rho(\lambda)$-summability of $f(\lambda)$ over K and of $\int_K f(\lambda)\, d\rho(\lambda)$ arise from those of $\rho(\lambda)$-

measurability and $\rho(\lambda)$-summability of $f(\lambda)$ and of $\int f(\lambda)\, d\rho(\lambda)$ by replacing $f(\lambda)$ by $1_K(\lambda) f(\lambda)$.

(γ) Let $\rho(\lambda), \sigma(\lambda)$ be two N-functions. $\rho(\lambda)$ is $\sigma(\lambda)$-*totally continuous* if $\mu_{\sigma(\lambda)}(K) = 0$ implies $\mu_{\rho(\lambda)}(K) = 0$ for Borel-sets K. (We formulate this for Borel sets, because they are always $\sigma(\lambda)$- and $\rho(\lambda)$-measurable.) By (11, p. 204, criterium (5)), this implies, that every $\sigma(\lambda)$-measurable K with $\mu_{\sigma(\lambda)}(K) = 0$ is $\rho(\lambda)$-measurable with $\mu_{\rho(\lambda)}(K) = 0$; and now the same criterium gives, that $\sigma(\lambda)$-measurability implies $\rho(\lambda)$-measurability in general.

The necessary and sufficient condition for a $\rho(\lambda)$ to be $\sigma(\lambda)$-totally-continuous is, that it be possible to write it in the form

$$\rho(\lambda) = a + \int_{\mu \leq \lambda} f(\mu)\, d\sigma(\mu), \tag{2.1}$$

where a is a real constant, and $f(\mu)$ a $\sigma(\lambda)$-summable function, $f(\mu) \geq 0$. $\rho(\lambda), \sigma(\lambda)$ determine a uniquely and $f(\lambda)$ uniquely up to an arbitrary set of $\sigma(\lambda)$-measure 0. a is equal to $\lim_{\lambda \to -\infty} \rho(\lambda) = \rho(-\infty)$, $f(\lambda)$ will be denoted by $d\rho(\lambda)/d\sigma(\lambda)$. (In (11) the $\sigma(\lambda)$-total continuity of $\rho(\lambda)$ is denoted by $\rho < \sigma$, and the roles of the definition and the necessary and sufficient criterium are interchanged. The decisive fact is stated on p. 215, Lemma 6.5, (1).)

It is worth while to state the following consequence of (2.1), which holds for all $\sigma(\lambda)$-measurable sets K:

$$\begin{aligned}
\mu_{\rho(\lambda)}(K) &= \int_K 1\, d\rho(\lambda) = \int 1_K(\lambda)\, d\rho(\lambda) \\
&= \int 1_K(\lambda)\, d\left[\int_{\mu \leq \lambda} f(\mu)\, d\sigma(\mu)\right] = \int 1_K(\lambda) f(\lambda)\, d\sigma(\lambda) \\
&= \int_K f(\lambda)\, d\sigma(\lambda) = \int_K \frac{d\rho(\lambda)}{d\sigma(\lambda)}\, d\sigma(\lambda).
\end{aligned}$$

(δ) Let $E(\lambda)$, $-\infty < \lambda < +\infty$, be a resolution of unity in the unitary space $\mathfrak{H}$, cf. (1, p. 91), or (11, p. 174). One verifies immediately, that

$$\rho_f(\lambda) = (E(\lambda) f, f) = \|E(\lambda) f\|^2$$

is an N-function for every $f \in \mathfrak{H}$. $E(\lambda)$ is $\sigma(\lambda)$-*totally-continuous* if every $\rho_f(\lambda)$ ($f \in \mathfrak{H}$) is $\sigma(\lambda)$-totally-continuous.

If $\varphi_1, \varphi_2, \ldots$ is a complete, normalized, orthogonal set in $\mathfrak{H}$, and K a Borel set with $\mu_{\rho_{\varphi_n}(\lambda)}(K) = 0$ for $n = 1, 2, \ldots$, then $\mu_{\rho_f(\lambda)}(K) = 0$ for every f, by (11, p.

227, Theorem 6.3). Therefore all $\rho_f(\lambda)$ are $\sigma(\lambda)$-totally-continuous, if the $\rho_{\varphi_n}(\lambda)$, $n = 1, 2, \ldots$, are such. In other words:

$E(\lambda)$ is $\sigma(\lambda)$-totally-continuous if and only if all $\rho_{\varphi_n}(\lambda)$, $n = 1, 2, \ldots$, are $\sigma(\lambda)$-totally-continuous.

Choose a sequence $a_1, a_2, \ldots$ of numbers > 0, with a finite $\sum_n a_n$. Considering $\rho_{\varphi_n}(\lambda) = \|E(\lambda)\varphi_n\|^2 \geq 0$ and $\leq \|\varphi_n\|^2 = 1$, we see, that $\sum_n a_n \rho_{\varphi_n}(\lambda)$ is (absolutely) majorised by $\sum_n a_n$. Thus

$$\sigma(\lambda) = \sum_n a_n \rho_{\varphi_n}(\lambda) \tag{2.2}$$

is an N-function. By (11, p. 215, end of Lemma 6.5), all $\rho_{\varphi_n}(\lambda)$ are $\sigma(\lambda)$-totally continuous. Therefore $E(\lambda)$ is $\sigma(\lambda)$-totally continuous.

Conversely: If $E(\lambda)$ is $\tau(\lambda)$-totally-continuous for some N-function $\tau(\lambda)$, then all $\rho_{\varphi_n}(\lambda)$ are $\tau(\lambda)$-totally continuous, and with them $\sigma(\lambda) = \sum_n a_n \rho_{\varphi_n}(\lambda)$ is it too. (Verify this by using the normal form (2.1)!) So we see:

$E(\lambda)$ is $\tau(\lambda)$-totally continuous if and only if $\sigma(\lambda)$ is $\tau(\lambda)$-totally continuous. Every N-function $\sigma(\lambda)$ with this property is equivalent to $E(\lambda)$.

Before we close this section we wish to emphasize, that it is important for our further purposes, to have defined the $\rho(\lambda)$-measures in such a way, that the possibility of a one-point-sets having a $\rho(\lambda)$-measure $\neq 0$ is not excluded. The $\rho(\lambda)$-measure of the set with the unique element λ_0 is obviously

$$\lim_{\lambda > \lambda_0,\, \lambda \to \lambda_0} \rho(\lambda) - \lim_{\lambda < \lambda_0,\, \lambda \to \lambda_0} \rho(\lambda) = \rho(\lambda_0 + 0) - \rho(\lambda_0 - 0) = \rho(\lambda_0) - \rho(\lambda_0 - 0),$$

so it is $\neq 0$ if and only if $\rho(\lambda)$ is discontinuous at $\lambda = \lambda_0$. This is the reason, why we required only right-semi-continuity, and not continuity, of our N-functions.

3 Definition and discussion of the generalized direct sum

After these preliminaries we are able to define the notion of the *generalized direct sum*, which is the basis of all our subsequent discussions. It is related to the common notion of direct sum (of linear spaces) in a similar way, as the general resolutions of unity (which may include continuous spectra) to those with pure point spectra. Therefore it is the appropriate tool for an "additive decomposition" of Hilbert space.

In this connection we will have to introduce postulatorially a certain Stieltjes integral in Hilbert space, which is similar to some integrals introduced by F. Maeda, (12).

We will first define the generalized notion of direct sum, and discuss only afterwards, how the common notion of direct sum arises as a special case from it.

Definition 1. *Let $\mathfrak{H}$ be a unitary space, $\mathfrak{H}^{(\lambda)}$ a unitary space for each $\lambda > -\infty, < +\infty$, and $\sigma(\lambda)$ an N-function. $\mathfrak{H}$ is the* generalized direct sum of the $\mathfrak{H}^{(\lambda)}$ with the weight-function $\sigma(\lambda)$, *if the following is the case:*

Consider all functions $f(\lambda)$ which are defined for all $\lambda > -\infty, < +\infty$, and the values of which are subject to this restriction: $f(\lambda) \in \mathfrak{H}^{(\lambda)}$.

For these the notion of $\sigma(\lambda)$-summability is defined in some way[1]*, and whenever $f(\lambda)$ is $\sigma(\lambda)$-summable, its $\sigma(\lambda)$-integral $\int_{-} f(\lambda)\sqrt{d\sigma(\lambda)}$ is defined, being necessarily $\in \mathfrak{H}$. The following postulates are satisfied:*

- *(i) In order that a given $f(\lambda)$ be $\sigma(\lambda)$-summable, it is necessary and sufficient, that it fulfill these two conditions:*
  - *$(i)_1$ For every $\sigma(\lambda)$-summable $g(\lambda)$ the (complex-valued, numerical) function of λ, $(f(\lambda), g(\lambda))$ is $\sigma(\lambda)$-measurable.*
  - *$(i)_2$ If the (non-negative, numerical) function of λ $\|f(\lambda)\|^2$ is $\sigma(\lambda)$-measurable at all*[2]*, then it is $\sigma(\lambda)$-summable too. In other words: Then $\int \|f(\lambda)\|^2 \, d\sigma(\lambda)$ is finite.*
- *(ii) If $f(\lambda), g(\lambda)$ are both $\sigma(\lambda)$-summable, and if the (complex-valued, numerical) function of λ $(f(\lambda), g(\lambda))$ is summable at all*[3]*, then we have for*

$$f = \int_{-} f(\lambda)\sqrt{d\sigma(\lambda)}, \qquad g = \int_{-} g(\lambda)\sqrt{d\sigma(\lambda)}$$

(observe $f, g \in \mathfrak{H}$):

$$(f, g) = \int (f(\lambda), g(\lambda)) \, d\sigma(\lambda).$$

- *(iii) Every $f \in \mathfrak{H}$*[4] *can be put in the form*

$$f = \int_{-} f(\lambda)\sqrt{d\sigma(\lambda)}$$

[1] This notion will not be defined explicitly at present, all we require is, that it must be defined in some way that satisfies the postulates which follow.

[2] We will see in Remark 1 in this section, that it is $\sigma(\lambda)$-measurable, provided that $(i)_1$ holds.

[3] We will see in Remark 1 in this section, that it is $\sigma(\lambda)$-summable, as a consequence of (i). (More precisely: Of the necessary character of the conditions $(i)_1$, $(i)_2$.)

[4] It would be sufficient to require, that the set $\mathfrak{S}$ of all $f \in \mathfrak{H}$ which can be put in this form determines the linear, closed set $\mathfrak{H}$. In other words: that the (finite) linear aggregates formed from elements of $\mathfrak{S}$ be everywhere dense. This is so, because we will prove in Lemmas 1 and 2 in §5, that $\mathfrak{S}$ is a linear, closed set, irrespectively of whether we require (iii) or not.

with some $\sigma(\lambda)$-summable $f(\lambda)$.

We will now prove the statements made in footnotes 2 and 3, and in §5 the statement of footnote 4. For this reason we will avoid the use of (iii) in the rest of this section as well as in §5.

Remark 1. Assume that $f(\lambda)$ satisfies $(\mathrm{i})_1$. Then it is either $\sigma(\lambda)$-summable, or it is not. If it is, then we can apply $(\mathrm{i})_1$ to $g(\lambda) = f(\lambda)$ obtaining the $\sigma(\lambda)$-measurability of $(f(\lambda), f(\lambda)) = \|f(\lambda)\|^2$. If it is not, then $(\mathrm{i})_2$ must be violated, which means that $\|f(\lambda)\|^2$ is $\sigma(\lambda)$-measurable (but not $\sigma(\lambda)$-summable). Thus $\|f(\lambda)\|^2$ is $\sigma(\lambda)$-measurable in any event, proving the statement of footnote 2. Now $(\mathrm{i})_1$, $(\mathrm{i})_2$ imply, that $\|f(\lambda)\|^2$ is $\sigma(\lambda)$-summable, whenever $f(\lambda)$ is $\sigma(\lambda)$-summable.

If $f(\lambda), g(\lambda)$ are both $\sigma(\lambda)$-summable, then $(f(\lambda), g(\lambda))$ is $\sigma(\lambda)$-measurable by $(\mathrm{i})_1$, and it is even $\sigma(\lambda)$-summable by what we proved above, considering

$$|(f(\lambda), g(\lambda))| \leq \tfrac{1}{2}\|f(\lambda)\|^2 + \tfrac{1}{2}\|g(\lambda)\|^2.$$

This proves the statement of footnote 3, and the unrestricted validity of the formula in (ii).

Remark 2. If $f(\lambda)$ fulfills $(\mathrm{i})_1$, $(\mathrm{i})_2$, $\alpha f(\lambda)$ does too. If $f(\lambda), g(\lambda)$ fulfill $(\mathrm{i})_1$, $f(\lambda) + g(\lambda)$ does it too, and owing to

$$\|f(\lambda) + g(\lambda)\|^2 \leq 2\|f(\lambda)\|^2 + 2\|g(\lambda)\|^2$$

the same is the case for $(\mathrm{i})_2$ (remembering Remark 1). Thus (i) implies: If $f(\lambda), g(\lambda)$ are $\sigma(\lambda)$-summable, $\alpha f(\lambda)$, $f(\lambda) + g(\lambda)$ are $\sigma(\lambda)$-summable too.

Consider now

$$\int_{-} \alpha f(\lambda)\sqrt{d\sigma(\lambda)} - \alpha \int_{-} f(\lambda)\sqrt{d\sigma(\lambda)}$$

and

$$\int_{-} (f(\lambda) + g(\lambda))\sqrt{d\sigma(\lambda)} - \int_{-} f(\lambda)\sqrt{d\sigma(\lambda)} - \int_{-} g(\lambda)\sqrt{d\sigma(\lambda)}.$$

By (ii), they are orthogonal (in $\mathfrak{H}$) to every $\int_{-} h(\lambda)\sqrt{d\sigma(\lambda)}$. (iii), or even its weakened form from footnote 4 would give at once, that they vanish, but we can prove this without using (iii) at all: Both expressions are linear aggregates from $\mathfrak{S}$ (cf. footnote 4), and orthogonal to all elements of $\mathfrak{S}$, thus they are orthogonal to all linear aggregates from $\mathfrak{S}$ too—therefore to themselves, and so they must both vanish. So we have proved:

$$\int_{-} \alpha f(\lambda)\sqrt{d\sigma(\lambda)} = \alpha \int_{-} f(\lambda)\sqrt{d\sigma(\lambda)},$$

$$\int_{-}(f(\lambda)+g(\lambda))\sqrt{d\sigma(\lambda)} = \int_{-} f(\lambda)\sqrt{d\sigma(\lambda)} + \int_{-} g(\lambda)\sqrt{d\sigma(\lambda)}.$$

Remark 3. Put in the formula of (ii) $f(\lambda) = g(\lambda)$, and then replace $f(\lambda)$ by $f(\lambda) - g(\lambda)$, using the formula of Remark 2. Then

$$\left\| \int_{-} f(\lambda)\sqrt{d\sigma(\lambda)} - \int_{-} g(\lambda)\sqrt{d\sigma(\lambda)} \right\|^2 = \int \|f(\lambda) - g(\lambda)\|^2 \, d\sigma(\lambda)$$

results. This proves: We have

$$\int_{-} f(\lambda)\sqrt{d\sigma(\lambda)} = \int_{-} g(\lambda)\sqrt{d\sigma(\lambda)}$$

if and only if $f(\lambda) = g(\lambda)$ everywhere, except perhaps for a λ-set of $\sigma(\lambda)$-measure 0. (The λ-set where $f(\lambda) \neq g(\lambda)$ is $\sigma(\lambda)$-measurable in any event, as it is characterized by $\|f(\lambda) - g(\lambda)\|^2 \neq 0$, and the left side is a $\sigma(\lambda)$-measurable function of λ.)

If we change a $\sigma(\lambda)$-summable $f(\lambda)$ on a λ-set of $\sigma(\lambda)$-measure 0, then the validity of $(\mathrm{i})_1$, $(\mathrm{i})_2$, and thus the $\sigma(\lambda)$-summability of $f(\lambda)$ is unimpaired. So we have proved:

If $f(\lambda)$ is $\sigma(\lambda)$-summable, then all other $\sigma(\lambda)$-summable $h(\lambda)$'s with

$$\int_{-} f(\lambda)\sqrt{d\sigma(\lambda)} = \int_{-} h(\lambda)\sqrt{d\sigma(\lambda)}$$

are obtained, by changing $f(\lambda)$ arbitrarily on any λ-set of $\sigma(\lambda)$-measure 0.

This proves, that from the point of view $\sigma(\lambda)$-summability and of the value of $\int_{-} f(\lambda)\sqrt{d\sigma(\lambda)}$ (in $\mathfrak{H}$) $f(\lambda)$ may even be undefined on a λ-set of $\sigma(\lambda)$-measure 0. Therefore even the $\mathfrak{H}^{(\lambda)}$'s themselves may be undefined for a fixed λ-set of $\sigma(\lambda)$-measure 0.

4 Totally discontinuous weight-functions and the discrete direct sum

It is instructive to discuss the behavior of our Definition 1 of generalized direct sums for *totally discontinuous* $\sigma(\lambda)$'s, before we continue the general discussion. These $\sigma(\lambda)$'s arise in the following way:

As the $\sigma(\lambda)$-measure of $-\infty < \lambda < +\infty$ is finite, the number of points λ with a $\sigma(\lambda)$-measure $\geq 1/i$, $i = 1, 2, \ldots$, is finite, and therefore the number of points λ with a $\sigma(\lambda)$-measure $\neq 0$ is finite or countably infinite. Therefore we may write them as a (finite or infinite) sequence $\lambda_1, \lambda_2, \ldots$. Let their respective $\sigma(\lambda)$-measures be $a_1, a_2, \ldots$ (by assumption all $a_n > 0$). As we saw at the end of §2,

$$a_n = \sigma(\lambda_n + 0) - \sigma(\lambda_n - 0) = \sigma(\lambda_n) - \sigma(\lambda_n - 0).$$

We say that $\sigma(\lambda)$ is totally discontinuous, if the complementary set to $(\lambda_1, \lambda_2, \ldots)$ has the $\sigma(\lambda)$-measure 0. This means, that $-\infty < \lambda < +\infty$ has the same $\sigma(\lambda)$-measure as $(\lambda_1, \lambda_2, \ldots)$, that is

$$\sigma(+\infty) - \sigma(-\infty) = a_1 + a_2 + \cdots . \tag{4.1}$$

Total discontinuity implies, that the μ-set $\mu \leq \lambda$ (for any given λ) has the same $\sigma(\lambda)$-measure as the set of all λ_n with $\lambda_n \leq \mu$. That is: $\sigma(\lambda) - \sigma(-\infty) = \sum_{\lambda_n \leq \lambda} a_n$, and therefore

$$\sigma(\lambda) = a + \sum_{\lambda_n \leq \lambda} a_n. \tag{4.2}$$

Conversely: If $a, a_1, a_2, \ldots$ are real constants, $a_n > 0$ and $\sum_n a_n$ is finite, then (4.2) defines an N-function, the discontinuities of which are at $\lambda_1, \lambda_2, \ldots$ with the respective "jumps" $a_1, a_2, \ldots$. (It suffices to verify

$$\sigma(\lambda + 0) = \sigma(\lambda) = a + \sum_{\lambda_n \leq \lambda} a_n, \qquad \sigma(\lambda - 0) = a + \sum_{\lambda_n < \lambda} a_n.)$$

One sees, that (4.1) holds. So we see:

All totally discontinuous N-functions $\sigma(\lambda)$ can be obtained by (4.2), if $a, a_1, a_2, \ldots$ are any real constants, with $a_n > 0$ and $\sum_n a_n$ finite.

If the number of λ_n's is finite, then $\sigma(\lambda)$ is everywhere interval-wise constant, except at the points λ_n; the same is the case for an infinite but isolated set of λ_n's. If the λ_n's have condensation points, and in particular if they are everywhere dense, the outward appearance of $\sigma(\lambda)$ is very different, but all this is unimportant from the point of view of total discontinuity.

Assume now that $\sigma(\lambda)$ is totally discontinuous. Every subset of $(\lambda_1, \lambda_2, \ldots)$ is finite or countably infinite, therefore a Borel-set, therefore $\sigma(\lambda)$-measurable. Every subset of the complement of $(\lambda_1, \lambda_2, \ldots)$ is a subset of a set of $\sigma(\lambda)$-measure 0, and therefore $\sigma(\lambda)$-measurable too. Thus every set K is $\sigma(\lambda)$-measurable, and therefore every (numerical) function $f(\lambda)$ is such too. (This, by the way, is characteristic for total discontinuity, but we need not convince ourselves of it now.)

Now we verify easily:

$$\mu_{\sigma(\lambda)}(K) = \sum_{\lambda_n \in K} a_n;$$

if $f(\lambda) \geq 0$ then

$$\int f(\lambda)\, d\sigma(\lambda) = \sum_n f(\lambda_n) a_n;$$

if $f(\lambda)$ is complex-valued, then it is $\sigma(\lambda)$-summable, if $\int |f(\lambda)|\, d\sigma(\lambda) = \sum_n |f(\lambda_n)| a_n$ is finite, and in this case $\int f(\lambda)\, d\sigma(\lambda) = \sum_n f(\lambda_n) a_n$.

Thus all statements of Definition 1 which refer to $\sigma(\lambda)$-measurability, become void; in particular (i)$_1$ drops out completely. Owing to Remark 3, we need only to consider the $\mathfrak{H}^{(\lambda_n)}$ and the $f(\lambda_n)$, $n = 1, 2, \ldots$. So Definition 1 becomes this:

$\int_- f(\lambda)\sqrt{d\sigma(\lambda)}$ exists (that is: $f(\lambda)$ is $\sigma(\lambda)$-summable) if and only if $\sum_n \|f(\lambda_n)\|^2 a_n$ is finite (by (i)$_2$). These $\int_- f(\lambda)\sqrt{d\sigma(\lambda)}$ exhaust $\mathfrak{H}$ by (iii), and we have for them

$$\left(\int_- f(\lambda)\sqrt{d\sigma(\lambda)}, \int_- g(\lambda)\sqrt{d\sigma(\lambda)}\right) = \sum_n (f(\lambda_n), g(\lambda_n)) a_n.$$

These statements can be transformed and simplified further. Choose an $f^{(m)} \in \mathfrak{H}^{(\lambda_m)}$, and define:

$$f(\lambda_n) = \begin{cases} \dfrac{1}{\sqrt{a_m}} f^{(m)}, & n = m, \\ 0, & n \neq m. \end{cases} \tag{4.3}$$

Then $\sum_n \|f(\lambda_n)\|^2 a_n$ is finite, and thus $\int_- f(\lambda)\sqrt{d\sigma(\lambda)}$ exists, denote it by $[f^{(m)}]_m$ ($\in \mathfrak{H}$). Now the inner-product-formula becomes

$$([f^{(m)}]_m, [g^{(l)}]_l) = \begin{cases} (f^{(m)}, g^{(m)}), & m = l, \\ 0, & m \neq l. \end{cases}$$

Denote the set of all $[f^{(m)}]_m$, $f^{(m)} \in \mathfrak{H}^{(\lambda_m)}$, by $\mathfrak{H}_m^*$. Then the above formula states for $m = l$: $f^{(m)} \longleftrightarrow [f^{(m)}]_m$ is a one-to-one isomorphic mapping of $\mathfrak{H}^{(\lambda_m)}$ on the subset $\mathfrak{H}_m^*$ of $\mathfrak{H}$. As $\mathfrak{H}^{(\lambda_m)}$ is a linear, complete space, $\mathfrak{H}_m^*$ is too; that is, it is a linear closed set in $\mathfrak{H}$. For $m \neq l$ the formula gives: $\mathfrak{H}_m^*$, $\mathfrak{H}_l^*$ are orthogonal. Thus the $\mathfrak{H}_1^*, \mathfrak{H}_2^*, \ldots$ are mutually orthogonal linear, closed sets in $\mathfrak{H}$.

The existence of $\int_- f(\lambda)\sqrt{d\sigma(\lambda)}$, that is the finiteness of $\sum_n \|f(\lambda_n)\|^2 a_n = \sum_n \|[f(\lambda_n)]_n\|^2 a_n$ is obviously equivalent to the convergence of the (orthogonal) series $\sum_n [f(\lambda_n)]_n \sqrt{a_n}$ in $\mathfrak{H}$. (Cf. (11, p. 9), or (1, p. 67).) Therefore it is natural to consider the correspondence

$$\int_{-} f(\lambda)\sqrt{d\sigma(\lambda)} \longleftrightarrow \sum_n [f(\lambda_n)]_n \sqrt{a_n}.$$

The inner-product-formula shows, that this leaves inner products invariant. Thus it is a one-to-one isomorphic mapping of a subset $\mathfrak{H}_1$ of $\mathfrak{H}$ on another one $\mathfrak{H}_2$. By (iii) $\mathfrak{H}_1 = \mathfrak{H}$; and as $[f(\lambda_n)]_n \in \mathfrak{H}_n^*$, $\mathfrak{H}_2 \subset [\mathfrak{H}_1^*, \mathfrak{H}_2^*, \ldots]$ = the linear, closed set determined by $\mathfrak{H}_1^*, \mathfrak{H}_2^*, \ldots$. Now the two sides of this correspondence are clearly equal, if $f(\lambda)$ is the expression from (4.3). Thus the correspondence is identical in each $\mathfrak{H}_m^*$, therefore in $[\mathfrak{H}_1^*, \mathfrak{H}_2^*, \ldots]$, and a fortiori in $\mathfrak{H}_2$. It maps $\mathfrak{H}_2$ on itself: Therefore $\mathfrak{H}_1 = \mathfrak{H}_2$, thus $[\mathfrak{H}_1^*, \mathfrak{H}_2^*, \ldots] = \mathfrak{H}$ and

$$\int_{-} f(\lambda)\sqrt{d\sigma(\lambda)} = \sum_n [f(\lambda_n)]_n \sqrt{a_n}. \tag{4.4}$$

Conversely: Let $\mathfrak{H}$ be decomposed into a (finite or infinite) sequence of mutually orthogonal linear, closed sets $\mathfrak{H}_1^*, \mathfrak{H}_2^*, \ldots$, which determine the linear, closed set $\mathfrak{H}$. Let $\mathfrak{H}_n^*$ be isomorphic to some unitary space $\mathfrak{H}^{(\lambda_n)}$ by an isomorphism $f^{(n)} \longleftrightarrow [f^{(n)}]_n$ ($f^{(n)} \in \mathfrak{H}^{(\lambda_n)}$, $[f^{(n)}]_n \in \mathfrak{H}_n^* \subset \mathfrak{H}$). Define $\int_{-} f(\lambda)\sqrt{d\sigma(\lambda)}$ by equation (4.4) (so that it is defined if and only if the right side of (4.4) converges, that is $\sum_n \|f(\lambda_n)\|^2 a_n = \sum_n \|[f(\lambda_n)]_n\|^2 a_n$ is finite). Then one verifies immediately that all parts of Definition 1 hold, in the modified form which we gave them for the totally discontinuous $\sigma(\lambda)$.

Now $\mathfrak{H}$ is what one usually calls the direct sum of $\mathfrak{H}_1^*, \mathfrak{H}_2^*, \ldots$, and these spaces are isomorphic to $\mathfrak{H}^{(\lambda_1)}, \mathfrak{H}^{(\lambda_2)}, \ldots$ respectively. Thus the totally discontinuous case permits us to replace the unitary spaces $\mathfrak{H}^{(\lambda)}$ (of course only the essential ones in the sense of Remark 3: $\lambda = \lambda_1, \lambda_2, \ldots$) by isomorphic linear, closed subsets of $\mathfrak{H}$, and leads back to the common notion of the direct sum, as we stated it at the beginning of §3. (If the number of λ_n's is finite, we can even get rid of the a_n, by putting $a_1 = a_2 = \cdots = 1$. If it is infinite, this cannot be done, as $\sum_n a_n$ must be finite, in order to safeguard the boundedness of $\sigma(\lambda)$. Cf. §2, condition (iv).)

If $\sigma(\lambda)$ is not totally discontinuous, we may expect new phenomena. Then our use of unitary spaces $\mathfrak{H}^{(\lambda)}$ outside of $\mathfrak{H}$ becomes essential, because there is no natural way, as there was one in the totally discontinuous case, to replace them by isomorphic images inside of $\mathfrak{H}$. We will now discuss the properties of this general case more thoroughly.

5 Elementary properties of generalized integrals

From now on $\sigma(\lambda)$ is again an arbitrary N-function. As it was pointed out in §3, we will not use (iii) for a while.

Let, as in footnote 4, $\mathfrak{S}$ be the set of all $f \in \mathfrak{H}$ of the form $f = \int_{-} f(\lambda)\sqrt{d\sigma(\lambda)}$ for some $\sigma(\lambda)$-summable $f(\lambda)$. We will deduce some properties of $\mathfrak{S}$ and of this representation. The latter ones are those, which are important for our purposes, we state the former ones only, because they follow from those easily. The analysis of $\mathfrak{S}$ would be altogether unnecessary, if we wanted to use (iii) (this prescribes $\mathfrak{S} = \mathfrak{H}$, and thus determines $\mathfrak{S}$ completely). But we wish to establish the statement of footnote 4, according to which $\mathfrak{S}$ is linear, closed in consequence of (i), (ii) alone, thus making a seemingly weaker but really equivalent formulation of (iii) possible (cf. there). This will be secured by the statements of Lemmas 1 and 2 on $\mathfrak{S}$.

We will prove three Lemmas. They are as follows:

Lemma 1. *$\mathfrak{S}$ is a linear set. If $f, g \in \mathfrak{S}$,*

$$f = \int_{-} f(\lambda)\sqrt{d\sigma(\lambda)}, \qquad g = \int_{-} g(\lambda)\sqrt{d\sigma(\lambda)},$$

then

$$\alpha f = \int_{-} \alpha f(\lambda)\sqrt{d\sigma(\lambda)}, \qquad f + g = \int_{-} (f(\lambda) + g(\lambda))\sqrt{d\sigma(\lambda)}.$$

Proof. Immediately by Remark 3 in §2. □

Lemma 2. *$\mathfrak{S}$ is a closed set. If $g_n \in \mathfrak{S}$, $n = 1, 2, \ldots$,*

$$g_n = \int_{-} g_n(\lambda)\sqrt{d\sigma(\lambda)},$$

and if $\lim_{n\to\infty} g_n$ exists (in the metric, "strong", topology of $\mathfrak{H}$), then a function $g(\lambda)$ (defined for all $\lambda > -\infty, < +\infty$, $g(\lambda) \in \mathfrak{H}^{(\lambda)}$), exists, which has this property:
(I) *A subsequence $p_1, p_2, \ldots$ exists, such that $\lim_{r\to\infty} g_{p_r}(\lambda) = g(\lambda)$ (in the metric, "strong", topology of $\mathfrak{H}^{(\lambda)}$) for all $\lambda > -\infty, < +\infty$, except perhaps for a λ-set of $\sigma(\lambda)$-measure 0.*

(I) *determines $g(\lambda)$ uniquely, except for an arbitrary λ-set of $\sigma(\lambda)$-measure 0, irrespective of what the sequence $p_1, p_2, \ldots$ happened to be. This $g(\lambda)$ is $\sigma(\lambda)$-summable and*

$$\lim_{n\to\infty} g_n = \int_{-} g(\lambda)\sqrt{d\sigma(\lambda)}.$$

Proof. The statement on $\mathfrak{S}$ follows from the rest of the Lemma, because the final equation shows, that if $\lim_{n\to\infty} g_n$ exists at all (all $g_n \in \mathfrak{S}$!), then it belongs to $\mathfrak{S}$. Thus $\mathfrak{S}$ is closed. We need therefore to consider only the other parts of the Lemma.

Assume now $g_n \in \mathfrak{S}$, $g_n = \int_{-} g_n(\lambda)\sqrt{d\sigma(\lambda)}$, $\lim_{n\to\infty} g_n$ exists. Then we have $\lim_{m,n\to\infty}\|g_m - g_n\| = 0$, or by Remark 3 in §2

$$\lim_{m,n\to\infty} \int \|g_m(\lambda) - g_n(\lambda)\|^2 \, d\sigma(\lambda) = 0.$$

Thus we have a "Convergence in the mean", and we can proceed in the customary way. The above equation means, that there exists for every $\epsilon > 0$ an $s = s(\epsilon)$, such that $m, n \geq s$ imply

$$\int \|g_m(\lambda) - g_n(\lambda)\|^2 \, d\sigma(\lambda) < \epsilon.$$

Denote the set of all λ with $\|g_m(\lambda) - g_n(\lambda)\| \geq \delta > 0$ by $K_1(m, n, \delta)$. Then we see: $\mu_{\sigma(\lambda)}(K_1(m, n, \delta)) < \epsilon/\delta^2$ if $m, n \geq s(\epsilon)$. Thus $\mu_{\sigma(\lambda)}(K_1(m, n, 1/2^r)) < 1/2^r$ if $m, n \geq s(1/8^r)$. Choose now a sequence $p_1 < p_2 < \cdots$ with $p_r \geq s(1/8^r)$; then we have $\mu_{\sigma(\lambda)}(K_1(p_r, p_{r+1}, 1/2^r)) < 1/2^r$. Put

$$K_2(r) = \sum_{t=r}^{\infty} K_1(p_t, p_{t+1}, 1/2^t), \qquad K_3 = \prod_{r=1}^{\infty} K_2(r),$$

then

$$\mu_{\sigma(\lambda)}(K_2(r)) < \frac{1}{2^r} + \frac{1}{2^{r+1}} + \cdots = \frac{1}{2^{r-1}}, \qquad \mu_{\sigma(\lambda)}(K_3) \leq \mu_{\sigma(\lambda)}(K_2(r)) < \frac{1}{2^{r-1}}$$

for every $r = 1, 2, \ldots$ and thus $\mu_{\sigma(\lambda)}(K_3) = 0$.

If $\lambda \notin K_3$, then there exists an r with $\lambda \notin K_2(r)$, and thus $\lambda \notin K_1(p_t, p_{t+1}, 1/2^t)$ for all $t \geq r$; this means $\|g_{p_t}(\lambda) - g_{p_{t+1}}(\lambda)\| < 1/2^t$ for all $t \geq r$. Then $\lim_{t\to\infty} g_{p_t}(\lambda)$ must exist (in the metric, "strong", topology of $\mathfrak{H}^{(\lambda)}$). So we see: $\lim_{r\to\infty} g_{p_r}(\lambda)$ exists for all $\lambda > -\infty, < +\infty$, except perhaps for a λ-set of $\sigma(\lambda)$-measure 0 (the above K_3). Define $g(\lambda) = \lim_{r\to\infty} g_{p_r}(\lambda)$ for all λ's for which this limit exists, for all other λ $g(\lambda)$ may be anything (their set has the $\sigma(\lambda)$-measure 0).

So we found a $g(\lambda)$ which fulfills the condition (I).

Let us now consider any such $g(\lambda)$, and see whether the final equation of our Lemma holds for it.

In what follows, we will always neglect λ-sets of $\sigma(\lambda)$-measure 0. We have always $\lim_{r\to\infty} g_{p_r}(\lambda) = g(\lambda)$, therefore

$$\lim_{r\to\infty}(g_{p_r}(\lambda) - g_n(\lambda)) = g(\lambda) - g_n(\lambda)$$

and thus

$$\lim_{r\to\infty}\|g_{p_r}(\lambda) - g_n(\lambda)\|^2 = \|g(\lambda) - g_n(\lambda)\|^2.$$

As $\|g_{p_r}(\lambda) - g_n(\lambda)\|^2 \geq 0$, $\|g(\lambda) - g_n(\lambda)\|^2 \geq 0$, this implies by a well-known property of Lebesgue integrals[5], that

$$\liminf_{r\to\infty} \int \|g_{p_r}(\lambda) - g_n(\lambda)\|^2 \, d\sigma(\lambda) \geq \int \|g(\lambda) - g_n(\lambda)\|^2 \, d\sigma(\lambda).$$

The left side is equal to $\liminf_{r\to\infty}\|g_{p_r} - g_n\|^2$ by Remark 3 in §2. If $n \geq s(\epsilon)$ then as for $r \to \infty$ finally $p_r \geq s(\epsilon)$ too, this expression is $\leq \epsilon$. So we see: For $n \geq s(\epsilon)$ we have

$$\int \|g(\lambda) - g_n(\lambda)\|^2 \, d\sigma(\lambda) \leq \epsilon.$$

If $f(\lambda)$ is $\sigma(\lambda)$-summable, then $(g_m(\lambda), f(\lambda))$ is $\sigma(\lambda)$-measurable for all $m = 1, 2, \ldots$ by $(\mathrm{i})_1$, thus $(g_{p_r}(\lambda) - g_n(\lambda), f(\lambda))$ is $\sigma(\lambda)$-measurable too, and so is its limit for $r \to \infty$, $(g(\lambda) - g_n(\lambda), f(\lambda))$. So $g(\lambda) - g_n(\lambda)$ fulfills $(\mathrm{i})_1$. As we saw it fulfills $(\mathrm{i})_2$ too (if n is great enough, for instance if $n \geq s(1)$). Thus (i) applies: $g(\lambda) - g_n(\lambda)$ is $\sigma(\lambda)$-summable, and with it $g(\lambda) = [g(\lambda) - g_n(\lambda)] + g_n(\lambda)$ too.

Put $g = \int_{-} g(\lambda)\sqrt{d\sigma(\lambda)}$. The inequality we had before becomes now: For $n \geq s(\epsilon)$ we have $\|g - g_n\|^2 \leq \epsilon$. Thus

$$\lim_{n\to\infty} g_n = g = \int_{-} g(\lambda)\sqrt{d\sigma(\lambda)},$$

[5] Cf. for instance (13, p. 443, Theorem 15). Using those notations, put $S(P) = 0$, then this obtains: If all $f_k(P) \leq 0$, then

$$\limsup_{k\to\infty}\left[\int f_k(P)\, dw\right] \leq \int \left[\limsup_{k\to\infty} f_k(P)\right] dw.$$

This is even so if the case $\limsup_{k\to\infty}[\int f_k(P)\, dw] = -\infty$ occurs, which was excluded loc. cit., in fact then it is trivial. Replacing $f_k(P)$ by $-f_k(P)$ and assuming the existence of $\lim_{k\to\infty} f_k(P)$, this becomes: If $f_k(P) \geq 0$, then

$$\liminf_{k\to\infty}\left[\int f_k(P)\, dw\right] \geq \int \left[\lim_{k\to\infty} f_k(P)\right] dw.$$

This is, disregarding the notation, the desired inequality. The entire argument holds unchanged for Lebesgue-Stieltjes-integrals, or their representation by Lebesgue integrals, cf. (4, p. 198), may be used to carry it over.

establishing the final equation of our Lemma.

This equation, together with Remark 3 in §2, proves, that $g(\lambda)$ is uniquely determined, except for an arbitrary λ-set of $\sigma(\lambda)$-measure 0. □

Lemma 3. *Let $f(\lambda)$ be an arbitrary function, defined for all $\lambda > -\infty, < +\infty$, $f(\lambda) \in \mathfrak{H}^{(\lambda)}$. Let $\mathfrak{S}'$ be the set of all those $g \in \mathfrak{S}$, $g = \int_{-} g(\lambda)\sqrt{d\sigma(\lambda)}$, for which the (complex-valued, numerical) function of λ $(f(\lambda), g(\lambda))$ is $\sigma(\lambda)$-measurable. (By Remark 3 in §2 it does not matter, which $g(\lambda)$ of a given g we choose.) Then $\mathfrak{S}'$ is a closed linear set.*

Proof. Assume $g, h \in \mathfrak{S}'$. Then

$$g = \int_{-} g(\lambda)\sqrt{d\sigma(\lambda)}, \qquad h = \int_{-} h(\lambda)\sqrt{d\sigma(\lambda)},$$

and $(f(\lambda), g(\lambda))$, $(f(\lambda), h(\lambda))$ are $\sigma(\lambda)$-measurable. From this $\alpha g = \int_{-} \alpha g(\lambda)\sqrt{d\sigma(\lambda)}$, $g + h = \int_{-}(g(\lambda) + h(\lambda))\sqrt{d\sigma(\lambda)}$ follow, and

$$(f(\lambda), \alpha g(\lambda)) = \overline{\alpha}(f(\lambda), g(\lambda)), \qquad (f(\lambda), g(\lambda) + h(\lambda)) = (f(\lambda), g(\lambda)) + (f(\lambda), h(\lambda))$$

are $\sigma(\lambda)$-measurable. Thus $\alpha g, g + h \in \mathfrak{S}'$, and so $\mathfrak{S}'$ is a linear set.

Assume now that g is a condensation point of $\mathfrak{S}'$. Choose $g_1, g_2, \ldots \in \mathfrak{S}'$ with $\lim_{n\to\infty} g_n = g$. Put

$$g_n = \int_{-} g_n(\lambda)\sqrt{d\sigma(\lambda)}, \qquad g = \int_{-} g(\lambda)\sqrt{d\sigma(\lambda)}.$$

Then we have by Lemma 2 a subsequence $p_1, p_2, \ldots$ of $1, 2, \ldots,$ so that $\lim_{r\to\infty} g_{p_r}(\lambda) = g(\lambda)$, except perhaps for a λ-set of $\sigma(\lambda)$-measure 0. Similarly $\lim_{r\to\infty}(f(\lambda), g_{p_r}(\lambda)) = (f(\lambda), g(\lambda))$. As all functions of λ, $(f(\lambda), g_n(\lambda))$ are $\sigma(\lambda)$-measurable, we have this now for $(f(\lambda), g(\lambda))$ too. Therefore $g \in \mathfrak{S}'$. Thus $\mathfrak{S}'$ is closed too. □

6 Measurable families and existence and uniqueness of generalized direct sums

As the linear, closed set character of $\mathfrak{S}$ is established, we can be satisfied concerning the observations of footnote 4, and we will use from now on (iii): That is $\mathfrak{S} = \mathfrak{H}$.

Let $\varphi_1, \varphi_2, \ldots$ be a complete normalized orthogonal set in $\mathfrak{H}$ (an everywhere dense sequence would do just as well). Put

$$\varphi_n = \int_{-} \varphi_n(\lambda)\sqrt{d\sigma(\lambda)}.$$

Consider now the sequence $\varphi_1(\lambda), \varphi_2(\lambda), \ldots$ in the unitary space $\mathfrak{H}^{(\lambda)}$. Apply to it the "orthogonalization process" of E. Schmidt, cf. (11, p. 13), or (1, p. 69). Then a normalized orthogonal set $\psi_1(\lambda), \psi_2(\lambda), \ldots$ (in $\mathfrak{H}^{(\lambda)}$) arises. It has $k = k(\lambda) = 0, 1, 2, \ldots, \infty$ elements, where k is the "rank" of $\varphi_1(\lambda), \varphi_2(\lambda), \ldots$ (the number of linearly independent elements in that sequence). If $k \neq \infty$, we define

$$\psi_{k+1}(\lambda) = \psi_{k+2}(\lambda) = \cdots = 0,$$

thus $\psi_1(\lambda), \psi_2(\lambda), \ldots$ is now an infinite sequence in any event—it is an orthogonal set, $\psi_n(\lambda)$ being normalized if $n \leq k$, and vanishing if $n > k$.

If one visualizes nature of the "orthogonalization process", one sees at once, that the $(\varphi_m(\lambda), \psi_n(\lambda))$ and the $\|\psi_n(\lambda)\|^2$ with $m, n \leq l$ are Baire-functions of the $(\varphi_p(\lambda), \varphi_q(\lambda))$ with $p, q \leq l$. These are $\sigma(\lambda)$-measurable, therefore the $(\varphi_m(\lambda), \psi_n(\lambda))$, and the $\|\psi_n(\lambda)\|^2$ are too, and so is

$$k(\lambda) = \sum_{1}^{\infty} \|\psi_n(\lambda)\|^2.$$

We now propose to prove:

If $f(\lambda)$ is a function, defined for all $\lambda > -\infty, < +\infty$, $f(\lambda) \in \mathfrak{H}^{(\lambda)}$, and if all (complex-valued-numerical) functions of λ $(f(\lambda), \psi_n(\lambda))$, $n = 1, 2, \ldots$, are $\sigma(\lambda)$-measurable, then $f(\lambda)$ fulfills (i)$_1$.

As we know, $\varphi_p(\lambda)$ is a linear aggregate of those $\psi_1(\lambda), \ldots, \psi_k(\lambda)$ $(k = k(\lambda))$ which have indices $\leq p$, thus of $\psi_1(\lambda), \ldots, \psi_l(\lambda)$ $(l = \min(k(\lambda), p))$. These form a normalized, orthogonal set, so

$$\varphi_p(\lambda) = \sum_{r=1}^{l} (\varphi_p(\lambda), \psi_r(\lambda)) \psi_r(\lambda).$$

Replacing $\sum_1^l$ by $\sum_1^p$ adds only vanishing terms, so we may do it. Now we have

$$(f(\lambda), \varphi_p(\lambda)) = \sum_{r=1}^{p} (f(\lambda), \psi_r(\lambda)) \overline{(\varphi_p(\lambda), \psi_r(\lambda))}$$

and so it is $\sigma(\lambda)$-measurable too. Form the $\mathfrak{S}'$ (in the sense of Lemma 3) for $f(\lambda)$, then we have proved $\varphi_p \in \mathfrak{S}'$, $p = 1, 2, \ldots$. By Lemma 3 this implies $\mathfrak{S}' = \mathfrak{H}$. Thus $(f(\lambda), g(\lambda))$ is $\sigma(\lambda)$-measurable for every $\sigma(\lambda)$-summable $g(\lambda)$, in other words: $f(\lambda)$ fulfills (i)$_1$. This completes the proof.

As

$$(\psi_m(\lambda), \psi_n(\lambda)) = \begin{cases} \|\psi_m(\lambda)\|^2, & m = n, \\ 0, & m \neq n, \end{cases}$$

it is $\sigma(\lambda)$-measurable in any event. Thus $\psi_m(\lambda)$ fulfills (i)$_1$. As $\|\psi_m(\lambda)\|^2 \leq 1$,

$$\int \|\psi_m(\lambda)\|^2 \, d\sigma(\lambda) \leq \int 1 \, d\sigma(\lambda) = \sigma(+\infty) - \sigma(-\infty)$$

is finite, and so (i)$_2$ holds too. Therefore $\psi_m(\lambda)$ is $\sigma(\lambda)$-summable by (i). Put

$$\psi_m = \int_{-} \psi_m(\lambda)\sqrt{d\sigma(\lambda)}, \qquad m = 1, 2, \ldots.$$

Let K_4 be the set of those $\lambda > -\infty, < +\infty$, for which the non-vanishing $\psi_n(\lambda)$'s (that is: the $k(\lambda)$ first ones) do not form a complete normalized orthogonal set in $\mathfrak{H}^{(\lambda)}$. For every $\lambda \in K_4$ choose an $f(\lambda)$ which is orthogonal to all $\psi_n(\lambda)$, and $\|f(\lambda)\|^2 = 1$. For all $\lambda \notin K_4$ put $f(\lambda) = 0$. Then we have always $(f(\lambda), \psi_n(\lambda)) = 0$, so this is $\sigma(\lambda)$-measurable. Thus $f(\lambda)$ fulfills (i)$_1$, and by Remark 1 in §2, $\|f(\lambda)\|^2$ must be $\sigma(\lambda)$-measurable. So K_4 is a $\sigma(\lambda)$-measurable set. Now

$$\int \|f(\lambda)\|^2 \, d\sigma(\lambda) \leq \int 1 \, d\sigma(\lambda) = \sigma(+\infty) - \sigma(-\infty)$$

is finite, so $f(\lambda)$ fulfills (i)$_2$ too. Therefore $f(\lambda)$ is $\sigma(\lambda)$-summable by (i), put $f = \int_{-} f(\lambda)\sqrt{d\sigma(\lambda)}$. The expression we used above for $(f(\lambda), \varphi_p(\lambda))$ shows, that in the present case this vanishes, because all $(f(\lambda), \psi_n(\lambda))$ vanish. Now apply (ii):

$$(f, \varphi_p) = \int (f(\lambda), \varphi_p(\lambda)) \, d\sigma(\lambda) = 0,$$

$$\|f\|^2 = \int \|f(\lambda)\|^2 \, d\sigma(\lambda) = \int 1_{K_4}(\lambda) \, d\sigma(\lambda) = \mu_{\sigma(\lambda)}(K_4).$$

f is orthogonal to all $\varphi_1, \varphi_2, \ldots$, therefore $f = 0$, $\|f\|^2 = 0$, and $\mu_{\sigma(\lambda)}(K_4) = 0$. So K_4 has the $\sigma(\lambda)$-measure 0.

Thus the non-vanishing $\psi_n(\lambda)$'s (that is: the $k(\lambda)$ first ones) form a complete normalized orthogonal set in $\mathfrak{H}^{(\lambda)}$ for every $\lambda > -\infty, < +\infty$, except perhaps for a λ-set of $\sigma(\lambda)$-measure 0. For these λ's however, we can redefine the $\psi_n(\lambda)$ and $k(\lambda)$ so that this should be true for them too, while the $\psi_n(\lambda)$ with $n > k(\lambda)$ again vanish. This will not effect the above derived criterium for (i)$_1$ in terms of the

$\psi_n(\lambda)$'s, nor the $\sigma(\lambda)$-summability of $\psi_n(\lambda)$, nor the value of ψ_n. And now $k(\lambda)$ is the dimensionality of the unitary space $\mathfrak{H}^{(\lambda)}$.

The results obtained so far can be best formulated with the help of this definition:

Definition 2. *Let $k(\lambda)$ be the dimensionality of $\mathfrak{H}^{(\lambda)}$. Denote the set of all $\lambda > -\infty$, $< +\infty$ with $k(\lambda) = n$ by Δ_n, and the set of those with $k(\lambda) \geq n$ by H_n. ($n = 1, 2, \ldots, \infty$, $H_n = \Delta_n + \Delta_{n+1} + \cdots + \Delta_\infty$.)*

Assume that for every $\lambda > -\infty$, $< +\infty$ a complete normalized orthogonal set $\psi_1(\lambda), \psi_2(\lambda), \ldots$ in $\mathfrak{H}^{(\lambda)}$ is given (the length of this sequence is of course $k(\lambda)$, that is n for $\lambda \in \Delta_n$). Then every function $f(\lambda)$ ($-\infty < \lambda < +\infty$, $f(\lambda) \in \mathfrak{H}^{(\lambda)}$) can be written in the form

$$f(\lambda) = \sum_m x_m(\lambda)\psi_m(\lambda), \qquad x_m(\lambda) = (f(\lambda), \psi_m(\lambda));$$

and the complex numbers $x_m(\lambda)$ are only subject to these two restrictions:

(α) $x_m(\lambda)$'s domain is characterized by $m \leq k(\lambda)$, that is by $\lambda \in H_m$,

(β) $\sum_m |x_m(\lambda)|^2$ is finite for every $\lambda > -\infty$, $< +\infty$.

If the $\sigma(\lambda)$-summability of $f(\lambda)$ is equivalent to the two further restrictions

(γ) $x_m(\lambda)$ is a measurable function of λ for every $m = 1, 2, \ldots$,

(δ) $\displaystyle\sum_{m=1}^{\infty} \int_{H_m} |x_m(\lambda)|^2 \, d\sigma(\lambda)$ is finite,

then the systems $\psi_m(\lambda)$ form a measurable family.

Remark. Considering the non-negativity of the integrands, we have

$$\begin{aligned}
\sum_{m=1}^{\infty} \int_{H_m} |x_m(\lambda)|^2 \, d\sigma(\lambda) &= \sum_{m=1}^{\infty} \sum_{n=m}^{\infty} \int_{\Delta_n} |x_m(\lambda)|^2 \, d\sigma(\lambda) \\
&= \sum_{n=1}^{\infty} \sum_{m=1}^{n} \int_{\Delta_n} |x_m(\lambda)|^2 \, d\sigma(\lambda) \\
&= \sum_{n=1}^{\infty} \int_{\Delta_n} \sum_{m=1}^{n} |x_m(\lambda)|^2 \, d\sigma(\lambda) \\
&= \int \sum_m |x_m(\lambda)|^2 \, d\sigma(\lambda).
\end{aligned}$$

Thus the finiteness of the expression at the extreme left implies the finiteness of $\sum_m |x_m(\lambda)|^2$ for all $\lambda > -\infty$, $< +\infty$, except perhaps for a set of $\sigma(\lambda)$-measure 0. But by Remark 3 in §3, such a set does not matter. So we see, that (δ) implies (β), and that the $\sigma(\lambda)$-summable $f(\lambda)$'s are characterized by (α), (γ), (δ) alone.

We now prove:

Theorem I. *(i)* $k(\lambda)$ is a measurable function, all Δ_n, H_n are measurable sets.*

(ii) Measurable families $\psi_m(\lambda)$ exist.*

(iii) If $\psi_m^0(\lambda)$ is a measurable family, then all others $\psi_m(\lambda)$ are obtained by choosing for each $\lambda > -\infty, < +\infty$ a unitary matrix $\{u_{mn}(\lambda)\}$ of degree $k(\lambda)$, such that $u_{mn}(\lambda)$ is a measurable function of λ for every pair $m, n = 1, 2, \ldots$, and putting*

$$\psi_m(\lambda) = \sum_n u_{mn}(\lambda)\psi_n^0(\lambda).$$

Then of course

$$u_{mn}(\lambda) = (\psi_m(\lambda), \psi_n^0(\lambda)).$$

Proof. Ad (i).* We defined at the beginning of this section a function $k(\lambda)$, which was then shown to be measurable. We saw later, that it coincides with the dimensionality of $\mathfrak{H}^{(\lambda)}$. Δ_n and H_n are the sets $k(\lambda) = n$ and $\geq n$, respectively.

Ad (ii).* Consider the $\psi_n(\lambda)$ defined before Definition 2. We proved that Definition 1, (i)$_1$ is equivalent to Definition 2, (γ), and Definition 1, (i)$_2$ is clearly identical with Definition 2, (δ). Thus Definition 1, (i) proves, that these $\psi_n(\lambda)$ form a measurable family.

Ad (iii). Sufficiency:* If the $\psi_n(\lambda)$ have this form, then the two representations

$$f(\lambda) = \sum_m x_m^0(\lambda)\psi_m^0(\lambda), \qquad f(\lambda) = \sum_m x_m(\lambda)\psi_m(\lambda)$$

are connected by the formulae

$$x_m(\lambda) = \sum_n \overline{u_{mn}(\lambda)}x_n^0(\lambda), \qquad x_n^0(\lambda) = \sum_m u_{mn}(\lambda)x_m(\lambda).$$

As $u_{mn}(\lambda)$ is measurable, (γ) for $\psi_m(\lambda)$ is equivalent to (γ) for $\psi_m^0(\lambda)$. As

$$\|f(\lambda)\|^2 = \sum_m |x_m^0(\lambda)|^2 = \sum_m |x_m(\lambda)|^2,$$

the same is the case for (δ). Thus $\psi_m(\lambda)$ is a measurable family along with $\psi_m^0(\lambda)$.

Necessity: Let $\psi_m(\lambda), \psi_m^0(\lambda)$ be two measurable families. As $\psi_n(\lambda)$ (n fixed, put $\psi_n(\lambda) = 0$ for $\lambda \notin H_n$) fulfills (γ), (δ) for the family $\psi_m(\lambda)$, it must also do this for the family $\psi_m^0(\lambda)$. Thus $u_{mn}(\lambda) = (\psi_m(\lambda), \psi_n^0(\lambda))$ is a measurable function of λ by (γ). The other statements follow from the fact, that $\psi_m(\lambda)$ and $\psi_m^0(\lambda)$ are both complete normalized orthogonal sets in $\mathfrak{H}^{(\lambda)}$. □

The theorem which follows may be considered as a converse of Theorem I:

Theorem II. *Let for every $\lambda > -\infty, < +\infty$ a unitary space $\mathfrak{H}^{(\lambda)}$ be given, and a complete normalized orthogonal set $\psi_1(\lambda), \psi_2(\lambda), \ldots$ in $\mathfrak{H}^{(\lambda)}$. Let $k(\lambda)$ be the dimensionality of $\mathfrak{H}^{(\lambda)}$, Δ_n and H_n the λ-sets with $k(\lambda) = n$ and $\geq n$, respectively ($n = 1, 2, \ldots, \infty$, $H_n = \Delta_n + \Delta_{n+1} + \cdots + \Delta_\infty$).*

Let $\sigma(\lambda)$ be an N-function, $k(\lambda)$ being $\sigma(\lambda)$-measurable. Therefore the sets Δ_n, H_n are such, too.

Consider all complex-valued functions $x_m(\lambda)$ of the two variables $m = 1, 2, \ldots$, $\lambda > -\infty, < +\infty$, which fulfill the conditions (α), (γ), (δ) of Definition 2. Defining for $x \equiv x_m(\lambda)$, $y \equiv y_m(\lambda)$

$$\alpha x \equiv \alpha x_m(\lambda), \qquad x + y \equiv x_m(\lambda) + y_m(\lambda),$$

$$(x, y) = \sum_{m=1}^{\infty} \int_{H_m} x_m(\lambda) \overline{y_m(\lambda)}\, d\sigma(\lambda),$$

we obtain a unitary space $\mathfrak{H}$[6].

If a function $f(\lambda)$ of $\lambda > -\infty, < +\infty$, $f(\lambda) \in \mathfrak{H}^{(\lambda)}$ is given, define $x_m(\lambda)$ by

$$f(\lambda) = \sum_m x_m(\lambda)\psi_m(\lambda), \qquad x_m(\lambda) = (f(\lambda), \psi_m(\lambda)).$$

Define $f(\lambda)$ to be $\sigma(\lambda)$-summable by $x \equiv x_m(\lambda) \in \mathfrak{H}$, and then define

$$\int_{-} f(\lambda)\sqrt{d\sigma(\lambda)} = x \equiv x_m(\lambda).$$

With these definitions $\mathfrak{H}$ is the generalized direct sum of the $\mathfrak{H}^{(\lambda)}$ with the weight-function $\sigma(\lambda)$, and the systems $\psi_m(\lambda)$ form a measurable family.

Proof. We must first verify the conditions (i), (ii), (iii) of Definition 1.

[6]Cf. (11, pp. 23–32) or (1, pp. 108–111). The most convenient way to look at this $\mathfrak{H}$ is, to consider the $x_m(\lambda)$ as functions in the m, λ-space defined by $m \leq k(\lambda)$, that is by $\lambda \in H_m$, in which $\sum_{m=1}^{\infty} \int_{H_m} \cdots d\sigma(\lambda)$ plays the role of the Lebesgue-(Stieltjes) integral.

Ad (i): Put $f(\lambda) = \sum_n x_n(\lambda)\psi_n(\lambda)$ and similarly $g(\lambda) = \sum_n y_n(\lambda)\psi_n(\lambda)$, the $\sigma(\lambda)$-summability of $g(\lambda)$ means $y \equiv y_m(\lambda) \in \mathfrak{H}$. (i)$_1$ requires the measurability of

$$(f(\lambda), g(\lambda)) = \sum_n x_n(\lambda)\overline{y_n(\lambda)}$$

for all $y \equiv y_m(\lambda) \in \mathfrak{H}$. As we can put

$$y_n(\lambda) = \begin{cases} 1, & n = m, \\ 0, & n \neq m, \end{cases}$$

the measurability of each $x_m(\lambda)$ is necessary; as all $y_m(\lambda)$ are measurable, this is sufficient. (i)$_2$ states, that

$$\int \|f(\lambda)\|^2 \, d\sigma(\lambda) = \int \sum_m |x_m(\lambda)|^2 \, d\sigma(\lambda) = \sum_m \int_{H_m} |x_m(\lambda)|^2 \, d\sigma(\lambda)$$

(cf. the Remark after Definition 2) is finite. So (i)$_1$, (i)$_2$ are equivalent to Definition 2, (γ), (δ), and as Definition 2, (α) is automatically fulfilled, so $x \equiv x_m(\lambda) \in \mathfrak{H}$. Thus they are equivalent to the $\sigma(\lambda)$-summability of $f(\lambda)$.

Ad (ii): This follows from the computation

$$\begin{aligned}(x, y) &= \sum_{m=1}^{\infty} \int_{H_m} x_m(\lambda)\overline{y_m(\lambda)} \, d\sigma(\lambda) \\ &= \int \sum_m x_m(\lambda)\overline{y_m(\lambda)} \, d\sigma(\lambda) \\ &= \int (f(\lambda), g(\lambda)) \, d\sigma(\lambda).\end{aligned}$$

The replacement of $\sum_m \int_{H_m}$ by $\int \sum_m$ is legitimate for $x = y$, because everything is non-negative (cf. above); replacing x by $\frac{1}{2}(x+y)$, $\frac{1}{2}(x-y)$, and subtracting gives its real part in the case of arbitrary x, y; replacing x, y by ix, y gives its imaginary part.

Ad (iii): Obvious.

Now the application of Definition 2 shows immediately that the $\psi_m(\lambda)$ form a measurable family. □

7 Operators, resolutions of unity, and characterization of the decomposition

We continue the analysis of an $\mathfrak{H}$, which is the generalized direct sum of a family of $\mathfrak{H}^{(\lambda)}$'s, with the weight function $\sigma(\lambda)$.

Let $\alpha(\lambda)$ be a numerical (complex-valued), $\sigma(\lambda)$-measurable function of λ, which is bounded, if a suitable set of $\sigma(\lambda)$-measure 0 is disregarded. Application of the criteria of Definition 1, (i) (or just as well of Definition 2, (α), (γ), (δ)), shows, that $\alpha(\lambda)f(\lambda)$ is $\sigma(\lambda)$-summable if $f(\lambda)$ is such. Remembering Remark 3 in §3, we see, that $f = \int_- f(\lambda)\sqrt{d\sigma(\lambda)}$ determines $f^* = \int_- \alpha(\lambda)f(\lambda)\sqrt{d\sigma(\lambda)}$ and thus an operator A is defined by $Af = f^*$. We denote this A by $\alpha(\Lambda)$:

$$\alpha(\Lambda)\int_- f(\lambda)\sqrt{d\sigma(\lambda)} = \int_- \alpha(\lambda)f(\lambda)\sqrt{d\sigma(\lambda)}.$$

This $\alpha(\Lambda)$ has the following properties:

(a) $\alpha(\Lambda)$ is a bounded operator.

(b) $\alpha(\Lambda)$ is unaffected if $\alpha(\lambda)$ is changed on a set of $\sigma(\lambda)$-measure 0.

(c) $\alpha(\lambda) \equiv a =$ Constant, or $\equiv \overline{\beta(\lambda)}$, or $\equiv \beta(\lambda)+\gamma(\lambda)$ or $\equiv \beta(\lambda)\gamma(\lambda)$ implies $\alpha(\Lambda) = a1$, or $= \beta(\Lambda)^*$ (the adjoint!), or $= \beta(\Lambda) + \gamma(\Lambda)$ or $= \beta(\Lambda)\gamma(\Lambda)$, respectively.

(d) If $\lim_{n\to\infty}\alpha_n(\lambda) = \alpha(\lambda)$ for every λ, and if the $\alpha_n(\lambda)$ are uniformly bounded for $n = 1, 2, \ldots$, then $\lim_{n\to\infty}\alpha_n(\Lambda) = \alpha(\Lambda)$. (In the strong sense, that is $\lim_{n\to\infty}\|\alpha_n(\Lambda)f - \alpha(\Lambda)f\| = 0$ for every $f \in \mathfrak{H}$, cf. (2, pp. 381–382).)

(a) results from Definition 1, (ii): If $|\alpha(\lambda)| \leq C$ disregarding a set of $\sigma(\lambda)$-measure 0, then

$$\|\alpha(\Lambda)f\|^2 = \int |\alpha(\lambda)|^2\|f(\lambda)\|^2\,d\sigma(\lambda) \leq C^2\int \|f(\lambda)\|^2\,d\sigma(\lambda) = C^2\|f\|^2.$$

(b) is obvious by Remark 3 in §3. The second statement of (c) again follows from Definition 1, (ii):

$$(\beta(\Lambda)^*f, g) = (f, \beta(\Lambda)g) = \int \overline{\beta(\lambda)}(f(\lambda), g(\lambda))\,d\sigma(\lambda)$$
$$= \int \alpha(\lambda)(f(\lambda), g(\lambda))\,d\sigma(\lambda) = (\alpha(\Lambda)f, g);$$

the other statements of (c) are obvious. (d) again follows from Definition 1, (ii), as

$$\|\alpha_n(\Lambda)f - \alpha(\Lambda)f\|^2 = \int |\alpha_n(\lambda) - \alpha(\lambda)|^2\|f(\lambda)\|^2\,d\sigma(\lambda).$$

(e) Let $\alpha(\lambda), \gamma(\lambda)$ be $\sigma(\lambda)$-measurable, $\gamma(\lambda)$ real, let $\beta(x)$ have the measurability-properties described loc. cit. below, and let it be bounded in every finite x-interval. If A is a bounded Hermitean operator, $\beta(A)$ denotes the operator-function in the sense of (11, p. 241) or (4, pp. 202–203). Then $\alpha(\lambda) \equiv \beta(\gamma(\lambda))$ implies $\alpha(\Lambda) = \beta(\gamma(\Lambda))$.

This is obvious for $\beta(x) = x$, and so by (c) for every polynomial $\beta(x)$. Now (d) extends it to every continuous- and then to every Baire-function $\beta(x)$, and then (b) to every $\beta(x)$. (Cf. (11, p. 234) or (4, p. 196).)

Remark. If we could put $\gamma(\lambda) = \lambda$, (e) would show, that $\alpha(\Lambda)$ is the operator-function of Λ in the sense loc. cit. But $\gamma(\lambda) = \lambda$ will in general violate our boundedness-requirements, and thereby exclude $\gamma(\lambda) = \lambda$. It would be easy, however, to extend our definitions to this case, and then the theory of functions of unbounded operators (cf. loc. cit. (11)) would be particularly useful. If $\sigma(\lambda)$ is constant outside of a finite λ-interval, then the complement of this interval has the $\sigma(\lambda)$-measure 0. Thus $\gamma(\lambda) = \lambda$ can be used, and we need even not change our present set-up if we desire to interpret $\alpha(\Lambda)$ as an operator-function. Of this we will make use in the proof of Theorem IV (cf. the part of the proof dealing with the "Relationship to equivalence").

If K is a $\sigma(\lambda)$-measurable set, define

$$1_K(\lambda) = \begin{cases} 1, & \lambda \in K, \\ 0, & \lambda \notin K. \end{cases}$$

Then $1_K(\Lambda)$ is a projection by (c); and if K' is a subset of K'', then we have $1_{K'}(\Lambda)1_{K''}(\Lambda) = 1_{K'}(\Lambda)$, that is $1_{K'}(\Lambda) \leq 1_{K''}(\Lambda)$ by (c). If $K', K'', \dots$ is a sequence of sets with $K' \supset K'' \supset \cdots$ and if $K' \cdot K'' \cdots = K$, then clearly $\lim_{n\to\infty} 1_{K^{(n)}}(\lambda) = 1_K(\lambda)$, and $|1_{K^{(n)}}(\lambda)| \leq 1$ so by (d) $\lim_{n\to\infty} 1_{K^{(n)}}(\Lambda) = 1_K(\Lambda)$.

If K is the set of all $\lambda > -\infty, \leq \mu$, then denote $1_K(\Lambda)$ by $E(\mu)$. The above statements prove that $E(\mu)$, $-\infty < \mu < +\infty$ is a resolution of unity. We have:

$$E(\mu) \int_- f(\lambda)\sqrt{d\sigma(\lambda)} = \int_- f_\mu(\lambda)\sqrt{d\sigma(\lambda)},$$

where

$$f_\mu(\lambda) = \begin{cases} f(\lambda), & \lambda \leq \mu, \\ 0, & \lambda > \mu. \end{cases}$$

We call these $E(\mu)$ the *resolution of unity which belongs to our generalized direct sum.*

We prove next:

Theorem III. *A given unitary space $\mathfrak{H}$ can be represented as a generalized direct sum to which a given weight function $\sigma(\lambda)$ and a given resolution of unity $E(\lambda)$ belong, if and only if $\sigma(\lambda)$ is equivalent to $E(\lambda)$ (cf. §2). In this case this representation is uniquely determined (up to replacements of the $\mathfrak{H}^{(\lambda)}$ by isomorphic images, and up to an arbitrary λ-set of $\sigma(\lambda)$-measure 0), by $\sigma(\lambda)$ and $E(\lambda)$.*

Proof. *Necessity of the condition:* Assume that $\mathfrak{H}$ is the generalized direct sum of the $\mathfrak{H}^{(\lambda)}$ with the weight function $\sigma(\lambda)$, and let $E(\lambda)$ be the resolution of unity which belongs to it. Form the $\rho_f(\mu)$ of §2. Put $f = \int_{-} f(\lambda)\sqrt{d\sigma(\lambda)}$, then

$$\rho_f(\mu) = \|E(\mu)f\|^2 = \int \|f_\mu(\lambda)\|^2 \, d\sigma(\lambda) = \int_{\lambda\leq\mu} \|f(\lambda)\|^2 \, d\sigma(\lambda).$$

Thus every $\rho_f(\lambda)$ is $\sigma(\lambda)$-totally continuous by the formula (2.1) in §2. Therefore if $\sigma(\lambda)$ is $\tau(\lambda)$-totally-continuous, then the same is true for every $\rho_f(\lambda)$, and so for $E(\lambda)$ too. Conversely: Choose $f(\lambda)$ with $\|f(\lambda)\| = 1$ (for instance $f(\lambda) = \psi_1(\lambda)$, that is $x_m(\lambda) = 1$ for $m = 1$ and 0 for $m \neq 1$); then our above formula gives $\rho_f(\mu) = \sigma(\mu)$. So if $E(\lambda)$ is $\tau(\lambda)$-totally-continuous, then the same is true for $\sigma(\lambda)$. Thus we have proved that $\sigma(\lambda)$ is equivalent to $E(\lambda)$.

Sufficiency of the condition: Assume that an N-function $\sigma(\lambda)$, and an equivalent resolution of unity $E(\lambda)$ are given. Under these assumptions a (finite or infinite) sequence of mutually orthogonal linear closed sets $\mathfrak{M}_1, \mathfrak{M}_2, \ldots$, which together determine the linear closed set $\mathfrak{H}$, can be found, so that the following requirement is fulfilled: Each $\mathfrak{M}_m$ reduces all $E(\lambda)$, and in each $\mathfrak{M}_m$ the resolution of unity $E(\lambda)$ is simple. Simplicity is defined in (11, p. 275, Definition 7.2), for self-adjoint transformations, and we will call a resolution of unity simple, if it belongs to a simple self-adjoint transformation. A definition of simplicity in terms of $E(\lambda)$ themselves results by combining Theorems 7.2 and 7.9 ((11, pp. 243, 275), respectively). The above statement is contained in Theorem 7.4 ((11, p. 247)), if we denote the $\mathfrak{M}(\varphi_1), \mathfrak{M}(\varphi_2), \ldots, \mathfrak{M}(g_1), \mathfrak{M}(g_2), \ldots$ of that theorem (in some order) by $\mathfrak{M}_1, \mathfrak{M}_2, \ldots$, and remember, that every $\mathfrak{M}(f)$ reduces the $E(\lambda)$ by a remark in (11, p. 244), and that in every $\mathfrak{M}(f)$ the $E(\lambda)$ are simple by Theorem 7.9, quoted above. In each $\mathfrak{M}_m$ we apply Theorem 6.2 ((11, p. 226)): We have $\mathfrak{M}_m = \mathfrak{M}(f_m)$ for a suitable $f_m \in \mathfrak{M}_m$, and $\mathfrak{M}_m$ is isomorphic to the unitary space $\mathfrak{H}^*_m$ which is defined as follows: $\rho_m(\lambda) = \rho_{f_m}(\lambda) = \|E(\lambda)f_m\|^2$ is an N-function, consider all

$\rho_m(\lambda)$-measurable (complex-valued) functions $\xi(\lambda)$, for which $\int |\xi(\lambda)|^2\, d\rho_m(\lambda)$ is finite, and define for $\xi \equiv \xi(\lambda)$, $\eta \equiv \eta(\lambda)$

$$\alpha\xi \equiv \alpha\xi(\lambda), \qquad \xi + \eta \equiv \xi(\lambda) + \eta(\lambda),$$

$$(\xi, \eta) = \int \xi(\lambda)\overline{\eta(\lambda)}\, d\rho_m(\lambda).$$

The set $\mathfrak{H}_m^*$ of all these $\xi \equiv \xi(\lambda)$ is then a unitary space. (Cf. loc. cit. footnote 5.)

As $\rho_m(\lambda)$ is $\sigma(\lambda)$-totally continuous, we can form $d\rho_m(\lambda)/d\sigma(\lambda)$ from §2, (γ). Let N_m be the set of all λ with $d\rho_m(\lambda)/d\sigma(\lambda) = 0$, owing to the final formula loc. cit. $\mu_{\rho_m(\lambda)}(N_m) = 0$. Thus we can restrict the $\xi(\lambda)$ of $\mathfrak{H}_m^*$ to the $\lambda \notin N_m$. Consider now all $\sigma(\lambda)$-measurable (complex-valued) functions $x(\lambda)$, defined for the $\lambda \notin N_m$, for which $\int_{\lambda \notin N_m} |x(\lambda)|^2\, d\sigma(\lambda)$ is finite, and define for $x \equiv x(\lambda)$, $y \equiv y(\lambda)$

$$\alpha x \equiv \alpha x(\lambda), \qquad x + y \equiv x(\lambda) + y(\lambda),$$

$$(x, y) = \int_{\lambda \notin N_m} x(\lambda)\overline{y(\lambda)}\, d\sigma(\lambda).$$

The set $\mathfrak{H}_m^{**}$ of all these $x \equiv x(\lambda)$ is again a unitary space. (Cf. as above.) But now the requirement $\lambda \notin N_m$ may be essential, because $\mu_{\sigma(\lambda)}(N_m)$ may be > 0. The correspondence

$$\xi(\lambda) \longleftrightarrow x(\lambda) = \sqrt{\frac{d\rho_m(\lambda)}{d\sigma(\lambda)}}\, \xi(\lambda)$$

is clearly an isomorphism of $\mathfrak{H}_m^*$ and $\mathfrak{H}_m^{**}$ (cf. the final formula of §2, (γ)).

In the $\mathfrak{M}_m \longleftrightarrow \mathfrak{H}_m^*$ isomorphism this is true (cf. Theorem 6.2 loc. cit. above, and a remark in (11, p. 243)): If $f \in \mathfrak{M}_m$ corresponds to $\xi(\lambda) \in \mathfrak{H}_m^*$ then $E(\mu)f$ corresponds to $1_\mu(\lambda)\xi(\lambda)$, where

$$1_\mu(\lambda) = \begin{cases} 1, & \lambda \leq \mu, \\ 0, & \lambda > \mu. \end{cases}$$

Obviously the same holds in $\mathfrak{H}_m^{**}$: $x(\lambda)$ becomes $1_\mu(\lambda)x(\lambda)$. So we have in the $\mathfrak{M}_m \longleftrightarrow \mathfrak{H}_m^{**}$-isomorphism:

$$E(\mu)x(\lambda) = \begin{cases} x(\lambda), & \lambda \leq \mu, \\ 0, & \lambda > \mu. \end{cases}$$

Let us now return to $\mathfrak{H}$ itself. Every $f \in \mathfrak{H}$ can be written uniquely as a sum $f = f_1 + f_2 + \cdots$, $f_m \in \mathfrak{M}_m$ (clearly $f_m = P_{\mathfrak{M}_m} f$). Conversely such a sum converges if and only if $\sum_m \|f_m\|^2$ is finite (cf. (1, p. 67), put in Theorem 5, $\varphi_m = (1/\|f_m\|) f_m$ for every $f_m \neq 0$). At the same time $f = f_1 + f_2 + \cdots$, $g = g_1 + g_2 + \cdots$, $f_m, g_m \in \mathfrak{M}_m$, imply

$$\alpha f = \alpha f_1 + \alpha f_2 + \cdots, \qquad f + g = (f_1 + g_1) + (f_2 + g_2) + \cdots,$$

$$(f, g) = \sum_m (f_m, g_m).$$

Now use for each f_m the $\mathfrak{H}_m^* \longleftrightarrow \mathfrak{H}_m^{**}$-isomorphism, replacing it by $x_m(\lambda)$. $x_m(\lambda)$ can be looked at as a complex-valued function of the two variables $m = 1, 2, \ldots$, $\lambda > -\infty, < +\infty$, subject to these two restrictions: (a) The inclusive domain $1, 2, \ldots$ of m may have to be replaced by a finite set $1, \ldots, m_0$. (b) $\lambda \notin N_m$. We can omit (a), by putting N_m equal to the entire interval $\lambda > -\infty, < +\infty$ for the forbidden m's (i.e. for $m = m_0 + 1, m_0 + 2, \ldots$). Denoting the complement of N_m by H_m, we obtain the condition $\lambda \in H_m$. Thus $\mathfrak{H}$ is isomorphic to the set of all $x_m(\lambda)$ satisfying the conditions (α), (γ), (δ), in Definition 2 and in the sense of Theorem II (in (α) of course $\lambda \in H_m$ alone must be used), with the definitions of $\alpha x, x + y, (x, y)$, given in Theorem II. At the same time

$$E(\mu) x_m(\lambda) = \begin{cases} x_m(\lambda), & \lambda \leq \mu, \\ 0, & \lambda > \mu. \end{cases}$$

Assume now, that a $\sigma(\lambda)$-measurable function $k(\lambda) = 1, 2, \ldots, \infty$ exists, so that $k(\lambda) \geq m$ is equivalent to $\lambda \in H_m$. Form then for each $\lambda > -\infty, < +\infty$ a $k(\lambda)$-dimensional unitary space $\mathfrak{H}^{(\lambda)}$, and choose in it a complete normalized orthogonal set $\psi_1(\lambda), \psi_2(\lambda), \ldots$. Then our above remarks establish, with the help of Theorem II, that $\mathfrak{H}$ is the generalized direct sum of the $\mathfrak{H}^{(\lambda)}$ with the weight-function $\sigma(\lambda)$, and with the help of the remark which precedes the present theorem, that the resolution of unity which belongs to this direct sum consists of our $E(\mu)$. Thus our only remaining task is to construct this $k(\lambda)$. But this requires certain changes in the H_m.

For each $\lambda > -\infty, < +\infty$ denote the number of m's with $\lambda \in H_m$ by $k_0(\lambda) = 0, 1, 2, \ldots, \infty$, let them be $m_1(\lambda) < m_2(\lambda) < \cdots$. Denote the domain of our $x_m(\lambda)$'s by Δ. It consists of all pairs (m, λ) with $\lambda \in H_m$, that is $m = m_n(\lambda)$, $n = 1, 2, \ldots, k_0(\lambda)$. Replace each (m, λ) by (n, λ), where $m = m_n(\lambda)$. Thus the set Δ_0 of all (n, λ) with $n = 1, 2, \ldots, k_0(\lambda)$ results, that is with $\lambda \in H_{0n}$, where H_{0n} is defined by $k_0(\lambda) \geq n$. We claim: This transformation carries every $\sigma(\lambda)$-measurable $x_m(\lambda)$ into a $\sigma(\lambda)$-measurable $x_{0n}(\lambda)$, and the $\sum_{m=1}^{\infty} \int_{H_m} \cdots d\sigma(\lambda)$ go over into $\sum_{n=1}^{\infty} \int_{H_{0n}} \cdots d\sigma(\lambda)$. This becomes obvious if we observe, that our transformation does not affect λ at all, and that the set of those λ's, for which a given m goes over into a given n is $\sigma(\lambda)$-measurable. (It is the set

$$H_m \cdot \sum_{m_1 < \cdots < m_{n-1} < m} H_{m_1} \cdots H_{m_{n-1}}.)$$

which is $\sigma(\lambda)$-measurable because all the H_l are such.) Thus all our previous statements remain true, if we replace our H_m by these H_{0n}, and then $k_0(\lambda)$ has all properties we required for $k(\lambda)$, except for the possibility of $k_0(\lambda) = 0$.

$k_0(\lambda) = 0$ means that $\lambda \in H_m$ holds for no $m = 1, 2, \ldots$, that is $\lambda \in \text{Complement}(H_1 + H_2 + \cdots) = N$. Choose any $f \in \mathfrak{H}$, it corresponds to an $x_m(\lambda)$, which may be defined to be 0 for $\lambda \notin H_m$, so that we can replace the integrals $\int_{\lambda \in H_m} \cdots d\sigma(\lambda)$ by $\int \cdots d\sigma(\lambda)$. In particular $x_m(\lambda) = 0$ for $\lambda \in N$. Now we have for $\rho_f(\lambda) = \|E(\lambda) f\|^2$:

$$\begin{aligned}
\mu_{\rho_f(\lambda)}(N) &= \int_N d(\|E(\mu) f\|^2) = \int 1_N(\mu)\, d(\|E(\mu) f\|^2) \\
&= \int 1_N(\mu)\, d\left(\sum_{m=1}^{\infty} \int_{\lambda \leq \mu} |x_m(\lambda)|^2\, d\sigma(\lambda)\right) \\
&= \sum_{m=1}^{\infty} \int 1_N(\mu)\, d\left(\int_{\lambda \leq \mu} |x_m(\lambda)|^2\, d\sigma(\lambda)\right) \\
&= \sum_{m=1}^{\infty} \int 1_N(\lambda) |x_m(\lambda)|^2\, d\sigma(\lambda) = 0,
\end{aligned}$$

as the integrand $1_N(\lambda)|x_m(\lambda)|^2$ is everywhere 0.

Now form the N-function $\tau(\lambda) = \sum_n a_n \rho_{\varphi_n}(\lambda)$ from §2, (δ) ($\varphi_1, \varphi_2, \ldots$ a complete, normalized, orthogonal set, $a_1, a_2, \ldots > 0$, $\sum_n a_n$ finite). We have

$$\mu_{\tau(\lambda)}(N) = \sum_n a_n \mu_{\rho_{\varphi_n}(\lambda)}(N) = 0,$$

but $E(\lambda)$ is $\tau(\lambda)$-totally-continuous and equivalent to $\sigma(\lambda)$ (cf. loc. cit.), so $\sigma(\lambda)$ is $\tau(\lambda)$-totally-continuous, and therefore $\mu_{\sigma(\lambda)}(N) = 0$. Thus $k_0(\lambda) = 0$ occurs only on a set of $\sigma(\lambda)$-measure 0, and therefore it can be neglected. (We can add N to H_1 without changing anything essential, and nevertheless excluding $k_0(\lambda) = 0$.)

Uniqueness of the representation: Assume that $\mathfrak{H}$ is at the same time generalized direct sum of the $\mathfrak{H}'^{(\lambda)}$ and of the $\mathfrak{H}''^{(\lambda)}$, in both cases with the weight-function $\sigma(\lambda)$, and that the same resolution of unity $E(\lambda)$ belongs to both of them. We will distinguish the notions relative to $\mathfrak{H}'^{(\lambda)}$ and to $\mathfrak{H}''^{(\lambda)}$, by affixing one and two primes: $'$, $''$, respectively.

The $f \in \mathfrak{H}$ are then in a one-to-one correspondence with the $'\sigma(\lambda)$-summable functions $f'(\lambda)$,

$$f = \int_{-}^{\prime} f'(\lambda)\sqrt{d\sigma(\lambda)}, \qquad f'(\lambda) \in \mathfrak{H}'^{(\lambda)},$$

and similarly with the $''\sigma(\lambda)$-summable functions $f''(\lambda)$,

$$f = \int_{-}^{\prime\prime} f''(\lambda)\sqrt{d\sigma(\lambda)}, \qquad f''(\lambda) \in \mathfrak{H}''^{(\lambda)}.$$

Thus a one-to-one correspondence between the $'\sigma(\lambda)$-summable functions $f'(\lambda)$ and the $''\sigma(\lambda)$-summable functions $f''(\lambda)$ results:

$$\int_{-}^{\prime} f'(\lambda)\sqrt{d\sigma(\lambda)} = \int_{-}^{\prime\prime} f''(\lambda)\sqrt{d\sigma(\lambda)}, \qquad f'(\lambda) \in \mathfrak{H}'^{(\lambda)}, \quad f''(\lambda) \in \mathfrak{H}''^{(\lambda)}.$$

Of course in all these correspondences sets of $\sigma(\lambda)$-measure 0 are disregarded.

If $f'(\lambda)$ and $g'(\lambda)$ correspond in this way to $f''(\lambda)$ and $g''(\lambda)$ respectively, put

$$f = \int_{-}^{\prime} f'(\lambda)\sqrt{d\sigma(\lambda)} = \int_{-}^{\prime\prime} f''(\lambda)\sqrt{d\sigma(\lambda)},$$

$$g = \int_{-}^{\prime} g'(\lambda)\sqrt{d\sigma(\lambda)} = \int_{-}^{\prime\prime} g''(\lambda)\sqrt{d\sigma(\lambda)}.$$

Considering that the resolution of unity $E(\lambda)$ belongs to both representations of $\mathfrak{H}$ as generalized direct sum, this implies

$$(E(\mu)f, g) = \int_{\lambda\leq\mu} (f'(\lambda), g'(\lambda))\, d\sigma(\lambda) = \int_{\lambda\leq\mu} (f''(\lambda), g''(\lambda))\, d\sigma(\lambda)$$

and so

$$\int_{\lambda\leq\mu} [(f'(\lambda), g'(\lambda)) - (f''(\lambda), g''(\lambda))]\, d\sigma(\lambda) = 0$$

for every $\mu > -\infty, < +\infty$. Thus $(f'(\lambda), g'(\lambda)) = (f''(\lambda), g''(\lambda))$ except perhaps for a λ-set of $\sigma(\lambda)$-measure 0[7], which may depend on the $f'(\lambda), g'(\lambda), f''(\lambda), g''(\lambda)$.

Let now $\varphi'_m(\lambda)$ be a measurable family in the $\mathfrak{H}'^{(\lambda)}$, defined for $m \leq k'(\lambda)$; and $\psi''_n(\lambda)$ a measurable family in the $\mathfrak{H}''^{(\lambda)}$, defined for $n \leq k''(\lambda)$. Define

$$f'_m(\lambda) = \begin{cases} \varphi'_m(\lambda), & m \leq k'(\lambda), \\ 0, & m > k'(\lambda), \end{cases}$$

this function $f'_m(\lambda)$ is clearly $'\sigma(\lambda)$-summable (m fixed!), let the corresponding $''$-function be $f''_m(\lambda)$. Define similarly

$$g''_n(\lambda) = \begin{cases} \psi''_n(\lambda), & n \leq k''(\lambda), \\ 0, & n > k''(\lambda), \end{cases}$$

this function $g''_n(\lambda)$ is clearly $''\sigma(\lambda)$-summable (n fixed!), let the corresponding $'$-function be $g'_n(\lambda)$. Then we have:

$$(f''_m(\lambda), f''_p(\lambda)) = (f'_m(\lambda), f'_p(\lambda)) = \begin{cases} 1, & m = p \leq k'(\lambda), \\ 0, & \text{otherwise}; \end{cases} \tag{1}$$

$$(g'_n(\lambda), g'_q(\lambda)) = (g''_n(\lambda), g''_q(\lambda)) = \begin{cases} 1, & n = q \leq k''(\lambda), \\ 0, & \text{otherwise}; \end{cases} \tag{2}$$

$$(f''_m(\lambda), g''_n(\lambda)) = (f'_m(\lambda), g'_n(\lambda)) = u_{mn}(\lambda), \tag{3}$$

except when $\lambda \in N$, where N is the sum of the exceptional sets of all these equations. Thus N is the sum of a countably infinite number of sets, each of $\sigma(\lambda)$-

[7] Put

$$\nu(\mu) = \int_{\lambda \leq \mu} [(f'(\lambda), g'(\lambda)) - (f''(\lambda), g''(\lambda))]\, d\sigma(\lambda)$$

and form for any $\sigma(\lambda)$-measurable function $\epsilon(\mu)$ the $\int \epsilon(\mu)\, d\nu(\mu)$. This is 0 and at the same time

$$\int \epsilon(\lambda)[(f'(\lambda), g'(\lambda)) - (f''(\lambda), g''(\lambda))]\, d\sigma(\lambda),$$

thus this expression must vanish. Choose

$$\epsilon(\lambda) = \begin{cases} |Z|/Z, & Z \neq 0, \\ 0, & Z = 0, \end{cases}$$

where Z is the $[\cdots]$ expression, then $\int |(f'(\lambda), g'(\lambda)) - (f''(\lambda), g''(\lambda))|\, d\sigma(\lambda) = 0$ results, and thus the statement in the text.

measure 0, therefore

$$\mu_{\sigma(\lambda)}(N) = 0.$$

Equation (1) proves that the $f''_m(\lambda)$, $m \leq k'(\lambda)$, form a normalized orthogonal set in $\mathfrak{H}''^{(\lambda)}$, while the $g''_n(\lambda)$, $n \leq k''(\lambda)$, form a complete normalized orthogonal set in $\mathfrak{H}''^{(\lambda)}$. Therefore $u_{mn}(\lambda) = (f''_m(\lambda), g''_n(\lambda))$ in (3) leads to

$$\sum_n u_{mn}(\lambda)\overline{u_{pn}(\lambda)} = \begin{cases} 1, & m = p \leq k'(\lambda), \\ 0, & \text{otherwise.} \end{cases} \tag{4}$$

Similarly equation (2) proves that the $g'_n(\lambda)$, $n \leq k''(\lambda)$, form a normalized orthogonal set in $\mathfrak{H}'^{(\lambda)}$, while the $f'_m(\lambda)$, $m \leq k'(\lambda)$, form a complete normalized orthogonal set in $\mathfrak{H}'^{(\lambda)}$. Therefore $u_{mn}(\lambda) = (f'_m(\lambda), g'_n(\lambda))$ in (3) leads to

$$\sum_m u_{mn}(\lambda)\overline{u_{mq}(\lambda)} = \begin{cases} 1, & n = p \leq k''(\lambda), \\ 0, & \text{otherwise.} \end{cases} \tag{5}$$

(4), (5) state together, that $k'(\lambda) = k''(\lambda)$ (both are $= \sum_{m,n} |u_{mn}(\lambda)|^2$) and that the matrix $\{u_{mn}(\lambda)\}_{m,n=1,2,\ldots}$ $(m, n \leq k'(\lambda) = k''(\lambda))$ is unitary. This holds for all $\lambda \notin N$. As N has the $\sigma(\lambda)$-measure 0, we can change the $\mathfrak{H}'^{(\lambda)}$, $\lambda \in N$, arbitrarily. We make use of this liberty by replacing these $\mathfrak{H}'^{(\lambda)}$ by the corresponding $\mathfrak{H}''^{(\lambda)}$, so that $k'(\lambda) = k''(\lambda)$, and choose

$$f'_m(\lambda) = f''_m(\lambda) = g'_m(\lambda) = g''_m(\lambda) = \begin{cases} \psi''_m(\lambda), & m \leq k'(\lambda), \\ 0, & m > k'(\lambda), \end{cases}$$

so that

$$u_{mn}(\lambda) = \begin{cases} 1, & m = n, \\ 0, & m \neq n. \end{cases}$$

(All this for $\lambda \in N$ only!) Thus we have $k'(\lambda) = k''(\lambda)$ and $\{u_{mn}(\lambda)\}$ unitary without exceptions.

Owing to its definition in (3) $u_{mn}(\lambda)$ is a $\sigma(\lambda)$-measurable function. Besides (3) gives

$$f''_m(\lambda) = \sum_n u_{mn}(\lambda) g''_n(\lambda), \qquad m, n \leq k'(\lambda).$$

As the $g''_n(\lambda) = \psi''_n(\lambda)$, $n \leq k'(\lambda)$, form a measurable family in the $''$-sense, we can apply Theorem I, (iii)*, and thus see, that the $f''_m(\lambda)$, $m \leq k'(\lambda)$, too form a measurable family in the $''$-sense. On the other hand the $f'_m(\lambda)$, $m \leq k'(\lambda)$, form a measurable family in the $'$-sense.

$\sum_m x_m f'_m(\lambda)$ is the general element of $\mathfrak{H}'^{(\lambda)}$, $\sum_m x_m f''_m(\lambda)$ is the general element of $\mathfrak{H}''^{(\lambda)}$ ($\sum_m |x_m|^2$ finite), if we let them correspond to each other, an isomorphism $\mathfrak{H}'^{(\lambda)} \longleftrightarrow \mathfrak{H}''^{(\lambda)}$ results.

Assume now

$$\int_{-}^{\prime} h'(\lambda)\sqrt{d\sigma(\lambda)} = \int_{-}^{\prime\prime} h''(\lambda)\sqrt{d\sigma(\lambda)}$$

and put

$$h'(\lambda) = \sum_m x_m(\lambda) f'_m(\lambda), \qquad h''(\lambda) = \sum_m y_m(\lambda) f''_m(\lambda).$$

Then we have, as we proved at the beginning of this discussion, always with the possible exception of a set of $\sigma(\lambda)$-measure 0: $(h'(\lambda), f'_m(\lambda)) = (h''(\lambda), f''_m(\lambda))$, $x_m(\lambda) = y_m(\lambda)$, and so $h'(\lambda) \longleftrightarrow h''(\lambda)$ in the above isomorphism $\mathfrak{H}'^{(\lambda)} \longleftrightarrow \mathfrak{H}''^{(\lambda)}$. Thus the isomorphisms $\mathfrak{H}'^{(\lambda)} \longleftrightarrow \mathfrak{H}''^{(\lambda)}$ carry all $'$-notions into the corresponding $''$-notions, and conversely. □

Part II: Abelian rings and the generalized direct sums

8 Effect of a change of the weight-function with the resolution of unity fixed

We have seen that a representation of a unitary space $\mathfrak{H}$ as the generalized direct sum of the unitary spaces $\mathfrak{H}^{(\lambda)}$ is essentially (that is, up to replacements of the $\mathfrak{H}^{(\lambda)}$ by isomorphic images and up to an arbitrary λ-set of $\sigma(\lambda)$-measure 0) characterized by its weight-function $\sigma(\lambda)$ and its resolution of unity $E(\lambda)$. The only restriction we had to impose on these was that $\sigma(\lambda)$ must be equivalent to $E(\lambda)$ (Theorem III). Our next task is to show that even less than this (that is, than $\sigma(\lambda)$ and $E(\lambda)$) suffices to determine the $\mathfrak{H}^{(\lambda)}$ up to unessential transformations.

We first get rid of $\sigma(\lambda)$. Let $E(\lambda)$ be a resolution of unity, and $\sigma_1(\lambda), \sigma_2(\lambda)$ two N-functions, both equivalent to $E(\lambda)$. Then $\sigma_2(\lambda)$ is $\sigma_1(\lambda)$-totally-continuous and conversely, therefore we can form $d\sigma_2(\lambda)/d\sigma_1(\lambda)$ and $d\sigma_1(\lambda)/d\sigma_2(\lambda)$ by §2, (γ). Now we have by (11),

$$\frac{d\sigma_2(\lambda)}{d\sigma_1(\lambda)}\frac{d\sigma_1(\lambda)}{d\sigma_2(\lambda)} = 1,$$

except perhaps for a set of $\sigma(\lambda)$-measure 0. But on this set we may redefine both differential quotients, giving them the value 1. Then we have the displayed identity everywhere, and thus both factors are $\neq 0$ everywhere.

Let now $\mathfrak{H}^{(\lambda)}$ be a representation of $\mathfrak{H}$ as a generalized direct sum with the weight-function $\sigma_1(\lambda)$ and the resolution of unity $E(\lambda)$. We speak correspondingly of $\sigma_1(\lambda)$-summability and of $f = \int_{-} f(\lambda)\sqrt{d\sigma_1(\lambda)}$. We now define: A function $f(\lambda) \in \mathfrak{H}^{(\lambda)}$ is $\sigma_2(\lambda)$-summable if the function

$$\sqrt{\frac{d\sigma_2(\lambda)}{d\sigma_1(\lambda)}}\, f(\lambda) \quad (\in \mathfrak{H}^{(\lambda)})$$

is $\sigma_1(\lambda)$-summable, and then (again by definition!)

$$\int_{-} f(\lambda)\sqrt{d\sigma_2(\lambda)} = \int_{-} \sqrt{\frac{d\sigma_2(\lambda)}{d\sigma_1(\lambda)}} f(\lambda)\sqrt{d\sigma_1(\lambda)}.$$

One verifies easily that $\mathfrak{H}$ is the generalized direct sum of the $\mathfrak{H}^{(\lambda)}$ with the weight-function $\sigma_2(\lambda)$ using these definitions, because it is the generalized direct sum of the $\mathfrak{H}^{(\lambda)}$ with the weight-function $\sigma_1(\lambda)$ using the original definitions. The essential fact to be used, in order to establish this, is

$$\begin{aligned}\int \left(\sqrt{\frac{d\sigma_2(\lambda)}{d\sigma_1(\lambda)}} f(\lambda), \sqrt{\frac{d\sigma_2(\lambda)}{d\sigma_1(\lambda)}} g(\lambda)\right) d\sigma_1(\lambda) \\ = \int (f(\lambda), g(\lambda)) \frac{d\sigma_2(\lambda)}{d\sigma_1(\lambda)} d\sigma_1(\lambda) \\ = \int (f(\lambda), g(\lambda)) d\sigma_2(\lambda).\end{aligned}$$

Denoting the resolution of unity which belongs to the $\sigma_2(\lambda)$-decomposition of $\mathfrak{H}$ by $E'(\lambda)$ we see that we have again

$$E'(\mu)\int_{-} f(\lambda)\sqrt{d\sigma_2(\lambda)} = \int_{-} f_\mu(\lambda)\sqrt{d\sigma_2(\lambda)},$$

where

$$f_\mu(\lambda) = \begin{cases} f(\lambda), & \lambda \le \mu, \\ 0, & \lambda > \mu, \end{cases}$$

because the factor $\sqrt{d\sigma_2(\lambda)/d\sigma_1(\lambda)}$ does not affect the corresponding rule which holds for the $\sigma_1(\lambda)$-decomposition. Thus $E'(\mu) = E(\mu)$. In other words: With the new definitions $\mathfrak{H}$ is the generalized direct sum of the $\mathfrak{H}^{(\lambda)}$ with the weight-function $\sigma_2(\lambda)$ and the resolution of unity $E(\lambda)$.

Remembering Theorem III, we can therefore state:

Let $E(\lambda)$ be a resolution of unity, and $\sigma_1(\lambda)$, $\sigma_2(\lambda)$ two N-functions equivalent to $E(\lambda)$. Let $\mathfrak{H}$ be the generalized direct sum of the $\mathfrak{H}_1^{(\lambda)}$ resp. of the $\mathfrak{H}_2^{(\lambda)}$ with the weight-function $\sigma_1(\lambda)$ resp. $\sigma_2(\lambda)$, and in both cases with the resolution of unity $E(\lambda)$. Then these two representations become identical if we replace every $f(\lambda) \in \mathfrak{H}_1^{(\lambda)}$ by $\sqrt{d\sigma_2(\lambda)/d\sigma_1(\lambda)}\, f(\lambda)$ and apply to this a suitable isomorphism $\mathfrak{H}_1^{(\lambda)} \rightleftarrows \mathfrak{H}_2^{(\lambda)}$ (depending on λ only), and disregard a (fixed) λ-set of $\sigma(\lambda)$-measure 0.

This result justifies our symbolic notation $\int_{-} f(\lambda)\sqrt{d\sigma(\lambda)}$: If we do not express the isomorphism $\mathfrak{H}_1^{(\lambda)} \rightleftarrows \mathfrak{H}_2^{(\lambda)}$ in our formulae (that is, if we use the same symbol for the corresponding elements of $\mathfrak{H}_1^{(\lambda)}$ and $\mathfrak{H}_2^{(\lambda)}$) it implies the relation

$$\int_{-} \sqrt{\frac{d\sigma_2(\lambda)}{d\sigma_1(\lambda)}}\, f(\lambda)\sqrt{d\sigma_1(\lambda)} = \int_{-} f(\lambda)\sqrt{d\sigma_2(\lambda)}.$$

The above result shows that if we have once given the resolution of unity $E(\lambda)$ then it is not very important which particular $\sigma(\lambda)$ (equivalent to $E(\lambda)$!) we choose: This choice does not affect the $\mathfrak{H}^{(\lambda)}$'s essentially, and within each $\mathfrak{H}^{(\lambda)}$ it has only the effect of multiplying everything with a constant factor $\sqrt{d\sigma_2(\lambda)/d\sigma_1(\lambda)}$ if we pass from $\sigma(\lambda) = \sigma_1(\lambda)$ to $\sigma(\lambda) = \sigma_2(\lambda)$.

9 Definition of the equivalence. Elementary properties

We next define a notion of equivalence for two generalized direct sum-representations of a unitary space $\mathfrak{H}$.

Definition 3. *Assume that $\mathfrak{H}$ is at the same time generalized direct sum of the $\mathfrak{H}_1^{(\lambda)}$ with the weight-function $\sigma_1(\lambda)$ and of the $\mathfrak{H}_2^{(\lambda)}$ with the weight-function $\sigma_2(\lambda)$. We call these two representations* equivalent *if this is the case:*

There exists a (real-valued) function $m(\lambda)$, a positive function $\gamma(\lambda)$, a set N_1 with the $\sigma_1(\lambda)$-measure 0, and a set N_2 with the $\sigma_2(\lambda)$-measure 0, which have the following properties:

(1) $\lambda \rightleftarrows m(\lambda)$ is a one-to-one mapping of the complement of N_1 on the complement of N_2.

(2) *If $\lambda \notin N_1$ then there exist an isomorphic mapping of $\mathfrak{H}_1^{(\lambda)}$ on $\mathfrak{H}_2^{(m(\lambda))}$. We denote this isomorphism $\mathfrak{H}_1^{(\lambda)} \rightleftarrows \mathfrak{H}_2^{(m(\lambda))}$ by $g^{(\lambda)}$.*

(3) *$f_2(\lambda)$ ($\in \mathfrak{H}_2^{(\lambda)}$) is $\sigma_2(\lambda)$-summable if and only if*

$$f_1(\lambda) = \gamma(\lambda)(g^{(\lambda)})^{-1} f_2(m(\lambda))$$

is $\sigma_1(\lambda)$-summable, and then

$$\int_{-} f_1(\lambda)\sqrt{d\sigma_1(\lambda)} = \int_{-} f_2(\lambda)\sqrt{d\sigma_2(\lambda)}.$$

We call $\lambda \rightleftarrows m(\lambda)$ the mapping *and $\gamma(\lambda)$ the* factor *of this equivalence.*

Remark 1. Let K_1 be an arbitrary set, and K_2 its $\lambda \rightleftarrows m(\lambda)$ image. Assume that K_2 is $\sigma_2(\lambda)$-measurable. Choose a measurable family $\varphi_{2m}(\lambda)$ in $\mathfrak{H}_2^{(\lambda)}$, and define

$$f_2(\lambda) = \begin{cases} \varphi_{21}(\lambda), & \lambda \in K_2, \\ 0, & \lambda \notin K_2. \end{cases}$$

This $f_2(\lambda)$ is clearly $\sigma_2(\lambda)$-summable; therefore $f_1(\lambda) = \gamma(\lambda)(g^{(\lambda)})^{-1} f_2(m(\lambda))$ is $\sigma_1(\lambda)$-summable. Thus the function

$$\|f_1(\lambda)\| = \gamma(\lambda)\|f_2(m(\lambda))\| = \begin{cases} \gamma(\lambda), & \lambda \in K_1, \\ 0, & \lambda \notin K_1 \end{cases}$$

(we now disregard sets of $\sigma_1(\lambda)$-measure 0) is $\sigma_1(\lambda)$-measurable, and so is the set of all λ with $\|f_1(\lambda)\| \neq 0$: K_1. Conversely: If K_1 is $\sigma_1(\lambda)$-measurable, then K_2 is $\sigma_2(\lambda)$-measurable, because the inverse $m^{-1}(\lambda)$ of $m(\lambda)$ (define $m^{-1}(\lambda)$ as the inverse of $m(\lambda)$ in the complement of N_2, and, as N_2 has the $\sigma_2(\lambda)$-measure 0, arbitrarily in N_2) has the same properties as $m(\lambda)$, if we interchange $\mathfrak{H}_1^{(\lambda)}$, $\sigma_1(\lambda)$ and $\mathfrak{H}_2^{(\lambda)}$, $\sigma_2(\lambda)$.

So we see: K_1 is $\sigma_1(\lambda)$-measurable if and only if K_2 is $\sigma_2(\lambda)$-measurable.

With the above notations we have

$$\left\|\int_{-} f_1(\lambda)\sqrt{d\sigma_1(\lambda)}\right\|^2 = \int \|f_1(\lambda)\|^2 d\sigma_1(\lambda) = \int_{K_1} \gamma(\lambda) d\sigma_1(\lambda),$$

$$\left\|\int_{-} f_2(\lambda)\sqrt{d\sigma_2(\lambda)}\right\|^2 = \int \|f_2(\lambda)\|^2 d\sigma_2(\lambda) = \int_{K_2} d\sigma_2(\lambda) = \mu_{\sigma_2(\lambda)}(K_2),$$

and as the left sides are equal,

$$\mu_{\sigma_2(\lambda)}(K_2) = \int_{K_1} \gamma(\lambda) d\sigma_1(\lambda). \tag{9.1}$$

So $\mu_{\sigma_1(\lambda)}(K_1) = 0$ implies $\mu_{\sigma_2(\lambda)}(K_2) = 0$, and of course conversely. Furthermore one easily proves from (9.1) that

$$\int \omega(\lambda) d\sigma_2(\lambda) = \int \omega(m(\lambda))\gamma(\lambda) d\sigma_1(\lambda) \tag{9.2}$$

for every $\sigma_2(\lambda)$-summable numerical function $\omega(\lambda)$. (If $\omega(\lambda) \geq 0$, then apply the definition

$$\int \omega(\lambda) d\sigma(\lambda) = \int_0^{+\infty} x\, d[\mu_{\sigma(\lambda)}(\text{set of all } \lambda \text{ with } \omega(\lambda) \leq x)]$$

of the Lebesgue–Stieltjes integral, and then extend to complex-valued $\omega(\lambda)$'s.)

Remark 2. That our notion of equivalence is reflexive and symmetric is obvious. The results of Remark 1 concerning the images of sets of measure 0 prove that it is transitive too. Thus our notion of equivalence possesses the three essential properties of an equivalence. Therefore all possible representations of $\mathfrak{H}$ as a generalized direct sum (of any $\mathfrak{H}^{(\lambda)}$'s with any weight-function $\sigma(\lambda)$) can be classified in the customary way: Form for each representation the set of all those which are equivalent to it, and call this set an *equivalence-class*. Then any two equivalence-classes are either identical or they have no common element, and so every representation belongs to one and only one equivalence-class.

The result of §8 can now be formulated as follows:

All representations of $\mathfrak{H}$ as a generalized direct sum to which the same resolution of unity belongs, are equivalent. These equivalences have all the identical mapping $\lambda \rightleftarrows \lambda$. The factor is $\gamma(\lambda) = \sqrt{d\sigma_2(\lambda)/d\sigma_1(\lambda)}$ if the weight-function $\sigma_1(\lambda)$ is changed into $\sigma_2(\lambda)$.

10 Discussion of the discrete special case (cf. §4)

We repeat the procedure of §4 after Definition 1: Before we continue the systematic investigation we first discuss the meaning of the new notion (now it is the equivalence, introduced by Definition 3) for totally discontinuous $\sigma(\lambda)$'s, that is for the case in which our generalized direct sum becomes a common direct sum (cf. §4).

Consider two equivalent representations of $\mathfrak{H}$ as a direct sum: $\mathfrak{H}_1^{(\lambda)}, \sigma_1(\lambda)$ and $\mathfrak{H}_2^{(\lambda)}, \sigma_2(\lambda)$. Assume that $\sigma_1(\lambda)$ is totally discontinuous. Let, as in §4, $\lambda_1, \lambda_2, \ldots$ be the (finite or infinite) sequence of all points λ with a $\sigma_1(\lambda)$-measure $\neq 0$. Apply now Definition 3. As $\mu_{\sigma_1(\lambda)}(N_1) = 0$, no λ_n can belong to N_1. Thus the $\lambda_1, \lambda_2, \ldots$ have distinct $\lambda \rightleftarrows m(\lambda)$-images $\mu_1, \mu_2, \ldots, \mu_n = m(\lambda_n)$. By Remark 1 in §9 each μ_n has a $\sigma_2(\lambda)$-measure $\neq 0$. As the complement of $(\lambda_1, \lambda_2, \ldots)$ has the $\sigma_1(\lambda)$-measure 0, the same Remark yields that its $\lambda \rightleftarrows m(\lambda)$-image has the $\sigma_2(\lambda)$-measure 0. Adding to it N_2, we see: The complement of $(\mu_1, \mu_2, \ldots)$ has the $\sigma_2(\lambda)$-measure 0. Thus $\sigma_2(\lambda)$, too, is totally discontinuous.

Forming the $a_n = \mu_{\sigma_1(\lambda)}(\lambda_n) > 0$ and the $b_n = \mu_{\sigma_2(\lambda)}(\mu_n) > 0$, as in §4, the Remark 2 of §9 gives for the sets $K_1 = (\lambda_n)$, $K_2 = (\mu_n)$ that $a_n = (\gamma(\lambda_n))^2 b_n$. That is:

$$\gamma(\lambda_n) = \sqrt{\frac{a_n}{b_n}}.$$

Summing up, we can state that an equivalence-class contains no representations with a totally discontinuous $\sigma(\lambda)$, or only such ones. In the latter case the common direct sums which arise (in the sense of §4) differ only by permutations of their λ_n's and the numerical values of their a_n's. The subspaces $\mathfrak{H}_1^*, \mathfrak{H}_2^*, \ldots$ of $\mathfrak{H}$ (cf. §4) are invariant.

Thus for the common direct sums our notion of equivalence states precisely the invariance of the essential features.

11 $E(\lambda)$ constant outside of $-2 < \lambda < -1$ can always be secured by equivalence. P is the set of all $\alpha(\Lambda)$, and is an equivalence-invariant

We now resume the general discussion.

Let $\mathfrak{H}$ be the direct sum of the $\mathfrak{H}^{(\lambda)}$ with the weight-function $\sigma(\lambda)$, and let the resolution of unity $E(\lambda)$ belong to this representation. Form the ring of bounded operators which the $E(\lambda)$, $-\infty < \lambda < +\infty$, generate: P, cf. (2, p. 388). We call P *the ring which belongs* to the above representation of $\mathfrak{H}$.

As $E(\lambda) \in P$ and $E(\lambda)$ converges (strongly) to 1 for $\lambda \to +\infty$, we have $1 \in P$. As the $E(\lambda)$ are all Hermitean operators and as they all commute, the set of all $E(\lambda)$, $-\infty < \lambda < +\infty$, is Abelian, and therefore the ring P is Abelian too, cf. (2, p. 389). So we have:

The ring P which belongs to a generalized direct sum is always Abelian and contains 1.

Consider now the operators $\alpha(\Lambda)$ defined in §7. We ask: For which functions

$\alpha(\lambda)$ is $\alpha(\Lambda) \in P$? For

$$\alpha(\lambda) = e_\mu(\lambda) = \begin{cases} 1, & \lambda \leq \mu, \\ 0, & \lambda > \mu, \end{cases}$$

we have $\alpha(\Lambda) = E(\mu) \in P$, thus $\alpha(\lambda) = e_\mu(\lambda)$ is such. Using (c) in §7 we see that

$$\alpha(\lambda) = e_\mu(\lambda) - e_\nu(\lambda) = \begin{cases} 1, & \nu < \lambda \leq \mu, \\ 0, & \text{otherwise} \end{cases} \qquad (\nu < \mu),$$

too, is such, and every finite linear aggregate of functions $\alpha(\lambda) = e_\mu(\lambda) - e_\nu(\lambda)$ and $\alpha(\lambda) = 1$ similarly,—that is every finite-valued intervalwise constant function. Now by (d) in §7 every continuous function $\alpha(\lambda)$ is such, as it is a (uniform) limit of the above functions. Renewed application of (d) in §7 extends this to all Baire-functions $\alpha(\lambda)$ and (b) in §7 to all $\sigma(\lambda)$-measurable $\alpha(\lambda)$'s. So we see: The ring P contains all $\alpha(\Lambda)$'s. Conversely: If a ring contains all $\alpha(\Lambda)$'s, then it contains all $e_\mu(\Lambda) = E(\mu)$ and therefore P must be a subset of it. Thus P is the ring generated by all $\alpha(\Lambda)$'s.

Consider now two equivalent representations of $\mathfrak{H}$: $\mathfrak{H}_1^{(\lambda)}, \sigma_1(\lambda)$ and $\mathfrak{H}_2^{(\lambda)}, \sigma_2(\lambda)$.

Denote their rings by P_1 and P_2 respectively and their $\alpha(\Lambda)$'s by $\alpha(\Lambda_1)$ and $\beta(\Lambda_2)$ respectively. If $\beta(\lambda)$ is $\sigma_2(\lambda)$-measurable, then we know from Remark 1, after Definition 3, that $\alpha(\lambda) = \beta(m(\lambda))$ is $\sigma_1(\lambda)$-measurable, and the definitions of §7 make it evident that $\alpha(\Lambda_1) = \beta(\Lambda_2)$. Thus the set of all $\alpha(\Lambda_1)$'s is a subset of the set of all $\beta(\Lambda_2)$'s. Interchanging $\mathfrak{H}_1^{(\lambda)}, \sigma_1(\lambda)$ and $\mathfrak{H}_2^{(\lambda)}, \sigma_2(\lambda)$, we see that the converse too is true, thus these two sets are identical. Therefore they generate the same ring, that is: $P_1 = P_2$.

In other words:

The set of all $\alpha(\Lambda)$'s is an equivalence-invariant, and so is the ring P which it generates.

Let $m(\lambda)$ be a function which maps the interval $-2 < \lambda < -1$ in a one-to-one a bicontinuous way on the interval $-\infty < \lambda < +\infty$ (For instance:

$$m(\lambda) = \frac{2\lambda + 3}{1 - |2\lambda + 3|}.)$$

If the $E(\lambda), \sigma(\lambda)$ are given, define

$$E'(\lambda) = \begin{cases} 0, & \lambda \leq -2, \\ E(m(\lambda)), & -2 < \lambda < -1, \\ 1, & \lambda \geq -1, \end{cases} \qquad \sigma'(\lambda) = \begin{cases} \sigma(-\infty), & \lambda \leq -2, \\ \sigma(m(\lambda)), & -2 < \lambda < -1, \\ \sigma(+\infty), & \lambda \geq -1. \end{cases}$$

Then $E'(\lambda)$ is a resolution of unity, $\sigma'(\lambda)$ is an N-function, and one sees that $\sigma'(\lambda)$ is equivalent to $E'(\lambda)$ because $\sigma(\lambda)$ is equivalent to $E(\lambda)$. The set of all λ with $\lambda \leq -2$ or $\lambda \geq -1$ has the $\sigma'(\lambda)$-measure 0. Thus $\lambda \rightleftarrows m(\lambda)$ is a one-to-one mapping of the complement of a set of $\sigma'(\lambda)$-measure 0 on the complement of a set of $\sigma(\lambda)$-measure 0 (of $-2 < \lambda < -1$ on $-\infty < \lambda < +\infty$) which maps $\sigma'(\lambda)$-measurable sets on $\sigma(\lambda)$-measurable sets and conversely, transforming the $\sigma'(\lambda)$ measure into the $\sigma(\lambda)$ measure.

Define next $\mathfrak{H}'^{(\lambda)} = \mathfrak{H}^{(m(\lambda))}$ (for $-2 < \lambda < -1$, the $\lambda \leq -2$ or ≥ -1 do not matter, as their set has the $\sigma'(\lambda)$-measure 0). Define $\sigma'(\lambda)$-summability for a function $f'(\lambda)$ ($\in \mathfrak{H}'^{(\lambda)}$) by the $\sigma(\lambda)$-summability of the $f(\lambda)$ ($\in \mathfrak{H}^{(\lambda)}$) for which

$$f(m(\lambda)) = f'(\lambda) \quad (\in \mathfrak{H}'^{(\lambda)} = \mathfrak{H}^{(m(\lambda))}),$$

and then

$$\int_{-} f'(\lambda)\sqrt{d\sigma'(\lambda)} = \int_{-} f(\lambda)\sqrt{d\sigma(\lambda)}.$$

Clearly $\mathfrak{H}$ is the generalized direct sum of the $\mathfrak{H}'^{(\lambda)}$ with the weight-function $\sigma'(\lambda)$ because it is so for $\mathfrak{H}^{(\lambda)}$ and $\sigma(\lambda)$. And these two representations are equivalent if we use the mapping $\lambda \rightleftarrows m(\lambda)$ and the factor 1.

The resolution of unity belonging to $\mathfrak{H}'^{(\lambda)}$, $\sigma'(\lambda)$ is $E'(\lambda)$.

Summing up we can state:

Every equivalence-class of representations of $\mathfrak{H}$ as a generalized direct sum contains at least one, for the resolution of unity of which $E(\lambda) = 0$, $\sigma(\lambda) = \sigma(-\infty)$ for $\lambda \leq -2$ and $E(\lambda) = 1$, $\sigma(\lambda) = \sigma(+\infty)$ for $\lambda \geq -1$.

We prove next:

Lemma 4. *P is the set of all $\alpha(\Lambda)$ where $\alpha(\lambda)$ is any $\sigma(\lambda)$-measurable function which is bounded if a suitable set of $\sigma(\lambda)$-measure 0 is disregarded. One may restrict $\alpha(\lambda)$ to bounded Baire-functions, without affecting the result.*

Proof. Let us first consider the first part of the Lemma. We have already seen that P contains the set of all $\alpha(\Lambda)$, thus it suffices to prove that it is contained in it. As both sets are equivalence-invariant, therefore the above result implies that we need to consider only the case where $E(\lambda) = 0$, $\sigma(\lambda) = \sigma(-\infty)$ for $\lambda \leq -2$ and $E(\lambda) = 1$, $\sigma(\lambda) = \sigma(+\infty)$ for $\lambda \geq -1$.

In this case the Remark in §7 applies: The operator Λ exists and is bounded, and $\alpha(\Lambda)$ is the operator-function of Λ in the usual sense. Thus by the theorem of (4, p. 213), the ring generated by Λ is contained in the set of all $\alpha(\Lambda)$.

$E(\lambda)$ is clearly the resolution of unity of Λ, thus Λ generates the same ring as the $E(\lambda)$ with $\lambda < 0$ and the $1 - E(\lambda)$ with $\lambda \geq 0$, by (2, pp. 389–390). As $E(\lambda) = 1$ for all $\lambda \geq -1$, we may as well speak of this ring generated by all $E(\lambda)$, $-\infty < \lambda < +\infty$, that is P. Thus P is contained in the set of all $\alpha(\Lambda)$. This completes the proof of the first half. The second half follows from (b) in §7 together with (4, p. 213). □

12 P can be any Abelian ring containing 1, but no other ring; P determines the generalized direct sum precisely up to equivalence

The foregoing results put us in the position to prove this theorem:

Theorem IV. *A given unitary space $\mathfrak{H}$ can be represented as a generalized direct sum to which a given ring P belongs, if and only if P is Abelian and contains* 1. *In this case these representations form precisely one equivalence-class.*

Proof. *Necessity of the condition:* We proved at the beginning of §11 that every ring P which belongs to a generalized direct sum is Abelian and contains 1.

Sufficiency of the condition: Let P be an Abelian ring containing 1. By (2, pp. 401, 403), a resolution of unity $E(\lambda)$ exists, so that $E(\lambda)$ with $\lambda < 0$ and the $1 - E(\lambda)$ with $\lambda \geq 0$ generate P. As $1 \in P$, we can add the 1 to these. If 1 is included, we can replace the $1 - E(\lambda)$ by the $E(\lambda)$ (for $\lambda \geq 0$), and then, as $\lim_{\lambda\to+\infty} E(\lambda) = 1$, we may omit 1. Thus P is generated by all $E(\lambda)$, $-\infty < \lambda < +\infty$.

Now choose an N-function $\sigma(\lambda)$ equivalent to $E(\lambda)$, and apply Theorem III to $E(\lambda), \sigma(\lambda)$. Then a representation of $\mathfrak{H}$ as a generalized direct sum results, to which the resolution of unity $E(\lambda)$, and thus the ring P belongs.

Relationship to equivalence: Let P be an Abelian ring containing 1, and $\mathfrak{H}_1^{(\lambda)}$ a representation of $\mathfrak{H}$ as a generalized direct sum with the weight function $\sigma_1(\lambda)$ and the ring P. Let $\mathfrak{H}_2^{(\lambda)}$ be another representation of $\mathfrak{H}$, with the weight-

function $\sigma_2(\lambda)$. If these two representations are equivalent, they must have, as we proved in §11, the same rings; therefore $\mathfrak{H}_2^{(\lambda)}, \sigma_2(\lambda)$ has the ring P. If we can prove conversely, that whenever $\mathfrak{H}_2^{(\lambda)}, \sigma_2(\lambda)$ has the ring P then $\mathfrak{H}_1^{(\lambda)}, \sigma_1(\lambda)$ and $\mathfrak{H}_2^{(\lambda)}, \sigma_2(\lambda)$ are equivalent, then we see: The $\mathfrak{H}_2^{(\lambda)}, \sigma_2(\lambda)$ with the ring P form precisely the equivalence-class of $\mathfrak{H}_1^{(\lambda)}, \sigma_1(\lambda)$.

Assume therefore that $\mathfrak{H}_1^{(\lambda)}, \sigma_1(\lambda)$ has the resolution of unity $E_1(\lambda)$, that $\mathfrak{H}_2^{(\lambda)}, \sigma_2(\lambda)$ has the resolution of unity $E_2(\lambda)$, and that both have the ring P. We want to prove their equivalence. Owing to the results of §11 we can restrict ourselves to the case where $E_1(\lambda)$ and $E_2(\lambda)$ are 0 for $\lambda < -2$ and 1 for $\lambda \geq -1$. And remembering those at the end of §9 we can replace $\sigma_1(\lambda)$ and $\sigma_2(\lambda)$ by any two N-functions $\tau_1(\lambda)$ and $\tau_2(\lambda)$ equivalent to $E_1(\lambda)$ and $E_2(\lambda)$ respectively. Let $\varphi_1, \varphi_2, \ldots$ be a complete normalized orthogonal set in $\mathfrak{H}$, and $a_1, a_2, \ldots$ a sequence of numbers > 0 with a finite $\sum_n a_n$, then we can put by (δ) in §2

$$\tau_1(\lambda) = \sum_n a_n \|E_1(\lambda)\varphi_n\|^2, \qquad \tau_2(\lambda) = \sum_n a_n \|E_2(\lambda)\varphi_n\|^2.$$

$\tau_1(\lambda), \tau_2(\lambda)$ are constant for $\lambda < -2$ and for $\lambda > -1$.

We form now the operators $\alpha(\Lambda)$ for both resolutions of unity $E_1(\lambda)$ and $E_2(\lambda)$, denoting them by $\alpha(\Lambda_1)$ and $\beta(\Lambda_2)$ respectively. As $\tau_1(\lambda), \tau_2(\lambda)$ are constant outside of the finite interval $-2 \leq \lambda \leq -1$, the Remark in §7 applies: The operators Λ_1, Λ_2 exist and are bounded, and $\alpha(\Lambda_1), \beta(\Lambda_2)$ are their respective operator-functions in the usual sense. Now apply Lemma 4: $\alpha(\lambda) = \lambda$ is bounded if we disregard the set $\lambda < -2$ or $\lambda > -1$, which has $\tau_1(\lambda)$- and $\tau_2(\lambda)$-measure 0, so $\Lambda_1, \Lambda_2 \in P$. Therefore $\Lambda_1 = m(\Lambda_2)$, where $m(\lambda)$ is a bounded Baire-function, and similarly $\Lambda_2 = n(\Lambda_1)$, where $n(\lambda)$ is a bounded Baire-function.

So $\Lambda_1 = n(m(\Lambda_1))$, therefore the set N_1' of all λ with $n(m(\lambda)) \neq \lambda$ has the $\tau_1(\lambda)$-measure 0 (this is best seen by applying $\Lambda_1 = n(m(\Lambda_1))$ to a $\int_{-} f(\lambda)\sqrt{d\tau_1(\lambda)}$ for which $f(\lambda) \neq 0$ for all λ, for instance $f(\lambda) = \varphi_1(\lambda)$, where $\varphi_m(\lambda)$ is a measurable family) and similarly the set N_2' of all λ with $m(n(\lambda)) \neq \lambda$ has the $\tau_2(\lambda)$-measure 0.

If $n(m(\lambda)) = \lambda$, then $m(n(m(\lambda))) = m(\lambda)$, that is: $\lambda \notin N_1'$ implies $m(\lambda) \notin N_2'$, thus $\lambda \rightleftarrows m(\lambda)$ maps the complement of N_1' on a subset of the complement of N_2', and has there the inverse $\lambda \rightleftarrows n(\lambda)$. The same is true if we interchange $N_1', m(\lambda)$ with $N_2', n(\lambda)$. Therefore $\lambda \rightleftarrows m(\lambda)$ is a one-to-one mapping of the complement of N_1' on the complement of N_2' and has there the inverse $\lambda \rightleftarrows n(\lambda)$.

Consider a non-negative, bounded Baire-function $\gamma(\lambda)$. We have

$$\int \gamma(\lambda) d\tau_1(\lambda) = \sum_n a_n \int \gamma(\lambda) d\big(\|E_1(\lambda)\varphi_n\|^2\big) = \sum_n a_n (\gamma(\Lambda_1)\varphi_n, \varphi_n)$$

$$\left(\text{observe that } \int \gamma(\lambda) d\big(\|E_1(\lambda)\varphi_n\|^2\big) = \int \gamma(\lambda) d\left[\int_{\mu \leq \lambda} \|\varphi_n(\mu)\|^2 d\tau_1(\mu)\right]\right.$$

$$\left. = \int \gamma(\lambda)\|\varphi_n(\lambda)\|^2 d\tau_1(\lambda) = \int (\gamma(\lambda)\varphi_n(\lambda), \varphi_n(\lambda)) d\tau_1(\lambda) = (\gamma(\Lambda_1)\varphi_n, \varphi_n)\right)$$

and similarly

$$\int \gamma(\lambda) d\tau_2(\lambda) = \sum_n a_n \int \gamma(\lambda) d\big(\|E_2(\lambda)\varphi_n\|^2\big) = \sum_n a_n(\gamma(\Lambda_2)\varphi_n, \varphi_n).$$

As $\gamma(\Lambda_1) = \gamma(n(\Lambda_2))$, these formulae imply

$$\int \gamma(\lambda) d\tau_1(\lambda) = \int \gamma(n(\lambda)) d\tau_2(\lambda).$$

Now let K_1 be a Borel-set,

$$\gamma(\lambda) = 1_{K_1}(\lambda) = \begin{cases} 1, & \lambda \in K_1, \\ 0, & \lambda \notin K_1. \end{cases}$$

Then

$$\gamma(n(\lambda)) = 1_{K_2'}(\lambda) = \begin{cases} 1, & \lambda \in K_2', \\ 0, & \lambda \notin K_2', \end{cases}$$

where K_2' is the inverse-image of K_1 under the mapping $\lambda \rightleftarrows n(\lambda)$. [8] So we have proved $\mu_{\tau_1(\lambda)}(K_1) = \mu_{\tau_2(\lambda)}(K_2')$. Assume now $K_1 \subset \mathrm{Complement}\, N_1'$. Then $K_2 = K_2' \cdot \mathrm{Complement}\, N_2'$ is a Borel set, because K_2', N_2' are such sets. K_2 is the $\lambda \rightleftarrows m(\lambda)$-image of K_1 by the properties of $m(\lambda), n(\lambda)$ derived above. As $\mu_{\tau_2(\lambda)}(N_2') = 0$, we have

$$\mu_{\tau_1(\lambda)}(K_1) = \mu_{\tau_2(\lambda)}(K_2). \tag{12.1}$$

If K_1 is an arbitrary Borel-set, then its $\lambda \rightleftarrows m(\lambda)$-image is the sum of those of $K_1 \cdot N_1'$ and of $K_1 \cdot \mathrm{Complement}\, N_1'$. The former is a subset of N_2' and thus has the $\tau_2(\lambda)$ measure 0. The latter is the image of a Borel-subset of $\mathrm{Complement}\, N_1'$, thus a Borel set, with the $\tau_2(\lambda)$-measure $\mu_{\tau_1(\lambda)}(K_1 \cdot \mathrm{Complement}\, N_1') = \mu_{\tau_1(\lambda)}(K_1)$. Thus the total image K_2 of K_1 is $\tau_2(\lambda)$-measurable, and (12.1) holds again. If K_1 is any set with $\mu_{\tau_1(\lambda)}(K_1) = 0$, then it is contained in a Borel-set K_1^0 with $\mu_{\tau_1(\lambda)}(K_1^0) = 0$. If their $\lambda \rightleftarrows m(\lambda)$-images are K_2 and K_2^0 respectively, then K_2 is a subset of K_2^0, and by (12.1) $\mu_{\tau_2(\lambda)}(K_2^0) = 0$, therefore $\mu_{\tau_2(\lambda)}(K_2) = 0$. Thus (12.1) holds for sets K_1 of $\tau_1(\lambda)$-measure 0, too. Combining this with the result for Borel-sets, we see: If K_1 is $\tau_1(\lambda)$-measurable, then its $\lambda \rightleftarrows m(\lambda)$-image K_2 is $\tau_2(\lambda)$-measurable, and (12.1) is true.

[8] Being an inverse-image of a Borel-set with respect to a Baire-function, K_2' is necessarily a Borel-set too. Observe that this is not true in general for images!

Consideration of the mapping $\lambda \rightleftarrows n(\lambda)$ proves that the converse, too, holds: K_1 is $\tau_1(\lambda)$-measurable if and only if K_2 is $\tau_2(\lambda)$-measurable, and in this case (12.1) holds. The same is true if we make the replacement

$$(\tau_1(\lambda), \tau_2(\lambda), m(\lambda)) \quad \text{by} \quad (\tau_2(\lambda), \tau_1(\lambda), n(\lambda)).$$

Let now $\varphi_m(\lambda)$ be a measurable family in the $E_1(\lambda), \tau_1(\lambda)$-representation of $\mathfrak{H}$. $\varphi_m(\lambda)$ is defined for $m \leq k_1(\lambda)$ ($k_1(\lambda)$ is the dimensionality of $\mathfrak{H}_1^{(\lambda)}$), define for $m > k_1(\lambda)$ $\varphi_m(\lambda) = 0$ (as in §6). $\varphi_m(\lambda)$ is clearly $\tau_1(\lambda)$-summable, define

$$f_m = \int_{-} \varphi_m(\lambda)\sqrt{d\tau_1(\lambda)},$$

and then in the $E_2(\lambda), \tau_2(\lambda)$-representation of $\mathfrak{H}$

$$f_m = \int_{-} \psi_m(\lambda)\sqrt{d\tau_2(\lambda)},$$

We have for any bounded Baire-function $\alpha(\lambda)$

$$\begin{aligned}
(\alpha(\Lambda_1)f_m, f_n) &= \left(\int_{-} \alpha(\lambda)\varphi_m(\lambda)\sqrt{d\tau_1(\lambda)}, \int_{-} \varphi_n(\lambda)\sqrt{d\tau_1(\lambda)}\right) \\
&= \int (\alpha(\lambda)\varphi_m(\lambda), \varphi_n(\lambda))d\tau_1(\lambda) \\
&= \int \alpha(\lambda)(\varphi_m(\lambda), \varphi_n(\lambda))d\tau_1(\lambda) \\
&= \begin{cases} 0, & m \neq n, \\ \int \alpha(\lambda)d\tau_1(\lambda), & m = n, \end{cases}
\end{aligned}$$

and at the same time

$$\begin{aligned}
(\alpha(\Lambda_2)f_m, f_n) &= \left(\int_{-} \alpha(\lambda)\psi_m(\lambda)\sqrt{d\tau_2(\lambda)}, \int_{-} \psi_n(\lambda)\sqrt{d\tau_2(\lambda)}\right) \\
&= \int (\alpha(\lambda)\psi_m(\lambda), \psi_n(\lambda))d\tau_2(\lambda) \\
&= \int \alpha(\lambda)(\psi_m(\lambda), \psi_n(\lambda))d\tau_2(\lambda).
\end{aligned}$$

Considering $\alpha(\Lambda_2) = \alpha(m(\Lambda_1))$ this means:

$$\int \alpha(\lambda)(\psi_m(\lambda), \psi_n(\lambda))d\tau_2(\lambda)$$
$$= \begin{cases} 0, & m \neq n, \\ \begin{aligned} &\int_{\{k_1(\lambda) \geq m\}} \alpha(m(\lambda))d\tau_1(\lambda) \\ &\quad = \int_{\{k_1(n(\lambda)) \geq m\}} \alpha(\lambda)d\tau_2(\lambda) \end{aligned}, & m = n. \end{cases}$$

Thus

$$\int \alpha(\lambda)\big((\psi_m(\lambda), \psi_n(\lambda)) - e_{mn}(\lambda)\big)d\tau_2(\lambda) = 0,$$
$$e_{mn}(\lambda) = \begin{cases} 1, & m = n \leq k_1(n(\lambda)), \\ 0, & \text{otherwise,} \end{cases}$$

for every bounded Baire-function $\alpha(\lambda)$, and therefore for every bounded $\tau_2(\lambda)$-measurable $\alpha(\lambda)$. This implies $(\psi_m(\lambda), \psi_n(\lambda)) - e_{mn}(\lambda) = 0$, except for a set $N_2^{\prime\prime\prime(m,n)}$ of $\tau_2(\lambda)$-measure 0. [9] The sum of all $N_2^{\prime\prime\prime(m,n)}$, say $N_2^{\prime\prime\prime}$, is a fixed set of $\tau_2(\lambda)$-measure 0, and with the same property. So for $\lambda \notin N_2^{\prime\prime\prime}$ the $\psi_m(\lambda)$, $m < k_1(n(\lambda))$ form a normalized orthogonal set in $\mathfrak{H}_2^{(\lambda)}$, therefore the dimensionality $k_2(\lambda)$ of $\mathfrak{H}_2^{(\lambda)}$ is then $\geq k_1(n(\lambda))$.

Similarly there exists a set $N_1^{\prime\prime\prime}$ of $\tau_1(\lambda)$-measure 0, so that $\lambda \notin N_1^{\prime\prime\prime}$ implies $k_1(\lambda) \geq k_2(m(\lambda))$. Let now N_1 be the sum of $N_1^{\prime}$, $N_1^{\prime\prime\prime}$, and the $\lambda \rightleftarrows n(\lambda)$-image of $N_2^{\prime\prime\prime}$, and N_2 the sum of $N_2^{\prime}, N_2^{\prime\prime\prime}$, and the $\lambda \rightleftarrows m(\lambda)$-image of $N_1^{\prime\prime\prime}$. Clearly N_1, N_2 are respective sets of $\tau_1(\lambda), \tau_2(\lambda)$-measure 0, and $\lambda \rightleftarrows m(\lambda)$ is a one-to-one mapping of the complement of N_1 on the complement of N_2, its inverse

being there $\lambda \rightleftarrows n(\lambda)$. Furthermore we have for $\lambda \notin N_1$, $k_1(\lambda) \geq k_2(m(\lambda))$ and for $\lambda \notin N_2$, $k_2(\lambda) \geq k_1(n(\lambda))$. $\lambda \notin N_1$ implies $m(\lambda) \notin N_2$, $k_2(m(\lambda)) \geq k_1(n(m(\lambda))) = k_1(\lambda)$ too, so $k_1(\lambda) = k_2(m(\lambda))$.

Choose a measurable family $\varphi_{1m}(\lambda)$ for the $E_1(\lambda), \tau_1(\lambda)$-representation, and one $\varphi_{2m}(\lambda)$ for the $E_2(\lambda), \tau_2(\lambda)$-representation. Define the isomorphism $g^{(\lambda)}$

[9] Choose $\alpha_1(\lambda) = \epsilon(Z)$ for the $\epsilon(Z)$ of footnote 7 and with Z equal to the $\{\cdots\}$-expression, then $\int |(\psi_m(\lambda), \psi_n(\lambda)) - e_{mn}(\lambda)| d\tau_2(\lambda) = 0$ results, and thus the statement in the text.

of $\mathfrak{H}_1^{(\lambda)}$ on $\mathfrak{H}_2^{(m(\lambda))}$, for $\lambda \notin N_1$, by

$$g^{(\lambda)}\left(\sum_m x_m \varphi_{1m}(\lambda)\right) = \sum_m x_m \varphi_{2m}(m(\lambda)) \qquad \text{(remember } k_1(\lambda) = k_2(m(\lambda))\text{!).}$$

Introduce now the functions $x_{1m}(\lambda)$ and $x_{2m}(\lambda)$ in the sense of Definition 2 for the respective $E_1(\lambda), \tau_1(\lambda)$- and $E_2(\lambda), \tau_2(\lambda)$-representation. Then the connection between $f_1(\lambda)$ and $f_2(\lambda)$ as described in (3) in Definition 3, and putting $\gamma(\lambda) = 1$, is equivalent to $x_{1m}(\lambda) = x_{2m}(m(\lambda))$ (disregarding a λ-set of $\tau_1(\lambda)$-measure 0). Now the equivalence of the $\tau_1(\lambda)$-summability-condition of (γ), (δ) in Definition 2, in terms of the $x_{1m}(\lambda)$ and of the $\tau_2(\lambda)$-summability condition in the same sense, in terms of the $x_{2m}(\lambda)$, can be established at once: It follows from the fact that if K_1 is the $\lambda \rightleftarrows m(\lambda)$-inverse-image of K_2 then the $\tau_1(\lambda)$-measurability of K_1 and the $\tau_2(\lambda)$-measurability of K_2 are equivalent, and $\mu_{\tau_1(\lambda)}(K_1) = \mu_{\tau_2(\lambda)}(K_2)$.

Summing up: Our $N_1, N_2, m(\lambda)$ and $\gamma(\lambda) = 1$ fulfill the conditions (1)–(3) in Definition 3. Therefore we have established the equivalence of the $E_1(\lambda), \tau_1(\lambda)$- and of the $E_2(\lambda), \tau_2(\lambda)$-representations of $\mathfrak{H}$, and thus completed our proof. □

By the preceding theorem we have found what is usually termed a "complete set of invariants" for the representations of a given unitary space as a generalized direct sum, with respect to the notion of equivalence, as defined in Definition 3 and Remark 2 in §9. It consists of the one invariant P, the ring which belongs to the representation, and the range of P is precisely the set of all Abelian rings containing 1.

13 Discussion of the decomposition of an operator $A \in P'$: $A \sim \sum A(\lambda)$. Necessary and sufficient conditions and uniqueness

Let $\mathfrak{H}$ be the generalized direct sum of the $\mathfrak{H}^{(\lambda)}$ with the weight-function $\sigma(\lambda)$, the resolution of unity $E(\lambda)$, and the ring P. Form the ring P' of all those bounded linear operators in $\mathfrak{H}$ which commute with every element of P, cf. (2, p. 388).

Consider an operator $A \in P'$, put

$$\underset{f \neq 0}{\text{l.u.b.}} \frac{\|Af\|}{\|f\|} = |\!|\!|A|\!|\!| = C.$$

Let $\varphi_m(\lambda)$, $m \leq k(\lambda)$, be a measurable family, and define again $\varphi_m(\lambda) = 0$ for $m > k(\lambda)$. $\varphi_m(\lambda)$ is $\sigma(\lambda)$-summable, put $\omega_m = \int_{-} \varphi_m(\lambda)\sqrt{d\sigma(\lambda)}$. Form now the

$f_m(\lambda)$ with

$$A^*\omega_m = \int_{-} f_m(\lambda)\sqrt{d\sigma(\lambda)}$$

(A^* is the adjoint of A, cf. (1, p. 112) and (5, p. 295)), and define $a_{mn}(\lambda) = (f_m(\lambda), \varphi_n(\lambda))$, so that $f_m(\lambda) = \sum_n a_{mn}(\lambda)\varphi_n(\lambda)$.

Now choose an arbitrary $f \in \mathfrak{H}$, and put

$$f = \int_{-} \bar{f}(\lambda)\sqrt{d\sigma(\lambda)}, \qquad Af = \int_{-} \bar{\bar{f}}(\lambda)\sqrt{d\sigma(\lambda)},$$

then

$$(f, A^*\omega_m) = (Af, \omega_m),$$

and so

$$\int \left(\bar{f}(\lambda), \sum_n \overline{a_{mn}(\lambda)}\varphi_n(\lambda)\right) d\sigma(\lambda) = \int (\bar{\bar{f}}(\lambda), \varphi_m(\lambda)) d\sigma(\lambda).$$

Replace f by

$$E(\mu)f = \int_{-} \bar{f}_\mu(\lambda)\sqrt{d\sigma(\lambda)}, \qquad \bar{f}_\mu(\lambda) = \begin{cases} \bar{f}(\lambda), & \lambda \leq \mu, \\ 0, & \lambda > \mu. \end{cases}$$

As $E(\mu) \in P$, $A \in P'$, so that $E(\mu)$ and A commute, this has the effect of replacing Af by

$$AE(\mu)f = E(\mu)Af = \int_{-} \bar{\bar{f}}_\mu(\lambda)\sqrt{d\sigma(\lambda)}, \qquad \bar{\bar{f}}_\mu(\lambda) = \begin{cases} \bar{\bar{f}}(\lambda), & \lambda \leq \mu, \\ 0, & \lambda > \mu. \end{cases}$$

Thus the above formula becomes

$$\int_{\lambda \leq \mu} \left(\bar{f}(\lambda), \sum_n \overline{a_{mn}(\lambda)}\varphi_n(\lambda)\right) d\sigma(\lambda) = \int_{\lambda \leq \mu} (\bar{\bar{f}}(\lambda), \varphi_m(\lambda)) d\sigma(\lambda),$$

$$\int_{\lambda \leq \mu} \left[\sum_n a_{mn}(\lambda)(\bar{f}(\lambda), \varphi_n(\lambda)) - (\bar{\bar{f}}(\lambda), \varphi_m(\lambda))\right] d\sigma(\lambda) = 0.$$

Thus

$$\sum_n a_{mn}(\lambda)(\bar{f}(\lambda), \varphi_n(\lambda)) = (\bar{\bar{f}}(\lambda), \varphi_m(\lambda)),$$

except perhaps for a set $\widetilde{N}^{(m)}$ of $\sigma(\lambda)$-measure 0 (cf. the argument of footnote 7). The sum of all $\widetilde{N}^{(m)}$, say $\widetilde{N}$, is a fixed set of $\sigma(\lambda)$-measure 0 (depending of course on the $\bar{f}(\lambda)$ and $\bar{\bar{f}}(\lambda)$) and with the same property. If we put

$$\bar{f}(\lambda) = \sum_m x_m(\lambda)\varphi_m(\lambda), \qquad \bar{\bar{f}}(\lambda) = \sum_m y_m(\lambda)\varphi_m(\lambda),$$

then we can formulate the above result as follows:

$$y_m(\lambda) = \sum_n a_{mn}(\lambda)x_n(\lambda), \tag{13.1}$$

except for a set of $\sigma(\lambda)$-measure 0. (The series on the right side converges absolutely in any event, because

$$\sum_n |a_{mn}(\lambda)|^2 = \|f_m(\lambda)\|^2, \qquad \sum_n |x_n(\lambda)|^2 = \|\bar{f}(\lambda)\|^2$$

are finite.) Observe that while the $a_{mn}(\lambda)$ are determined numbers (depending only on A), the exceptional set in (13.1) depends on the choice of the $\bar{f}(\lambda)$ and $\bar{\bar{f}}(\lambda)$ too. If, however, the $\bar{f}(\lambda)$ (that is, the $x_n(\lambda)$) are given, then (13.1) can be used to define the $\bar{\bar{f}}(\lambda)$ (that is the $y_n(\lambda)$), and then no exceptions occur at all.

Consider now the function

$$\operatorname*{l.u.b.}_{\substack{\sum_m |x_m|^2 < +\infty \\ \sum_m |x_m|^2 > 0}} \frac{\sum_m \left|\sum_n a_{mn}(\lambda)x_n\right|^2}{\sum_m |x_m|^2} = C(\lambda).$$

The $\sum_n$ in the numerator are absolutely convergent (because $\sum_n |a_{mn}(\lambda)|^2$, $\sum_n |x_n|^2$ are finite), the $\sum_m$ in the numerator has non-negative terms and thus is either absolutely convergent or properly divergent to $+\infty$, the $\sum_m$ in the denominator is absolutely convergent and > 0. Thus $C(\lambda)$ is ≥ 0, $\leq +\infty$.

Let S be the set of all λ with $C(\lambda) > C^2$, and for a given sequence $x_1, x_2, \ldots$ with $\sum_m |x_m|^2 = 1$ let $S_p(x_1, x_2, \ldots)$ be the set of all λ with $k(\lambda) = p$ and

$$\sum_m \left|\sum_n a_{mn}(\lambda)x_n\right|^2 > C^2.$$

Form all sequences $x_1, x_2, \ldots$ of length of p with $\sum_m |x_m|^2 = 1$, in which only a finite number of $x_n \neq 0$ occur, and in which the ratios $\Re x_1 : \Im x_1 : \Re x_2 : \Im x_2 : \cdots$

are all rational. They form a countable set, and we can therefore enumerate them: $x_1^{(\nu)}, x_2^{(\nu)}, \ldots$ for $\nu = 1, 2, \ldots$. Now clearly

$$\begin{aligned}\operatorname*{l.u.b.}_{\substack{\sum_m |x_m|^2<+\infty \\ \sum_m |x_m|^2>0}} \frac{\sum_m |\sum_n a_{mn}(\lambda)x_n|^2}{\sum_m |x_m|^2} &= \operatorname*{l.u.b.}_{\sum_m |x_m|^2=1} \sum_m \left|\sum_n a_{mn}(\lambda)x_n\right|^2 \\ &= \operatorname*{l.u.b.}_{\nu=1,2,\ldots} \sum_m \left|\sum_n a_{mn}(\lambda)x_n^{(\nu)}\right|^2.\end{aligned}$$

Therefore

$$S = \sum_{p,\nu=1,2,\ldots} S_p(x_1^{(\nu)}, x_2^{(\nu)}, \ldots).$$

We will prove below $\mu_{\sigma(\lambda)}(S_p(x_1, x_2, \ldots)) = 0$ in general, then this relation gives $\mu_{\sigma(\lambda)}(S) = 0$.

Assume in fact that a sequence $x_1, x_2, \ldots$ with $\sum_m |x_m|^2 = 1$ is given. The set $S_p(x_1, x_2, \ldots)$ is $\sigma(\lambda)$-measurable (because the functions $k(\lambda)$, $a_{mn}(\lambda)$ are such), and we can therefore define

$$\bar{f}(\lambda) = \begin{cases} \sum_m x_m \varphi_m(\lambda), & \lambda \in S_p(x_1, x_2, \ldots), \\ 0, & \text{otherwise.} \end{cases}$$

We have $\|\bar{f}(\lambda)\|^2 = \sum_m |x_m|^2 = 1$ resp. 0. This and the $\sigma(\lambda)$-measurability of $S_p(x_1, x_2, \ldots)$, give, by (γ), (δ) in Definition 2, at once that $\bar{f}(\lambda)$ is $\sigma(\lambda)$-summable. Now put $f = \int_- \bar{f}(\lambda)\sqrt{d\sigma(\lambda)}$. Thus $\|f\|^2 = \int \|\bar{f}(\lambda)\|^2 d\sigma(\lambda) = \mu_{\sigma(\lambda)}(S_p(x_1, x_2, \ldots))$. For $Af = \int_- \bar{\bar{f}}(\lambda)\sqrt{d\sigma(\lambda)}$ we have

$$\bar{\bar{f}}(\lambda) = \begin{cases} \sum_m y_m(\lambda)\varphi_m(\lambda), \quad y_m(\lambda) = \sum_n a_{mn}(\lambda)x_n, & \lambda \in S_p(x_1, x_2, \ldots), \\ 0, & \text{otherwise.} \end{cases}$$

Thus $\|\bar{\bar{f}}(\lambda)\|^2 = \sum_m |y_m(\lambda)|^2 = \sum_m |\sum_n a_{mn}(\lambda)x_n|^2 > C^2$ resp. $= 0$, and so if $\mu_{\sigma(\lambda)}(S_p(x_1, x_2, \ldots)) > 0$, then $\|Af\|^2 = \int \|\bar{\bar{f}}(\lambda)\|^2 d\sigma(\lambda) > C^2\mu_{\sigma(\lambda)}(S_p(x_1, x_2, \ldots))$. This gives $\|Af\|^2 > C^2\|f\|^2$, therefore $f \neq 0$ and $\|Af\|/\|f\| > C$. Considering the definition of C, this is impossible. Thus there must be $\mu_{\sigma(\lambda)}(S_p(x_1, x_2, \ldots)) = 0$, and this proves, as we saw above, $\mu_{\sigma(\lambda)}(S) = 0$.

We have proved $C(\lambda) \leq C^2$, except on a set of $\sigma(\lambda)$-measure 0. As such a set does not matter, we can replace in it all $a_{mn}(\lambda)$'s by 0, thus securing $C(\lambda) \leq C^2$ without exception. In other words:

$$\frac{\sum_m |\sum_n a_{mn}(\lambda)x_n|^2}{\sum_m |x_m|^2} \leq C^2 \quad \text{if} \quad \sum_m |x_m|^2 \begin{cases} < +\infty, \\ > 0. \end{cases}$$

This enables us to define an operation $A(\lambda)$ in $\mathfrak{H}^{(\lambda)}$ by

$$f(\lambda) = \sum_m x_m \varphi_m(\lambda), \qquad A(\lambda)f(\lambda) = \sum_m y_m \varphi_m(\lambda), \qquad y_m = \sum_n a_{mn}(\lambda)x_n.$$

Our inequality proves first of all that the series for $A(\lambda)f(\lambda)$ converges, so that $A(\lambda)$ is an everywhere defined operator, obviously linear. Second, it proves $\|A(\lambda)f(\lambda)\|/\|f(\lambda)\| \leq C$, thus establishing the boundedness of $A(\lambda)$, and even

$$\underset{f(\lambda)\neq 0}{\text{l.u.b.}} \frac{\|A(\lambda)f(\lambda)\|}{\|f(\lambda)\|} = |\!|A(\lambda)|\!| \leq C \qquad \text{(cf. (2, p. 385))}.$$

Our previous results concerning f and Af can now be formulated as follows:

$$\text{If } f = \int_- f(\lambda)\sqrt{d\sigma(\lambda)}, \quad \text{then} \quad Af = \int_- A(\lambda)f(\lambda)\sqrt{d\sigma(\lambda)}.$$

We now define:

Definition 4. *Let $\mathfrak{H}$ be the generalized direct sum of the $\mathfrak{H}^{(\lambda)}$, with the weight function $\sigma(\lambda)$, the resolution of unity $E(\lambda)$, and the ring P. Let A be a linear, bounded operator in $\mathfrak{H}$, and $A(\lambda)$ one in $\mathfrak{H}^{(\lambda)}$, for each $\lambda > -\infty, < +\infty$. We say that A is the* generalized direct sum *of the $A(\lambda)$, in symbols $A \sim \sum A(\lambda)$, if*

$$A \int_- f(\lambda)\sqrt{d\sigma(\lambda)} = \int_- A(\lambda)f(\lambda)\sqrt{d\sigma(\lambda)}$$

holds for all $\sigma(\lambda)$-summable $f(\lambda)$'s.

The results obtained in the first part of this section put us in the position to prove:

Theorem V. *If A is given, then the necessary and sufficient condition for the existence of a system of $A(\lambda)$'s with $A \sim \sum A(\lambda)$ is $A \in P'$. Then these $A(\lambda)$ are uniquely determined (up to an arbitrary λ-set of $\sigma(\lambda)$-measure 0).*

If the $A(\lambda)$ are given, then the necessary and sufficient conditions for the existence of a A with $A \sim \sum A(\lambda)$ are these:

(i) Choose a measurable family $\varphi_m(\lambda)$. Each (numerical) function

$$(A(\lambda)\varphi_m(\lambda), \varphi_n(\lambda))$$

is $\sigma(\lambda)$-measurable.

(ii) The numbers $\|A(\lambda)\|$ are bounded, if a suitable λ-set of $\sigma(\lambda)$-measure 0 is disregarded. Then this A is uniquely determined.

Proof. *A is given: Necessity of the condition:*

The formula

$$A \int_{-} f(\lambda)\sqrt{d\sigma(\lambda)} = \int_{-} A(\lambda) f(\lambda)\sqrt{d\sigma(\lambda)}$$

implies

$$\begin{aligned} A\alpha(\Lambda) \int_{-} f(\lambda)\sqrt{d\sigma(\lambda)} &= A \int_{-} \alpha(\lambda) f(\lambda)\sqrt{d\sigma(\lambda)} \\ &= \int_{-} A(\lambda)\alpha(\lambda) f(\lambda)\sqrt{d\sigma(\lambda)} \\ &= \int_{-} \alpha(\lambda)A(\lambda) f(\lambda)\sqrt{d\sigma(\lambda)} \\ &= \alpha(\Lambda) \int_{-} A(\lambda) f(\lambda)\sqrt{d\sigma(\lambda)} \\ &= \alpha(\Lambda)A \int_{-} f(\lambda)\sqrt{d\sigma(\lambda)}, \end{aligned}$$

that is $A\alpha(\Lambda)f = \alpha(\Lambda)Af$. Thus A commutes with every $\alpha(\Lambda)$, that is with every element of P. Therefore $A \in P'$.

Sufficiency of the condition: This was proved in the first part of this section.

Uniqueness: Assume $A \sim \sum A(\lambda)$ and $A \sim \sum B(\lambda)$. Then we have for every $\sigma(\lambda)$-summable $f(\lambda)$

$$A \int_{-} f(\lambda)\sqrt{d\sigma(\lambda)} = \int_{-} A(\lambda) f(\lambda)\sqrt{d\sigma(\lambda)} = \int_{-} B(\lambda) f(\lambda)\sqrt{d\sigma(\lambda)},$$

and so $A(\lambda)f(\lambda) = B(\lambda)f(\lambda)$ except for a λ-set of $\sigma(\lambda)$-measure 0 (depending on the $f(\lambda)$). Let $\varphi_m(\lambda)$ be a measurable family, extend again the definition of these $\varphi_m(\lambda)$, $m \leq k(\lambda)$: $\varphi_m(\lambda) = 0$ for $m > k(\lambda)$. $\varphi_m(\lambda)$ is summable, therefore we see: $A(\lambda)\varphi_m(\lambda) = B(\lambda)\varphi_m(\lambda)$ except for a set N_m^* of $\sigma(\lambda)$-measure 0. So the sum of all N_m^*, N^{**}, is a fixed set of $\sigma(\lambda)$-measure 0, and for $\lambda \notin N^{**}$ we have $A(\lambda)\varphi_m(\lambda) = B(\lambda)\varphi_m(\lambda)$ for all $m = 1, 2, \ldots$. But the $\varphi_m(\lambda)$, $m \leq k(\lambda)$ form already a complete normalized orthogonal set in $\mathfrak{H}^{(\lambda)}$, and so we have even $A(\lambda) = B(\lambda)$ for $\lambda \notin N^{**}$.

The $A(\lambda)$ are given: Necessity of the condition:

Assume $A \sim \sum A(\lambda)$, and form the $\varphi_m(\lambda)$ as above, define $\omega_m = \int_{-} \varphi_m(\lambda)\sqrt{d\sigma(\lambda)}$. Then as $A\omega_m = \int_{-} A(\lambda)\varphi_m(\lambda)\sqrt{d\sigma(\lambda)}$, both $A(\lambda)\varphi_m(\lambda)$ and $\varphi_n(\lambda)$ are $\sigma(\lambda)$-summable, and so $(A(\lambda)\varphi_m(\lambda), \varphi_n(\lambda))$ is $\sigma(\lambda)$-measurable, fulfilling condition (i).

Constructing the $A(\lambda)$ of A with the method of the first part of this section, we obtain $|A(\lambda)| \leq |A|$ everywhere. Therefore we have in general $|A(\lambda)| \leq |A|$, except for a set of $\sigma(\lambda)$-measure 0.

Sufficiency of the condition: All functions $(A(\lambda)\varphi_m(\lambda), \varphi_n(\lambda))$ are $\sigma(\lambda)$-measurable, and $|A(\lambda)| \leq C$, except for a set of $\sigma(\lambda)$-measure 0. Let $f(\lambda)$ be $\sigma(\lambda)$-summable, then all $(f(\lambda), \varphi_m(\lambda))$ are $\sigma(\lambda)$-measurable, and so

$$
\begin{aligned}
(A(\lambda)f(\lambda), \varphi_n(\lambda)) &= (f(\lambda), A^*(\lambda)\varphi_n(\lambda)) \\
&= \sum_m (f(\lambda), \varphi_m(\lambda))(\varphi_m(\lambda), A^*(\lambda)\varphi_n(\lambda)) \\
&= \sum_m (f(\lambda), \varphi_m(\lambda))(A(\lambda)\varphi_m(\lambda), \varphi_n(\lambda))
\end{aligned}
$$

is $\sigma(\lambda)$-measurable too. Furthermore $\int \|f(\lambda)\|^2 d\sigma(\lambda)$ is finite, and so

$$
\int \|A(\lambda)f(\lambda)\|^2 d\sigma(\lambda) \leq \int C^2 \|f(\lambda)\|^2 d\sigma(\lambda) = C^2 \int \|f(\lambda)\|^2 d\sigma(\lambda)
$$

is finite too.

Thus $A(\lambda)f(\lambda)$ is $\sigma(\lambda)$-summable, and we can define an operator A by

$$
A \int_{-} f(\lambda)\sqrt{d\sigma(\lambda)} = \int_{-} A(\lambda)f(\lambda)\sqrt{d\sigma(\lambda)}.
$$

This A is clearly linear, and as

$$\begin{aligned}\left\| A \int_{-} f(\lambda)\sqrt{d\sigma(\lambda)} \right\|^2 &= \left\| \int_{-} A(\lambda) f(\lambda)\sqrt{d\sigma(\lambda)} \right\|^2 \\ &= \int \|A(\lambda) f(\lambda)\|^2 d\sigma(\lambda) \\ &\leq C^2 \int \|f(\lambda)\|^2 d\sigma(\lambda) \\ &= C^2 \left\| \int_{-} f(\lambda)\sqrt{d\sigma(\lambda)} \right\|^2 ,\end{aligned}$$

that is $\|Af\|^2 \leq C^2\|f\|^2$, $\|Af\| \leq C\|f\|$, therefore A is bounded too. $A \sim \sum A(\lambda)$ is obvious from A's definition.

Uniqueness: Obvious, as the $A(\lambda)$ determine each $A \int_{-} f(\lambda)\sqrt{d\sigma(\lambda)}$, that is each Af. □

14 Computation rules for $A \sim \sum A(\lambda)$. Convergence criteria. Operator functions. Behavior under equivalences

We now derive some properties of the decomposition $A \sim \sum A(\lambda)$, which are immediate consequences of the existence-and-uniqueness-criteria of the preceding theorem.

In the fourth and fifth part of the proof of this theorem we saw, that $|A(\lambda)| \leq |A|$, except for a set of $\sigma(\lambda)$-measure 0, and that if $|A(\lambda)| \leq C$, except for a set of $\sigma(\lambda)$-measure 0, then $|A| \leq C$. This means:

$|A|$ is the smallest number C for which $|A(\lambda)| \leq C$ holds, disregarding a suitable set of $\sigma(\lambda)$-measure 0. In other words: $|A|$ is the "least upper bound up to a set of $\sigma(\lambda)$-measure 0" of the $|A(\lambda)|$.

If $A \sim \sum \alpha(\lambda)1$ (1 is the "unit operator": $1f = f$, this time in $\mathfrak{H}^{(\lambda)}$), then we see: $(\alpha(\lambda)\varphi_1(\lambda), \varphi_1(\lambda)) = \alpha(\lambda)$ is $\sigma(\lambda)$-measurable, $|\alpha(\lambda)1| = |\alpha(\lambda)|$ is bounded up to a set of $\sigma(\lambda)$-measure 0, so $\alpha(\Lambda)$ can be formed. Comparison of the respective definitions shows, that $A = \alpha(\Lambda)$. Conversely $A = \alpha(\Lambda)$ obviously implies $A \sim \sum \alpha(\lambda)1$. As P is the set of all $\alpha(\Lambda)$, we have:

$A \in P$ means (observe that we consider only the $A \in P'$, and that $P \subset P'$, because P is Abelian), that $A \sim \sum \alpha(\lambda)1$ for some (complex-valued) function $\alpha(\lambda)$.

Finally we prove:

If $A \sim \sum A(\lambda)$, $B \sim \sum B(\lambda)$, then $\alpha A \sim \sum \alpha A(\lambda)$, $A^* \sim \sum A^*(\lambda)$,

$AB \sim \sum A(\lambda)B(\lambda)$. *Any one of the following properties is possessed by A if and only if it is possessed by all $A(\lambda)$, except for a set of $\sigma(\lambda)$-measure 0: To be a projection, to be definite, to be Hermitean, to be normal, to be unitary, to be partially unitary. (Cf. (1, pp. 74, 74, 70, 115, 71), and (10, p. 141).)*

The statements concerning $\alpha A, A+B, AB$ are obvious. The one concerning A^* (the adjoint) is proved as follows: As

$$(A(\lambda)^*\varphi_m(\lambda), \varphi_n(\lambda)) = \overline{(A(\lambda)\varphi_n(\lambda), \varphi_m(\lambda))}, \qquad \| A(\lambda)^* \| = \| A(\lambda) \|,$$

there exists an $A_1 \sim \sum A(\lambda)^*$ owing to the existence of $A \sim \sum A(\lambda)$. Now one verifies immediately $A_1 = A^*$. All other statements follow by applying these results to the equations which characterize the respective notions. These are respectively $A^* = A$, $A = A^2$, an X with $A = X^*X$ exists, [10] $A^* = A$, $A^*A = AA^*$, $A^*A = AA^* = 1$, $AA^*A = A$.

The convergence-properties of the decomposition $A \sim \sum A(\lambda)$ are contained in this criterium:

If $A \sim \sum A(\lambda)$, $A_n \sim \sum A_n(\lambda)$, then A is the strong limit of $A_1, A_2, \ldots$ (cf. (2, pp. 381–382)) if and only if the two following conditions are fulfilled:

(I) The $\| A_n(\lambda) \|$ are uniformly bounded, disregarding a suitable λ-set of $\sigma(\lambda)$-measure 0.

(II) Every subsequence $n_1, n_2, \ldots$ of $1, 2, \ldots$ has a subsequence $m_1, m_2, \ldots$, such that $A(\lambda)$ is the strong limit of $A_{m_1}(\lambda), A_{m_2}(\lambda), \ldots$, disregarding a suitable λ-set of $\sigma(\lambda)$-measure 0 (that depends on $n_1, n_2, \ldots$).

If A is the strong limit of $A_1, A_2, \ldots$, then the $\| A_n \|$ are uniformly bounded (2, p. 382), $\| A_n \| \leq C$. Therefore $\| A_n(\lambda) \| \leq C$, except for a set of $\sigma(\lambda)$-measure 0, and so (I) is necessary. Thus we have to prove, that (II) is characteristic for the strong convergence of $A_1, A_2, \ldots$ to A if $\| A_n \| \leq C$, $\| A_n(\lambda) \| \leq C$ (exceptions as above) are assumed. In this situation strong convergence of A_n to A is equivalent to strong limit $A_n f = Af$ and strong convergence of $A_n(\lambda)$ to $A(\lambda)$ is equivalent to strong limit $A_n(\lambda)f(\lambda) = A(\lambda)f(\lambda)$, for all elements f or $f(\lambda)$ of a complete normalized orthogonal system in $\mathfrak{H}$ or $\mathfrak{H}^{(\lambda)}$ (2, pp. 385–386). Let the $\varphi_m(\lambda)$ be a measurable family, and put $\varphi_m = \int_- \varphi_m(\lambda)\sqrt{d\sigma(\lambda)}$; let ψ_m be a complete normalized orthogonal set in $\mathfrak{H}$, and put $\psi_m = \int_- \psi_m(\lambda)\sqrt{d\sigma(\lambda)}$. Let the $f_p = \int_- f_p(\lambda)\sqrt{d\sigma(\lambda)}$ be a sequence which contains these φ_m and ψ_m. Then we can replace strong limit $A_n = A$

[10] $A = X^*X$ is clearly definite: $(Af, f) = (X^*Xf, f) = (Xf, Xf) \geq 0$. If A is definite we can form the operator function $X = \sqrt{A}$, then $X = X^*$, $A = X^2$, so $A = X^*X$.

by strong limit $A_n f_p = A f_p$ for all $p = 1, 2, \ldots$; and strong limit $A_n(\lambda) = A(\lambda)$ by strong limit $A_n(\lambda) f_p(\lambda) = A(\lambda) f_p(\lambda)$ for all $p = 1, 2, \ldots$. So we can make this substitution both in the statement of our theorem and in (II). (But we continue assuming $|A_n| \leq C$, $|A_n(\lambda)| \leq C$.) Observe, that the sequence $m_1, m_2, \ldots$ (and the set of exceptions) in (II) may even depend on f_p: we can always get rid of this dependence, and obtain a new sequence $m_1, m_2, \ldots$ which works for all f_p, $p = 1, 2, \ldots$, simultaneously, by applying the diagonal process. Now we will prove the equivalence in question for each f_p separately, in other words: If $|A_n| \leq C$, $|A_n(\lambda)| \leq C$, then Af is the strong limit of $A_1 f, A_2 f, \ldots$, if and only if every subsequence $n_1, n_2, \ldots$ of $1, 2, \ldots$ has a subsequence $m_1, m_2, \ldots$, such that $A(\lambda) f(\lambda)$ is the strong limit of $A_{m_1}(\lambda) f(\lambda), A_{m_2}(\lambda) f(\lambda), \ldots$ ($f = \int_- f(\lambda)\sqrt{d\sigma(\lambda)}$), disregarding a suitable λ-set of $\sigma(\lambda)$-measure 0 (depending on $n_1, n_2, \ldots$ and f).

Observe that

$$Af = \int_- A(\lambda) f(\lambda)\sqrt{d\sigma(\lambda)}, \qquad A_n f = \int_- A_n(\lambda) f(\lambda)\sqrt{d\sigma(\lambda)}.$$

Application of Lemma 2 in §5, to the $A_{m_1} f, A_{m_2} f, \ldots$ and Af (for $g_1, g_2, \ldots$ and $\lim_{n\to\infty} g_n$) shows, that our condition is necessary. In order to prove its sufficiency, assume that it is fulfilled. We have then

$$\lim_{r\to\infty} \|A_{m_r}(\lambda) f(\lambda) - A(\lambda) f(\lambda)\|^2 = 0,$$

except for a λ-set of $\sigma(\lambda)$-measure 0, and

$$\|A_{m_r}(\lambda) f(\lambda) - A(\lambda) f(\lambda)\|^2 \leq (2C\|f(\lambda)\|)^2,$$

with the same exception. As

$$\int (2C\|f(\lambda)\|)^2 d\sigma(\lambda) = 4C^2 \int \|f(\lambda)\|^2 d\sigma(\lambda) = 4C^2\|f\|^2$$

is finite, this implies

$$\lim_{r\to\infty} \int \|A_{m_r}(\lambda) f(\lambda) - A(\lambda) f(\lambda)\|^2 d\sigma(\lambda) = 0, \qquad \lim_{r\to\infty} \|A_{m_r} f - Af\|^2 = 0.$$

Now if Af were not the strong limit of $A_1 f, A_2 f, \ldots$, there would exist an $\epsilon > 0$, such that $\|A_n f - Af\|^2 \geq \epsilon$ occurs for infinitely many n's. Then a subsequence $n_1, n_2, \ldots$ of $1, 2, \ldots$ with $\|A_{n_r} f - Af\|^2 \geq \epsilon$ would exist, and so there could not be

$\lim_{r\to\infty}\|A_{m_r}f - Af\|^2 = 0$ for any subsequence $m_1, m_2, \ldots$ of $n_1, n_2, \ldots$. Thus the proof is complete.

Replacing (II) by the stronger condition

(II′) $A(\lambda)$ is the strong limit of $A_1(\lambda), A_2(\lambda), \ldots$, disregarding a suitable λ-set of $\sigma(\lambda)$-measure 0,

makes our conditions sufficient (but not necessary).

We are now in a position to prove:

If $A \sim \sum A(\lambda)$ and $\alpha(A)$ is an operator-function of A (in the sense of (4, p. 204), or (11, p. 241)), then $\alpha(A) \sim \sum \alpha(A(\lambda))$.

This is obvious for $\alpha(x) = 1$ and $\alpha(x) = x$. Our rules of computation concerning $A, A+B, AB$ extend it to all polynomials $\alpha(x)$; the above convergence criterium (preferably in its sufficient form (I), (II′)) to all continuous functions, and then to all Baire-functions $\alpha(x)$. For a general $\alpha(x)$ apply (11, p. 231): Two Baire-function $\alpha_1(x), \alpha_2(x)$ can be found, for which $\alpha_1(x) \le \alpha(x) \le \alpha_2(x)$, and which differ from each other (and thus from $\alpha(x)$) only on a set, which, using the terminology of (11, p. 227), is a "null set with respect to A". Thus $\alpha_1(A) = \alpha_2(A) = \alpha(A)$. Besides $\alpha_1(A) \sim \sum \alpha_1(A(\lambda))$, $\alpha_2(A) \sim \sum \alpha_2(A(\lambda))$, therefore both $\alpha(A) \sim \sum \alpha_1(A(\lambda))$ and $\alpha(A) \sim \sum \alpha_2(A(\lambda))$. Thus $\alpha_1(A(\lambda)) = \alpha_2(A(\lambda))$, except for a set of $\sigma(\lambda)$-measure 0. For these λ's therefore $\alpha_1(x), \alpha_2(x)$ differ from each other only on a "null set with respect to $A(\lambda)$" (cf. above). As $\alpha_1(x) \le \alpha(x) \le \alpha_2(x)$, this means that $\alpha(A(\lambda))$ too can be formed, and that $\alpha_1(A(\lambda)) = \alpha(A(\lambda))$. Thus $\alpha(A) \sim \sum \alpha(A(\lambda))$, completing the proof.

The behavior of the decomposition $A \sim \sum A(\lambda)$ under equivalence is this:

If $A \sim \sum A(\lambda)$, and the representation of $\mathfrak{H}$ as a generalized direct sum is replaced by an equivalent one, with the mapping $\lambda \rightleftarrows m(\lambda)$, then the decomposition of A in this new representation will be $A \sim \sum B(\lambda)$, where $B(\lambda)$ is defined as follows: $B(m(\lambda))$ is the operator which corresponds to $A(\lambda)$ under the isomorphism $g^{(\lambda)}$ of $\mathfrak{H}_1^{(\lambda)}$ with $\mathfrak{H}_2^{(m(\lambda))}$, that is $B(m(\lambda)) = g^{(\lambda)}A(\lambda)(g^{(\lambda)})^{-1}$.

One verifies this statement immediately, by substituting it into condition (3) of Definition 3. Observe that the factor of the equivalence has no effect on the $B(\lambda)$.

PART III: DECOMPOSITION OF RINGS

15 $\sigma(\lambda)$-measurable dependence of λ: for $f(\lambda)$, $A(\lambda)$, $M(\lambda)$. Discussion of $f(\lambda)$, $A(\lambda)$

In what follows we will have to consider quantities of various sorts, which depend on the parameter λ of a generalized direct sum. It will be of importance to have a notion of measurability for such a dependence. It is given by the definition which follows:

Definition 5. *Let $\mathfrak{H}$ be the generalized direct sum of the $\mathfrak{H}^{(\lambda)}$ with the weight-function $\sigma(\lambda)$. We will consider an element $f(\lambda)$ of $\mathfrak{H}^{(\lambda)}$ or a linear bounded operator $A(\lambda)$ in $\mathfrak{H}^{(\lambda)}$, or a ring (containing 1) $M(\lambda)$ in $\mathfrak{H}^{(\lambda)}$, depending on the parameter λ. The dependence on λ is $\sigma(\lambda)$-measurable if the following conditions are fulfilled:*

For $f(\lambda)$: *For every $\sigma(\lambda)$-summable $g(\lambda)$ the (numerical) function $(f(\lambda), g(\lambda))$ is $\sigma(\lambda)$-measurable.*

For $A(\lambda)$: *For every $\sigma(\lambda)$-measurable $f(\lambda)$ the (element of $\mathfrak{H}^{(\lambda)}$-) function $A(\lambda)f(\lambda)$ is $\sigma(\lambda)$-measurable.*

For $M(\lambda)$: *A (finite or infinite) sequence of $\sigma(\lambda)$-measurable (operator in $\mathfrak{H}^{(\lambda)}$-) functions $A_n(\lambda)$, $n = 1, 2, \ldots$, exists, so that $M(\lambda)$ is the ring generated by the $A_n(\lambda)$, $n = 1, 2, \ldots$ (for that λ, in $\mathfrak{H}^{(\lambda)}$).*

Remark 1. *Ad $f(\lambda)$:* Our definition is clearly the condition $(\mathrm{i})_1$ in the characterization (i) of the $\sigma(\lambda)$-summability in Definition 1. If $\varphi_m(\lambda)$ is a measurable family, it suffices to postulate the $\sigma(\lambda)$-measurability of all $(f(\lambda), \varphi_m(\lambda))$, $m = 1, 2, \ldots$. This is clearly necessary, and it is sufficient too, because if $g(\lambda)$ is $\sigma(\lambda)$-summable, then

$$(f(\lambda), g(\lambda)) = \sum_m (f(\lambda), \varphi_m(\lambda))\overline{(g(\lambda), \varphi_m(\lambda))}$$

will also be $\sigma(\lambda)$-measurable.

The above formula shows, also, that if $f(\lambda), g(\lambda)$ are $\sigma(\lambda)$-measurable, then the (numerical) function $(f(\lambda), g(\lambda))$ has that property too, and with it $\|f(\lambda)\| = \sqrt{(f(\lambda), f(\lambda))}$.

It is obvious, that if $f(\lambda), g(\lambda)$ are $\sigma(\lambda)$-measurable, then $f(\lambda) + g(\lambda)$ has that property too; and if the (numerical) function $\alpha(\lambda)$ is $\sigma(\lambda)$-measurable, then $\alpha(\lambda)f(\lambda)$ has that property too.

Remark 2. *Ad $A(\lambda)$:* Considering Remark 1, we may as well require the $\sigma(\lambda)$-measurability of the (numerical) function $(A(\lambda)f(\lambda), g(\lambda))$ for all $\sigma(\lambda)$-measurable $f(\lambda), g(\lambda)$. If $\varphi_m(\lambda)$ is a measurable family, it suffices to postulate the $\sigma(\lambda)$-measurability of all $(A(\lambda)\varphi_m(\lambda), \varphi_n(\lambda))$, $m, n = 1, 2, \ldots$. This is clearly necessary, and it is sufficient too, because if $f(\lambda), g(\lambda)$ are $\sigma(\lambda)$-measurable, then

$$(A(\lambda)f(\lambda), g(\lambda)) = \sum_n \left[\sum_m (A(\lambda)\varphi_m(\lambda), \varphi_n(\lambda))(f(\lambda), \varphi_m(\lambda))\overline{(g(\lambda), \varphi_n(\lambda))} \right]$$

will also be $\sigma(\lambda)$-measurable. This form of our definition is clearly the condition (i) in the criterium for the existence of an $A \sim \sum A(\lambda)$ in Theorem V.

It is obvious that if $A(\lambda), B(\lambda)$ are $\sigma(\lambda)$-measurable, then $A(\lambda) + B(\lambda)$ has that property too; and if the (numerical) function $\alpha(\lambda)$ is $\sigma(\lambda)$-measurable, then $\alpha(\lambda)A(\lambda)$ has that property too. As

$$(A^*(\lambda)f(\lambda), g(\lambda)) = \overline{(A(\lambda)g(\lambda), f(\lambda))},$$

the $\sigma(\lambda)$-measurability of $A(\lambda)$ implies that one of $A^*(\lambda)$. If $A(\lambda), B(\lambda)$ are $\sigma(\lambda)$-measurable, then $A(\lambda)B(\lambda)$ has that property too: Let $f(\lambda), g(\lambda)$ be $\sigma(\lambda)$-measurable, then $B(\lambda)f(\lambda)$ is $\sigma(\lambda)$-measurable, and hence $A(\lambda)B(\lambda)f(\lambda)$ is $\sigma(\lambda)$-measurable too.

If $A_1(\lambda), A_2(\lambda), \ldots$ converge strongly (or even only weakly) to $A(\lambda)$, then their $\sigma(\lambda)$-measurability implies that one of $A(\lambda)$, because $\lim_{n\to\infty}(A_n(\lambda)f(\lambda), g(\lambda)) = (A(\lambda)f(\lambda), g(\lambda))$. All these facts permit us to conclude in the usual way that if $A(\lambda)$ is $\sigma(\lambda)$-measurable, then any operator-function $\varphi(A(\lambda))$ of it ($A(\lambda)$ Hermitian and $\varphi(x)$ a Baire-function) is $\sigma(\lambda)$-measurable too.

If $A(\lambda)$ is $\sigma(\lambda)$-measurable, so is the (numerical) function $|\!|A(\lambda)|\!|$: We know from the considerations preceding Definition 4, that

$$|\!|A(\lambda)|\!| = \sqrt{\operatorname*{l.u.b.}_{\nu=1,2,\ldots} \sum_m \left| \sum_n a_{mn}(\lambda) x_n^{(\nu)} \right|^2},$$

where the $x_1^{(\nu)}, x_2^{(\nu)}, \ldots$, $\nu = 1, 2, \ldots$, are the sequences described there, and $a_{mn}(\lambda) = (A(\lambda)\varphi_n(\lambda), \varphi_m(\lambda))$. All these expressions are $\sigma(\lambda)$-measurable.

Our next task is to discuss the consequences of $\sigma(\lambda)$-measurability for rings (containing 1) $M(\lambda)$, in $\mathfrak{H}^{(\lambda)}$. Here however a more elaborate argument is necessary.

16 The operators of norm at most 1 in the weak topology form a complete and separable metric space. A choice lemma for analytic sets

We need a lemma about the sets of M. Suslin and W. Lusin (cf. the monograph (14), and the exposition in (15, pp. 90–93, 179–196, 208–212, 276–277)), which we will call "analytic sets" following the prevalent usage. But first we make some remarks of a general topological nature.

Observe that if S is the set of all linear operators A of a unitary space $\mathfrak{H}$ with $\|A\| \leq 1$, then there exists a metric in S which is equivalent to its weak topology. If $\varphi_1, \varphi_2, \ldots$ is a complete normalized orthogonal set in $\mathfrak{H}$ define

$$(\mathrm{Dist}(A,B))^2 = \sum_{m,n=1}^{\infty} \frac{1}{2^{m+n}} |(A\varphi_m, \varphi_n) - (B\varphi_m, \varphi_n)|^2.$$

This is clearly a metric, and as the weak topology in S is separable (cf. (2, p. 386, footnote 38)), we need only to prove, in order to establish its equivalence to the weak topology, that $\lim_{p\to\infty} \mathrm{Dist}(A_p, A) = 0$ is equivalent to weak $\lim_{p\to\infty} A_p = A$. The latter means $\lim_{p\to\infty}(A_p\varphi_m, \varphi_n) = (A\varphi_m, \varphi_n)$ for all $m, n = 1, 2, \ldots$ (cf. (2, pp. 385–386)) and this is equivalent to the former, as all series for $(\mathrm{Dist}(A,B))^2$ converge uniformly:

$$\frac{1}{2^{m+n}} |(A\varphi_m, \varphi_n) - (B\varphi_m, \varphi_n)|^2 \leq \frac{4}{2^{m+n}}, \qquad \sum_{m,n=1}^{\infty} \frac{4}{2^{m+n}} = 4.$$

This metric space is separable, as S in the weak topology is separable (cf. loc. cit. above), and it is complete too. If $\lim_{p,q\to\infty} \mathrm{Dist}(A_p, A_q) = 0$, then $\lim_{p,q\to\infty} |(A_p\varphi_m, \varphi_n) - (A_q\varphi_m, \varphi_n)| = 0$, and so $\lim_{p\to\infty}(A_p\varphi_m, \varphi_n) = a_{mn}$ exists.

If $\sum_{m=1}^{\infty} |x_m|^2 \leq 1$ then

$$\begin{aligned}
\sum_{n=1}^{M} \left| \sum_{m=1}^{N} a_{mn} x_m \right|^2 &= \lim_{p\to\infty} \sum_{n=1}^{M} \left| \sum_{m=1}^{N} (A_p \varphi_m, \varphi_n) x_m \right|^2 \\
&\leq \limsup_{p\to\infty} \sum_{n=1}^{\infty} \left| \sum_{m=1}^{N} (A_p \varphi_m, \varphi_n) x_m \right|^2 \\
&= \limsup_{p\to\infty} \sum_{n=1}^{\infty} \left| \left(A_p \left\{ \sum_{m=1}^{N} x_m \varphi_m \right\}, \varphi_n \right) \right|^2 \\
&= \limsup_{p\to\infty} \left\| A_p \left\{ \sum_{m=1}^{N} x_m \varphi_m \right\} \right\|^2 \\
&\leq \limsup_{p\to\infty} \left\| \sum_{m=1}^{N} x_m \varphi_m \right\|^2 \\
&= \limsup_{p\to\infty} \sum_{m=1}^{N} |x_m|^2 \leq 1,
\end{aligned}$$

and $N \to \infty$ and then $M \to \infty$ give $\sum_{n=1}^{\infty} | \sum_{m=1}^{\infty} a_{mn} x_m|^2 \leq 1$.

Thus

$$A \left(\sum_{m=1}^{\infty} x_m \varphi_m \right) = \sum_{m=1}^{\infty} y_m \varphi_m, \qquad y_n = \sum_{m=1}^{\infty} a_{mn} x_m$$

defines a linear operator A with $|A| \leq 1$, so $A \in S$. Now as $\lim_{p\to\infty}(A_p \varphi_m, \varphi_n) = (A\varphi_m, \varphi_n)$, therefore weak $\lim_{p\to\infty} A_p = A$, $\lim_{p\to\infty} \mathrm{Dist}(A_p, A) = 0$. So we see that S in the weak topology may be looked at as a complete and separable metric space.

Observe finally that if S_1, S_2 are two complete and separable metric spaces, then their direct product $S_1 \times S_2$ is another one. [11] This generalizes by repetition to any finite number of factors.

Our lemma on analytic sets is this:

[11] The direct product $S_1 \times S_2$ is the set of all (ordered) pairs $(a^{(1)}, a^{(2)})$, $a^{(1)} \in S_1$, $a^{(2)} \in S_2$, the neighborhoods of $(a_0^{(1)}, a_0^{(2)})$ being the sets of all $(a^{(1)}, a^{(2)})$ with $a^{(1)} \in O_1$, $a^{(2)} \in O_2$ where O_1 is a neighborhood of $a_0^{(1)}$, and O_2 one of $a_0^{(2)}$. If a metric $\mathrm{Dist}_1(a^{(1)}, b^{(1)})$ exists in S_1 and another one $\mathrm{Dist}_2(a^{(2)}, b^{(2)})$ in S_2, then the topology in the direct product $S_1 \times S_2$ can be described by the metric $\mathrm{Dist}((a^{(1)}, a^{(2)}), (b^{(1)}, b^{(2)})) = \max(\mathrm{Dist}_1(a^{(1)}, b^{(1)}), \mathrm{Dist}_2(a^{(2)}, b^{(2)}))$ in $S_1 \times S_2$.

Lemma 5. *Let S be a complete and separable metric space, A an analytic set in S (cf. (15, pp. 91, 211)), and $F(a)$ a function which is defined and continuous for all $a \in A$ and has (real) numerical values. Let K be the set of all values, which $F(a)$, $a \in S$, assumes. Let $\sigma(\lambda)$ be an arbitrary N-function.*

Under these assumptions we have:

(i) K is a $\sigma(\lambda)$-measurable set.

(ii) There exists a function $f(\lambda)$, which is defined for all $\lambda \in K$ and assumes values in S, such that:

(ii)$_1$. If O is an open subset of S, then the set of all $\lambda \in K$ with $f(\lambda) \in O$ is $\sigma(\lambda)$-measurable.

(ii)$_2$. $F(f(\lambda)) = \lambda$ for every $\lambda \in K$.

Proof. Denote the points of discontinuity of $\sigma(\lambda)$ by $\lambda_1, \lambda_2, \ldots$ (their number is finite or countably infinite), and the "jumps" of $\sigma(\lambda)$ in these points by resp. $a_n = \sigma(\lambda_n) - \sigma(\lambda_n - 0)$. Form $\tau_1(\lambda) = \sum_{\lambda_n \leq \lambda} a_n$. We know from the discussion at the beginning of §4, that every set is $\tau_1(\lambda)$-measurable. As $\sigma(\lambda) - \tau_1(\lambda)$ is monotonous (if $\lambda \leq \mu$, then

$$[\sigma(\mu) - \tau_1(\mu)] - [\sigma(\lambda) - \tau_1(\lambda)] = \sigma(\mu) - \sigma(\lambda) - \sum_{\lambda < \lambda_n \leq \mu} a_n \geq 0),$$

therefore we see that $\sigma(\lambda) - \tau_1(\lambda)$-measurability implies $(\sigma(\lambda) - \tau_1(\lambda)) + \tau_1(\lambda) = \sigma(\lambda)$- measurability. Besides $\sigma(\lambda) - \tau_1(\lambda)$ is obviously continuous. Let now $\tau_2(\lambda)$ be a continuous and nowhere constant N-function (for instance: $\tau_2(\lambda) = \lambda/(1 + |\lambda|)$). Then $\sigma(\lambda) - \tau_1(\lambda) + \tau_2(\lambda) = \tau(\lambda)$ is a continuous and nowhere constant N-function too. It is clear from (γ) in §2 that $\sigma(\lambda) - \tau_1(\lambda)$ is $\sigma(\lambda) - \tau_1(\lambda) + \tau_2(\lambda) = \tau(\lambda)$-totally continuous. Therefore $\tau(\lambda)$-measurability implies $\sigma(\lambda) - \tau_1(\lambda)$-measurability, and thus $\sigma(\lambda)$-measurability too. Thus we may prove our lemma for $\tau(\lambda)$ instead of $\sigma(\lambda)$.

Since $\tau(\lambda)$ is a continuous and nowhere constant N-function, therefore $\lambda \rightleftarrows \mu = \tau(\lambda)$ is a topological mapping of the infinite interval $-\infty < \lambda < +\infty$ on the finite interval $a < \mu < b$, where $a = \tau(-\infty)$, $b = \tau(+\infty)$. Denote the inverse mapping by $\mu \rightleftarrows \lambda = \tau^{-1}(\mu)$.

Let us now replace $F(a)$, $f(\lambda)$ by $\tau(F(a))$, $f(\tau^{-1}(\mu))$, and then write again λ for μ. This leaves the structure of our assertions unaffected, but there are these changes regarding λ. The domain of λ is now $a < \lambda < b$ and not $-\infty < \lambda < +\infty$. In this new domain we have to use the notion of common (Lebesgue)-measurability, and not that one of $\tau(\lambda)$-measurability. (Our mapping $\lambda \rightleftarrows \tau(\lambda)$ transformed the latter into the former.)

We now introduce the "0-space of Baire", S_0, which is the set of all sequences $\alpha = [n_1, n_2, \ldots]$, $n_1, n_2, \ldots = 1, 2, \ldots$. It is metrized by the definition

$$\operatorname{Dist}([m_1, m_2, \ldots], [n_1, n_2, \ldots]) = \begin{cases} 0, & m_\nu = n_\nu \text{ for all } \nu = 1, 2, \ldots, \\ \dfrac{1}{\nu_0}, & \nu_0 \text{ is the smallest } \nu = 1, 2, \ldots \text{ for which } m_\nu \neq n_\nu. \end{cases}$$

Thus S_0 is a complete and separable metric space. (The countable set of all those $\alpha = [n_1, n_2, \ldots]$, for which $n_\nu \neq 1$ occurs for a finite number of ν's only, is obviously everywhere dense in S_0.)

On the other hand the definition

$$[m_1, m_2, \ldots] < [n_1, n_2, \ldots] \quad \begin{cases} \text{if a } \nu_0 \text{ exists, such that } m_{\nu_0} < n_{\nu_0}, \\ \text{and } m_\nu = n_\nu \text{ for } \nu < \nu_0 \end{cases}$$

establishes an "ordering" of S_0. If α is fixed, then the set of all $\beta < \alpha$ is open. If $\beta < \alpha$, and ν_0 is defined as above, then we have $\gamma < \alpha$ in the entire sphere $\operatorname{Dist}(\gamma, \beta) < 1/\nu_0$. Every sphere in S_0 is a set of the form $\alpha_1 \leq \beta < \alpha_2$ (with fixed α_1, α_2). If its center is $[m_1, m_2, \ldots]$ and its radius $\epsilon > 0$, then $[n_1, n_2, \ldots] = \beta$ belongs to it, if and only if, $\operatorname{Dist}([m_1, m_2, \ldots], [n_1, n_2, \ldots]) < \epsilon$, that is $m_\nu = n_\nu$ for $\nu = 1, \ldots, \nu_1$, where ν_1 is the greatest integer $\leq 1/\epsilon$. This may be written in the form $\alpha_1 \leq \beta < \alpha_2$ if we put

$$\alpha_1 = [m_1, \ldots, m_{\nu_1 - 1}, m_{\nu_1}, 1, 1, \ldots], \quad \alpha_2 = [m_1, \ldots, m_{\nu_1 - 1}, m_{\nu_1} + 1, 1, 1, \ldots].$$

Apply now (15, p. 211). As A is an analytic set in a complete and separable metric space, therefore there exists a function $\varphi(\alpha)$, which is defined and continuous for every $\alpha \in S_0$, and which maps S_0 on A. (The mapping may be many-to-one.) Then K is clearly the set of all $F(\varphi(\alpha))$, $\alpha \in S_0$. As $F(\varphi(\alpha))$ also is defined and continuous for every $\alpha \in S_0$, and the real numbers also form a complete and separable metric space, we can therefore now apply (15, p. 211), in the opposite sense: K is an analytic set, and therefore measurable by (14, pp. 151–152).

Consider now a $\lambda \in K$. Denote the set of all $\alpha \in S_0$ with $F(\varphi(\alpha)) = \lambda$ by $T(\lambda)$. As $\lambda \in K$, $T(\lambda)$ is not empty, and as $F(\varphi(\alpha))$ is continuous, $T(\lambda)$ is closed.

Denote the smallest n_1 with $[n_1, n_2, n_3, \ldots] \in T(\lambda)$ (for any $n_2, n_3, \ldots$) by n_1^0, the smallest n_2 with $[n_1^0, n_2, n_3, \ldots] \in T(\lambda)$ (for any $n_3, n_4, \ldots$) by n_2^0, the smallest n_3 with $[n_1^0, n_2^0, n_3, \ldots] \in T(\lambda)$ (for any $n_4, n_5, \ldots$) by n_3^0, etc. Put $\alpha(\lambda) = [n_1^0, n_2^0, \ldots]$. It is clearly a condensation point of $T(\lambda)$, and therefore $\alpha(\lambda) \in T(\lambda)$; besides its definition guarantees $\alpha(\lambda) \leq \beta$ for every $\beta \in T(\lambda)$. In other words: $T(\lambda)$ has a first element, and this is $\alpha(\lambda)$.

Choose now a fixed α_0 and consider the set $M(\alpha_0)$ of all λ's with $\alpha(\lambda) < \alpha_0$. $\lambda \in M(\alpha_0)$, that is $\alpha(\lambda) < \alpha_0$, is clearly equivalent to the existence of a $\beta < \alpha_0$ with $\beta \in T(\lambda)$, that is of a $\beta < \alpha_0$ with $F(\varphi(\beta)) = \lambda$. In other words: $M(\alpha_0)$ is the $\beta \rightleftarrows F(\varphi(\beta))$-image of the open set $\beta < \alpha_0$ in S_0. As $F(\varphi(\beta))$ is continuous, we can infer by (15, p. 209, Theorem III), that $M(\alpha_0)$ is an analytic set, and therefore measurable by (14).

If T is any subset of S_0, denote the set of all λ's with $\alpha(\lambda) \in T$ by $N(T)$. If T is a sphere, then it has the form $\alpha_1 \leq \beta < \alpha_2$, therefore $N(T) = M(\alpha_2) - M(\alpha_1)$. Our above result proves therefore, that $N(T)$ is measurable. If T is merely an open set, then, as S_0 is separable, we can write T as the sum of a sequence of spheres: $T = T_1 + T_2 + \cdots$. Now $N(T) = N(T_1) + N(T_2) + \cdots$, and as $N(T_1), N(T_2), \ldots$ are all measurable, so is $N(T)$.

Let finally O be an open subset of S, and denote by $T(O)$ the set of all $\alpha \in S_0$ with $\varphi(\alpha) \in O$. As O is open and $\varphi(\alpha)$ is continuous, therefore $T(O)$ is open too. Thus $N(T(O))$ is measurable. $\lambda \in N(T(O))$ means $\alpha(\lambda) \in T(O)$, that is $\varphi(\alpha(\lambda)) \in O$.

Observe finally, that as $\alpha(\lambda) \in T(\lambda)$ (assuming $\lambda \in K$), therefore $F(\varphi(\alpha(\lambda))) = \lambda$.

The measurability of K being already established, we have proved statement (i) of our lemma. And in order to prove statement (ii), it suffices to define $f(\lambda) = \varphi(\alpha(\lambda))$. Thus the proof is completed. □

17 $\sigma(\lambda)$-measurable dependence and the operations on rings; measurable ring relations and measurable separating projections

We now return to the discussion of the $\sigma(\lambda)$-measurable rings $M(\lambda)$:

Lemma 6. *Let the $M(\lambda)$ be rings containing 1, depending $\sigma(\lambda)$-measurably on λ. Let $k(\lambda)$, Δ_p, $p = 1, 2, \ldots, \infty$, be as in Definition 2, let $\varphi_m(\lambda)$ be a measurable family. Choose a $p = 1, 2, \ldots, \infty$ and a p-dimensional unitary space $\mathfrak{H}'_p$, and in it a complete normalized orthogonal system $\psi'_1, \psi'_2, \ldots$. For each $\lambda \in \Delta_p$ denote the isomorphism of $\mathfrak{H}^{(\lambda)}$ and $\mathfrak{H}'_p$,* *in which $\varphi_m(\lambda)$ corresponds to ψ'_m, by $g_p^{(\lambda)}$.*

Then there exists a set $N_p \subset \Delta_p$ of $\sigma(\lambda)$-measure 0, such that $\Delta_p - N_p$ is a Borel set and that the set of all pairs (λ, A) with $\lambda \in \Delta_p - N_p$, $(g_p^{(\lambda)})^{-1} A g_p^{(\lambda)} \in M(\lambda)'$ is a Borel-set in the direct product space $r \times S_p$. (r is the set of all real numbers λ in their usual topology, S_p the set of linear operators A in $\mathfrak{H}'_p$ with $|A| \leq 1$ in their weak topology, cf. the beginning of §16.)

Proof. Choose the $A_i(\lambda)$, $i = 1, 2, \ldots$, in the sense of Definition 5: $A_i(\lambda)$ is $\sigma(\lambda)$-measurable in λ, and the $A_i(\lambda)$, $i = 1, 2, \ldots$, determine $M(\lambda)$. Each function $(A_i(\lambda)\varphi_m(\lambda), \varphi_n(\lambda))$, $i, m, n = 1, 2, \ldots$ (of course $m, n \leq p$) is $\sigma(\lambda)$-measurable. Therefore there exists a set $N_1^{(i,m,n,p)}$ of $\sigma(\lambda)$-measure 0, such that $(A_i(\lambda)\varphi_m(\lambda), \varphi_n(\lambda))$ agrees with a Baire-function for all $\lambda \notin N_1^{(i,m,n,p)}$ (cf. (4, p. 171)). $N_2^{(p)} = \sum_{i,m,n} N_1^{(i,m,n,p)}$ has clearly the same properties, and is even independent of i, m, n. As Δ_p and $N_2^{(p)}$ are both $\sigma(\lambda)$-measurable, $\Delta_p - \Delta_p N_2^{(p)}$ is $\sigma(\lambda)$-measurable too, and so a Borel-set $\Delta'_p \subset \Delta_p - \Delta_p N_2^{(p)}$ exists, for which $(\Delta_p - \Delta_p N_2^{(p)}) - \Delta'_p$ has the $\sigma(\lambda)$-measure 0. Thus $N_p = \Delta_p - \Delta'_p \subset ((\Delta_p - \Delta_p N_2^{(p)}) - \Delta'_p) + N_2^{(p)}$ also has the $\sigma(\lambda)$-measure 0, and we have $\Delta'_p = \Delta_p - N_p$. As Δ'_p is a Borel-set, the function

$$1_{\Delta'_p}(\lambda) = \begin{cases} 1, & \lambda \in \Delta'_p, \\ 0, & \lambda \notin \Delta'_p \end{cases}$$

is obviously a Baire-function of λ.

Assume $\lambda \in r$, $A \in S_p$, we wish to find an analytic expression for $\lambda \in \Delta'_p$, $(g_p^{(\lambda)})^{-1} A g_p^{(\lambda)} \in M(\lambda)'$. The first requirement means

$$1_{\Delta'_p}(\lambda) - 1 = 0. \tag{17.1}$$

The second means that $(g_p^{(\lambda)})^{-1} A g_p^{(\lambda)}$ commute with all elements of $M(\lambda)$. In order that this be the case, it suffices for $(g_p^{(\lambda)})^{-1} A g_p^{(\lambda)}$ to commute with the $A_i(\lambda)$, $i = 1, 2, \ldots$. In other words:

$$(g_p^{(\lambda)})^{-1} A g_p^{(\lambda)} A_i(\lambda) = A_i(\lambda) (g_p^{(\lambda)})^{-1} A g_p^{(\lambda)}$$

for $i = 1, 2, \ldots$, or:

$$\big((g_p^{(\lambda)})^{-1} A g_p^{(\lambda)} A_i(\lambda)\varphi_m(\lambda), \varphi_n(\lambda)\big) = \big(A_i(\lambda)(g_p^{(\lambda)})^{-1} A g_p^{(\lambda)} \varphi_m(\lambda), \varphi_n(\lambda)\big)$$

for $i, m, n = 1, 2, \ldots$.

We have however:

$$\begin{aligned}
&\big((g_p^{(\lambda)})^{-1} A g_p^{(\lambda)} A_i(\lambda)\varphi_m(\lambda), \varphi_n(\lambda)\big) \\
&\quad = (A_i(\lambda)\varphi_m(\lambda), (g_p^{(\lambda)})^{-1} A^* g_p^{(\lambda)} \varphi_n(\lambda)) \\
&\quad = \sum_{l=1}^{p} (A_i(\lambda)\varphi_m(\lambda), \varphi_l(\lambda))(\varphi_l(\lambda), (g_p^{(\lambda)})^{-1} A^* g_p^{(\lambda)} \varphi_n(\lambda)) \\
&\quad = \sum_{l=1}^{p} (A_i(\lambda)\varphi_m(\lambda), \varphi_l(\lambda))(A g_p^{(\lambda)} \varphi_l(\lambda), g_p^{(\lambda)} \varphi_n(\lambda)) \\
&\quad = \sum_{l=1}^{p} (A_i(\lambda)\varphi_m(\lambda), \varphi_l(\lambda))(A\psi'_l, \psi'_n),
\end{aligned}$$

and similarly:

$$\begin{aligned}
&\big(A_i(\lambda)(g_p^{(\lambda)})^{-1} A g_p^{(\lambda)} \varphi_m(\lambda), \varphi_n(\lambda)\big) \\
&\quad = \sum_{l=1}^{p} (A\psi'_m, \psi'_l)(A_i(\lambda)\varphi_l(\lambda), \varphi_n(\lambda)).
\end{aligned}$$

So we obtain the following conditions:

$$\begin{aligned}
&\sum_{l=1}^{p} (A_i(\lambda)\varphi_m(\lambda), \varphi_l(\lambda))(A\psi'_l, \psi'_n) \\
&\quad - \sum_{l=1}^{p} (A\psi'_m, \psi'_l)(A_i(\lambda)\varphi_l(\lambda), \varphi_n(\lambda)) = 0, \\
&\qquad \text{for } i, m, n = 1, 2, \ldots.
\end{aligned} \tag{17.2}$$

The left side of (17.1) is clearly a Baire-function in λ, and independent of A. In the left side of (17.2) the factors $(A_i(\lambda)\varphi_m(\lambda), \varphi_l(\lambda))$ are Baire-functions in λ, and independent of A; while the factors $(A\psi'_m, \psi'_l)$ are (weakly) continuous in A, and independent of λ. Thus all our equations have Baire-functions in λ, A as left sides. Therefore each of them has a Borel-set of pairs (λ, A) in $r \times S_p$ as its domain of validity (cf. (15, p. 260)). As they are countably infinite in number, the intersection of their domains of validity is a Borel-set too. But this is the set of all pairs (λ, A) with $\lambda \in \Delta'_p$, $(g_p^{(\lambda)})^{-1} A g_p^{(\lambda)} \in M(\lambda)'$.

Thus the statement of our lemma holds for our present choice of N_p. □

Lemma 7. *Let the $M(\lambda)$ be rings containing 1, depending $\sigma(\lambda)$-measurably on λ. Then the $M'(\lambda)$ too depend $\sigma(\lambda)$-measurably on λ.*

Proof. Introduce $k(\lambda)$, $p = 1, 2, \dots, \infty$, Δ_p, $\mathfrak{H}'_p$, $g_p^{(\lambda)}$, S_p, as in the previous lemma, and form the corresponding sets N_p. Let $O_{p1}, O_{p2}, \dots$ be an equivalent set of neighborhoods in the separable space S_p. Denote the set of all pairs (λ, A) with $\lambda \in r$, $A \in O_{pj}$ by O^*_{pj}. As O_{pj} is open, therefore O^*_{pj} is open too. Let M^*_p be the set of all pairs (λ, A) with $\lambda \in \Delta_p - N_p$, $(g_p^{(\lambda)})^{-1} A g_p^{(\lambda)} \in M(\lambda)'$; we know from Lemma 6, that M^*_p is a Borel-set. Thus $O^*_{pj} M^*_p$ is a Borel-set too.

As $O^*_{pj} M^*_p$ is a Borel-set in $r \times S_p$, it is *a fortiori* an analytic set. Therefore Lemma 5 applies to it: Define $F(\lambda, A) = \lambda$, which is obviously a continuous

function. We denote the set K and the function $f(\lambda)$, as defined there, by K_{pj}, $f_{pj}(\lambda)$ (because their dependence on p, j is now essential). Clearly $K_{pj} \subset \Delta_p - N_p$. $f_{pj}(\lambda)$ is a pair (μ, A), as $F(f_{pj}(\lambda)) = \lambda$, there is $\mu = \lambda$. Thus we can put: $f_{pj}(\lambda) = (\lambda, B^0_{pj}(\lambda))$.

Let U be an open set in the complex-number-plane. The set of all $A \in S_p$ with $(A\psi'_m, \psi'_n) \in U$ is obviously open, and thus the set of all $(\lambda, A) \in r \times S_p$ with $(A\psi'_m, \psi'_n) \in U$ is open too. Therefore the set of all those $\lambda \in K_{pj}$, for which $f_{pj}(\lambda) = (\lambda, B^0_{pj}(\lambda))$ belongs to this set, is $\sigma(\lambda)$-measurable by Lemma 5. This set is characterized by $(B^0_{pj}(\lambda)\psi'_m, \psi'_n) \in U$. As this is true for all open U's, it means that the function $(B^0_{pj}(\lambda)\psi'_m, \psi'_n)$ is $\sigma(\lambda)$-measurable. As

$$\begin{aligned}(B^0_{pj}(\lambda)\psi'_m, \psi'_n) &= \big(B^0_{pj}(\lambda) g_p^{(\lambda)} \varphi_m(\lambda), g_p^{(\lambda)} \varphi_n(\lambda)\big) \\ &= \big((g_p^{(\lambda)})^{-1} B^0_{pj}(\lambda) g_p^{(\lambda)} \varphi_m(\lambda), \varphi_n(\lambda)\big),\end{aligned}$$

we can apply Remark 2 after Definition 5: The operator $(g_p^{(\lambda)})^{-1} B^0_{pj}(\lambda) g_p^{(\lambda)}$ (in $\mathfrak{H}^{(\lambda)}$) is $\sigma(\lambda)$-measurable.

Now define

$$C_{pj}(\lambda) = \begin{cases} (g_p^{(\lambda)})^{-1} B^0_{pj}(\lambda) g_p^{(\lambda)}, & \lambda \in K_{pj}, \\ 0, & \lambda \notin K_{pj}. \end{cases}$$

as K_{pj} is $\sigma(\lambda)$-measurable, the operator $C_{pj}(\lambda)$ (in $\mathfrak{H}^{(\lambda)}$) is $\sigma(\lambda)$-measurable, too.

$\lambda \in K_{pj}$ means that an $a \in O^*_{pj} M^*_p$ with $F(a) = \lambda$ exists, that is:

$$a = (\lambda, A), \quad \lambda \in \Delta_p - N_p, \quad A \in O_{pj}, \quad (g_p^{(\lambda)})^{-1} A g_p^{(\lambda)} \in M(\lambda)',$$

in other words: $\lambda \in \Delta_p - N_p$, $A \in O_{pj}$, and $M(\lambda)'$ possesses elements in $(g_p^{(\lambda)})^{-1} O_{pj} g_p^{(\lambda)}$. If this is the case, then $B^0_{pj}(\lambda)$ is defined, and $C_{pj}(\lambda) = (g_p^{(\lambda)})^{-1} B^0_{pj}(\lambda) g_p^{(\lambda)}$. Thus $f(\lambda) = (\lambda, B^0_{pj}(\lambda)) = (\lambda, g_p^{(\lambda)} C_{pj}(\lambda)(g_p^{(\lambda)})^{-1}) \in O^*_{pj} M^*_p$, that is: $g_p^{(\lambda)} C_{pj}(\lambda)(g_p^{(\lambda)})^{-1} \in O_{pj}$, $C_{pj}(\lambda) \in M(\lambda)'$, in other words: $C_{pj}(\lambda)$ is an element of $M(\lambda)'$ in $(g_p^{(\lambda)})^{-1} O_{pj} g_p^{(\lambda)}$. If this is not the case, then $C_{pj}(\lambda) = 0$. Assuming $\lambda \in \Delta_p - N_p$ we have furthermore $C_{qj}(\lambda) = 0$ for all $q \neq p$, as there cannot be a $\lambda \in \Delta_q - N_q$.

Assume now $\lambda \in \sum_{p=1,2,\ldots,\infty}(\Delta_p - N_p) = r - \sum_{p=1,2,\ldots,\infty} N_p$. Then we have $C_{qj}(\lambda) = 0$, except if $q = p$ and $M(\lambda)'$ possesses elements in $(g_p^{(\lambda)})^{-1} O_{pj} g_p^{(\lambda)}$, in which case $C_{qj}(\lambda)$ itself is such an element. Remembering that the O_{pj}, $j = 1, 2, \ldots$, form an equivalent set of neighborhoods for the A with $\|A\| \leq 1$ in $\mathfrak{H}'_p$, and thus the $(g_p^{(\lambda)})^{-1} O_{pj} g_p^{(\lambda)}$, $j = 1, 2, \ldots$, do the same in $\mathfrak{H}^{(\lambda)}$, this proves: The $C_{qj}(\lambda)$, $q, j = 1, 2, \ldots$, are everywhere dense in that part of $M(\lambda)'$ which lies in $\|A\| \leq 1$.

The ring generated by the $C_{qj}(\lambda)$ is contained in $M(\lambda)'$, because the $C_{qj}(\lambda)$ are. But as it contains all (weak) condensation points of the $C_{qj}(\lambda)$, it must contain all $A \in M(\lambda)'$ with $\|A\| \leq 1$. If $A \in M(\lambda)'$, $\|A\| > 1$, then this ring will contain $1/\|A\| \cdot A$ and thus A too. Thus it coincides with $M(\lambda)'$. In other words: The $C_{qj}(\lambda)$, $q, j = 1, 2, \ldots$, generate the ring $M(\lambda)'$.

Write $N = \sum_{p=1,2,\ldots,\infty} N_p$, and put the $C_{qj}(\lambda)$, $q, j = 1, 2, \ldots$, in some way in a simple sequence: $D_i(\lambda)$, $i = 1, 2, \ldots$. Then we have proved: N has the $\sigma(\lambda)$-measure 0, each $D_i(\lambda)$ (in $\mathfrak{H}^{(\lambda)}$) is $\sigma(\lambda)$-measurable, if $\lambda \notin N$ then the $D_i(\lambda)$,

$i = 1, 2, \ldots$, generate the ring $M(\lambda)'$. Now change the $D_i(\lambda)$, $i = 1, 2, \ldots$, for each $\lambda \in N$ in an arbitrary way into a sequence which generates the ring $M(\lambda)'$. (For instance again into a [weakly] everywhere dense set in the part of $M(\lambda)'$ in $\|A\| \leq 1$ [which is separable].) As the $\sigma(\lambda)$-measure of N is 0, this does not affect $\sigma(\lambda)$-measurability of the $D_i(\lambda)$; but now the $D_i(\lambda)$, $i = 1, 2, \ldots$, generate the ring $M(\lambda)'$ for each λ. Thus $M(\lambda)'$ is $\sigma(\lambda)$-measurable in the sense of Definition 5, and the proof is completed. □

Lemma 8. *Let the $M_s(\lambda)$, $s = 1, 2, \ldots$, (the sequence of the s may be finite or infinite) be rings containing 1, depending for each s $\sigma(\lambda)$-measurably on λ. Let $N(\lambda) = M_1(\lambda) M_2(\lambda) \cdots$ be their intersection, $P(\lambda) = R(M_1(\lambda), M_2(\lambda), \ldots)$ the ring generated by them, both rings containing 1. Then $N(\lambda), P(\lambda)$ depend $\sigma(\lambda)$-measurably on λ too.*

Proof. Apply Definition 5 to $M_s(\lambda)$: Let $A_{si}(\lambda)$, $i = 1, 2, \ldots$, be a sequence of $\sigma(\lambda)$-

measurable operators, such that $A_{si}(\lambda)$, $i = 1, 2, \ldots$, generate the ring $M_s(\lambda)$. Then all $A_{si}(\lambda)$, $s, i = 1, 2, \ldots$, generate the ring $P(\lambda) = R(M_1(\lambda), M_2(\lambda), \ldots)$. Thus if we write the $A_{si}(\lambda)$, $s, i = 1, 2, \ldots$, as a simple sequence: $B_j(\lambda)$, $j = 1, 2, \ldots$, then the $B_j(\lambda)$ are $\sigma(\lambda)$-measurable, and the $B_j(\lambda)$, $j = 1, 2, \ldots$, generate the ring $P(\lambda)$. So $P(\lambda)$ is $\sigma(\lambda)$-measurable by Definition 5.

This, together with Lemma 7, gives now our result for $N(\lambda)$ too, because

$$N(\lambda) = M_1(\lambda)M_2(\lambda)\cdots = \left(R(M_1(\lambda)', M_2(\lambda)', \ldots)\right)'.$$

[12] ◻

Lemma 9. *Let $M(\lambda), N(\lambda)$ be rings containing 1, both depending $\sigma(\lambda)$-measurably on λ. Assume $M(\lambda) \subset N(\lambda)$ always. Let $\mathcal{M}$ be the set of all λ's for which $M(\lambda) \neq N(\lambda)$.*

Under these assumptions we have:

(i) $\mathcal{M}$ is a $\sigma(\lambda)$-measurable set.

(ii) There exists a $\sigma(\lambda)$-measurable $E(\lambda)$, which is a projection for each λ, and for which $\lambda \in \mathcal{M}$ implies $E(\lambda) \notin M(\lambda)$, $E(\lambda) \in N(\lambda)$.

Proof. Apply Lemma 6 to $M(\lambda)'$ and to $N(\lambda)'$ (in place of the $M(\lambda)$ of that lemma). Introduce $k(\lambda)$, Δ_p, $p = 1, 2, \ldots, \infty$, as there, and the sets of $\sigma(\lambda)$-measure 0, $\widetilde{N}_p$ and $\widetilde{\widetilde{N}}_p$ respectively. Now $\Delta_p - \widetilde{N}_p - \widetilde{\widetilde{N}}_p = (\Delta_p - \widetilde{N}_p)(\Delta_p - \widetilde{\widetilde{N}}_p)$ is a Borel-set in r. Forming $r \times S_p$ again, the set M_p^* of all (λ, A) with $\lambda \in \Delta_p - \widetilde{N}_p$, $(g_p^{(\lambda)})^{-1}Ag_p^{(\lambda)} \in M(\lambda)'' = M(\lambda)$ is a Borel-set, and so is the set M_p^{**} of all (λ, A) with $\lambda \in \Delta_p - \widetilde{\widetilde{N}}_p$, $(g_p^{(\lambda)})^{-1}Ag_p^{(\lambda)} \in N_p(\lambda)'' = N(\lambda)$. The set N_p of all (λ, A) with $\lambda \in \Delta_p - \widetilde{N}_p - \widetilde{\widetilde{N}}_p$ is clearly a Borel-set too.

The projections E in $\mathfrak{H}_p'$ form a Borel-set in S_p. They are characterized by $E^* = E$, $E^2 = E$, that is: $(E^*\psi_m', \psi_n') = (E\psi_m', \psi_n')$, $(E^2\psi_m', \psi_n') = (E\psi_m', \psi_n')$, that is (cf. the analogous computations in the proof of Lemma 6):

$$\overline{(E\psi_n', \psi_m')} - (E\psi_m', \psi_n') = 0,$$

[12]The general relation $M_1M_2\cdots = (R(M_1', M_2', \ldots))'$ is proved as follows: The operators which commute with a given A form a ring, so if they include $M_1, M_2, \ldots$ they must also include $R(M_1, M_2, \ldots)$. The converse is obvious, as $M_1, M_2, \ldots$ are subsets of $R(M_1, M_2, \ldots)$. Thus A commutes with all of $M_1, M_2, \ldots$ if, and only if, it commutes with all of $R(M_1, M_2, \ldots)$. In other words: $M_1'M_2'\cdots = (R(M_1, M_2, \ldots))'$. If we now replace $M_1, M_2, \ldots$ by $M_1', M_2', \ldots$ and apply $'$ to the resulting equation then $(M_1''M_2''\cdots)' = (R(M_1', M_2', \ldots))''$ obtains. Next $N'' = N$ for every ring N containing 1 by (2, p. 372), and so the above equation becomes $(M_1M_2\cdots)' = R(M_1', M_2', \ldots)$. Thus M' is an involutory operation, which dualizes the operations $M_1M_2\cdots$ and $R(M_1, M_2, \ldots)$.

$$\sum_{l=1}^{p}(E\psi'_m,\psi'_l)(E\psi'_l,\psi'_n)-(E\psi'_m,\psi'_n)=0.$$

As the left sides are Baire-functions of E (in the weak topology of S_p, cf. the considerations loc. cit. above), we conclude as loc. cit., that they determine a Borel-set in S_p. Now we can infer, that the set $\mathfrak{E}_p$ of all (λ, E) where E is a projection, is a Borel-set too (in $r\times S_p$).

Thus the set $D_p = N_p\cap(M_p^{**}\setminus M_p^{*})\cap\mathfrak{E}_p$ is a Borel-set too (in $r\times S_p$), its elements are all pairs (λ, E) with $\lambda\in\Delta_p-\widetilde{N}_p-\overline{\overline{\widetilde{N}}}_p$, $E\notin M(\lambda)$, $E\in N(\lambda)$, E a projection.

As D_p is a Borel-set in $r\times S_p$ it is *a fortiori* an analytic set. Therefore Lemma 5 applies to it: Define $F(\lambda, A)=\lambda$, which is obviously a continuous function. We denote the set K and the function $f(\lambda)$, as defined there, by K_p, $f_p(\lambda)$ (because their dependence on p is now essential). Clearly $K_p\subset\Delta_p-\widetilde{N}_p-\overline{\overline{\widetilde{N}}}_p$. $f_p(\lambda)$ is a pair (μ, A), as $F(f_p(\lambda))=\lambda$, there is $\mu=\lambda$. Thus we can put: $f_p(\lambda)=(\lambda, E_p^0(\lambda))$.

Literally as in the corresponding part of the proof of Lemma 7, we prove that $E_p^0(\lambda)$ (which is defined in the $\sigma(\lambda)$-measurable set K_p) is $\sigma(\lambda)$-measurable.

Now define

$$E(\lambda)=\begin{cases}E_p^0(\lambda), & \lambda\in K_p,\quad p=1,2,\ldots,\infty,\\ 0, & \text{otherwise.}\end{cases}$$

(The $K_p\subset\Delta_p$ are mutually exclusive.)

$\lambda\in K_p$ means that an $a\in D_p$ with $F(a)=\lambda$ exists, that is:

$$a=(\lambda,E),\quad \lambda\in\Delta_p-\widetilde{N}_p-\overline{\overline{\widetilde{N}}}_p,\quad E\notin M(\lambda),\quad E\in N(\lambda),$$

E a projection. In other words, $\lambda\notin K_p$ means this: Either $\lambda\in\widetilde{N}_p+\overline{\overline{\widetilde{N}}}_p$, or $\lambda\in\Delta_p-\widetilde{N}_p-\overline{\overline{\widetilde{N}}}_p$, and every projection $E\in N(\lambda)$ belongs to $M(\lambda)$ too. Using the terminology of (2, p. 392), this latter statement means: $N(\lambda)^P\subset M(\lambda)^P$. By (2, p. 393), this is equivalent to $N(\lambda)\subset M(\lambda)$, and as $M(\lambda)\subset N(\lambda)$, it is further equivalent to $M(\lambda)=N(\lambda)$. Thus K_p is the set of all $\lambda\in\Delta_p-\widetilde{N}_p-\overline{\overline{\widetilde{N}}}_p$ with $M(\lambda)\neq N(\lambda)$.

$\lambda\in K_p$ implies $E(\lambda)=E_p^0(\lambda)$, and this is $\notin M(\lambda)$, $\in N(\lambda)$, and a projection. If $\lambda\notin K_p$ for every $p=1,2,\ldots,\infty$, then $E(\lambda)=0$ is a projection again. Thus $E(\lambda)$ is always a projection.

As the $E^0_p(\lambda)$ are all $\sigma(\lambda)$-measurable, and the sets K_p are $\sigma(\lambda)$-measurable too, therefore $E(\lambda)$ is also $\sigma(\lambda)$-measurable.

Put $N = \widetilde{N}_1 + \overline{\overline{N}}_1 + \widetilde{N}_2 + \overline{\overline{N}}_2 + \cdots + \widetilde{N}_\infty + \overline{\overline{N}}_\infty$. N has the $\sigma(\lambda)$-measure 0, and

$$(\Delta_1 - \widetilde{N}_1 - \overline{\overline{N}}_1) + (\Delta_2 - \widetilde{N}_2 - \overline{\overline{N}}_2) + \cdots + (\Delta_\infty - \widetilde{N}_\infty - \overline{\overline{N}}_\infty) = r - N.$$

Thus if $\lambda \notin N$, then λ belongs to some $\Delta_p - \widetilde{N}_p - \overline{\overline{N}}_p$, $p = 1, 2, \ldots, \infty$, and thus $M(\lambda) \neq N(\lambda)$ implies $\lambda \in K_p$, for some $p = 1, 2, \ldots, \infty$, and with it $E(\lambda) \notin M(\lambda)$, $E(\lambda) \in N(\lambda)$.

Now change the $E(\lambda)$ for each $\lambda \in N$ with $M(\lambda) \neq N(\lambda)$ in an arbitrary way in a projection $E \notin M(\lambda)$, $E \in N(\lambda)$ (as to the existence of such an E, cf. above). As the $\sigma(\lambda)$-measure of N is 0, this does not affect the $\sigma(\lambda)$-measurability of $E(\lambda)$; but now $E(\lambda) \notin M(\lambda)$, $E(\lambda) \in N(\lambda)$ holds whenever $M(\lambda) \neq N(\lambda)$.

The set $\mathcal{M}$ of the λ's, for which $M(\lambda) \neq N(\lambda)$, differs from the $\sigma(\lambda)$-measurable set $K_1 + K_2 + \cdots$ only by a subset of N (which therefore has the $\sigma(\lambda)$-measure 0). Thus $\mathcal{M}$ is $\sigma(\lambda)$-measurable. This completes the proof. □

Remark. As the $E(\lambda)$'s are projections, $\|E(\lambda)\| \leq 1$. Thus they fulfill condition (ii) in Theorem V, while condition (i) in that theorem follows from their $\sigma(\lambda)$-measurability (cf. Remark 2 after Definition 5). Thus an operator $E \sim \sum E(\lambda)$ in $\mathfrak{H}$ exists. As all $E(\lambda)$'s are projections, it follows from the results of §14, that E is a projection too.

Lemma 10. *Let $M(\lambda), N(\lambda)$ be rings containing 1, both depending $\sigma(\lambda)$-measurably on λ. Denote the set of all λ's for which $M(\lambda), N(\lambda)$ are respectively in the relation $=, \neq, \subset, \supset, \subsetneqq, \supsetneqq$, respectively by $K_1, \ldots, K_6$. All these sets are $\sigma(\lambda)$-measurable.*

Proof. $M(\lambda)N(\lambda)$ too depends $\sigma(\lambda)$-measurably on λ by Lemma 8. Thus we can apply Lemma 9 to the rings $M(\lambda)N(\lambda)$ and $N(\lambda)$ (as we have always $M(\lambda)N(\lambda) \subset N(\lambda)$). So we see that the set of all λ's with $M(\lambda)N(\lambda) = N(\lambda)$ is a $\sigma(\lambda)$-measurable. But this equation means $M(\lambda) \subset N(\lambda)$, and so the set in question is K_3.

Interchanging $M(\lambda), N(\lambda)$ proves the $\sigma(\lambda)$-measurability of K_4. Now $K_1 = K_3K_4$ too must be $\sigma(\lambda)$-measurable, and similarly $K_2 = r - K_1$, $K_5 = K_3 - K_1$, $K_6 = K_4 - K_1$. □

There exists (in $\mathfrak{H}$) a smallest and a greatest ring containing 1: The set of all operators of the form $\alpha 1$, resp. the set of all bounded linear operators. We will denote them respectively by 0 and 1. In $\mathfrak{H}^{(\lambda)}$ (instead of $\mathfrak{H}$) we will denote them by $0(\lambda)$ and $1(\lambda)$ respectively.

Put $A_i(\lambda) = 1$ for every $i = 1, 2, \ldots$, and every λ with $-\infty < \lambda < +\infty$. These are clearly $\sigma(\lambda)$-measurable, and so the rings $0(\lambda)$, which are generated by the $A_i(\lambda)$, $i = 1, 2, \ldots$, are $\sigma(\lambda)$-measurable too. By Lemma 7 the rings $1(\lambda) = 0(\lambda)'$ too are $\sigma(\lambda)$-measurable. So we see:

The rings $0(\lambda)$ and $1(\lambda)$ depend both $\sigma(\lambda)$-measurably on λ.

18 Definition and discussion of the relation $M \approx \sum M(\lambda)$

The information obtained in the last section about rings $M(\lambda)$ which depend $\sigma(\lambda)$-measurably on λ, permit us to establish and to discuss the following definition:

Definition 6. *Let M be a ring in $\mathfrak{H}$ and $M(\lambda)$ a ring containing 1 in $\mathfrak{H}^{(\lambda)}$ for each $\lambda > -\infty, < +\infty$, depending $\sigma(\lambda)$-measurably on λ. We write $M \approx \sum M(\lambda)$, if $A \in M$ is equivalent to $A \sim \sum A(\lambda)$, $A(\lambda) \in M(\lambda)$.*

If the $M(\lambda)$ are given, it is clear that there cannot be $M \approx \sum M(\lambda)$ for two different M's; so there exists exactly one such M, or none at all.

We prove:

Lemma 11. *Let $M(\lambda)$ depend $\sigma(\lambda)$-measurably on λ. Then the operators $A_i(\lambda)$, $i = 1, 2, \ldots$, from Definition 5 can be so chosen that all $\|A_i(\lambda)\| \leq 1$. In this case operators A_i (in $\mathfrak{H}$) exist, so that $A_i \sim \sum A_i(\lambda)$.*

Proof. By Remark 2 after Definition 5 the numerical functions $\|A_i(\lambda)\|$ are $\sigma(\lambda)$-measurable along with $A_i(\lambda)$. Thus the operators

$$A_i^0(\lambda) = \begin{cases} \dfrac{1}{\|A_i(\lambda)\|} A_i(\lambda), & \text{for } \|A_i(\lambda)\| \neq 0, \\ 0, & \text{for } \|A_i(\lambda)\| = 0 \quad (\text{that is } A_i(\lambda) = 0) \end{cases}$$

are $\sigma(\lambda)$-measurable too. Clearly the $A_i^0(\lambda)$, $i = 1, 2, \ldots$, generate the same ring as the $A_i(\lambda)$, $i = 1, 2, \ldots$, that is $M(\lambda)$; and $\|A_i^0(\lambda)\| \leq 1$. This proves the first statement.

As to the second statement, observe that the $\sigma(\lambda)$-measurability of $A_i(\lambda)$ means that it fulfills condition (i) in Theorem V, while condition (ii) is satisfied owing to $\|A_i(\lambda)\| \leq 1$. Thus an $A_i \sim \sum A_i(\lambda)$ exists.

Lemma 12. *Let $M(\lambda)$ depend $\sigma(\lambda)$-measurably on λ, and form the $A_i \sim \sum A_i(\lambda)$, $i = 1, 2, \ldots$, of Lemma 11. Then the ring generated by P*[13] *and by the A_i, $i = 1, 2, \ldots$, is $M \approx \sum M(\lambda)$.*

Proof. Let $\widetilde{M}$ be the ring generated by P and by the A_i, $i = 1, 2, \ldots$. $P \subset P'$, and by Theorem V all $A_i \in P'$, so $\widetilde{M} \subset P'$. Clearly $\widetilde{M} \supset P$, so $P \subset \widetilde{M} \subset P'$. This implies $P = P'' \subset \widetilde{M}' \subset P'$[14], so $\widetilde{M}, \widetilde{M}'$ are both $\supset P$ and $\subset P'$. As the space of all operators A (in $\mathfrak{H}$) with $\|A\| \leq 1$ is separable (cf. (2, p. 386)), therefore the part of $\widetilde{M}'$ in it is separable too. Thus there exists a sequence of operators B_j, $j = 1, 2, \ldots$, which is dense in it. As all $B_j \in \widetilde{M}'$, therefore $R(B_1, B_2, \ldots) \subset \widetilde{M}'$; and by our construction $R(B_1, B_2, \ldots)$ contains all $A \in \widetilde{M}'$, $\|A\| \leq 1$, thus all $A \in \widetilde{M}'$, that is $R(B_1, B_2, \ldots) \supset \widetilde{M}'$. So $R(B_1, B_2, \ldots) = \widetilde{M}'$, that is the B_j, $j = 1, 2, \ldots$, determine the ring $\widetilde{M}'$. As $\widetilde{M}' \subset P'$, all $B_j \in P'$, thus by Theorem V, $B_j \sim \sum B_j(\lambda)$.

As $1 \in \widetilde{M}$, we have $\widetilde{M}'' = \widetilde{M}$ (cf.[14]), so that $A \in \widetilde{M}$ means that A commutes with every element of $\widetilde{M}' = R(B_1, B_2, \ldots)$, or, which is equivalent, with all B_j, B_j^*, $j = 1, 2, \ldots$. As $\widetilde{M} \subset P'$ we have $A \sim \sum A(\lambda)$. Thus our condition is that $A \sim \sum A(\lambda)$ must commute with all $B_j \sim \sum B_j(\lambda)$, $B_j^* \sim \sum B_j(\lambda)^*$. By §14, this means that $A(\lambda)$ commutes with every $B_j(\lambda)$ and $B_j(\lambda)^*$, except for a set of $\sigma(\lambda)$-measure 0. This set depends on j and the choice of $B_j(\lambda)$ or $B_j(\lambda)^*$, but by summing all these (countably many) sets, we obtain a fixed set of exceptions which has $\sigma(\lambda)$-measure 0. On this λ-set we may replace $A(\lambda)$ by 0, so we see: The condition is $A \sim \sum A(\lambda)$, where $A(\lambda)$ commutes with all $B_j(\lambda), B_j(\lambda)^*$.

Denote the ring generated by $1, B_j(\lambda), B_j(\lambda)^*$ (in $\mathfrak{H}^{(\lambda)}$) by $N(\lambda)$, then our condition is: $A(\lambda) \in N(\lambda)'$. The $N(\lambda)$ are by definition $\sigma(\lambda)$-measurable, and so by Lemma 7 $N(\lambda)'$ are $\sigma(\lambda)$-measurable too. Our above result states, remembering Definition 6, that $\widetilde{M} \approx \sum N(\lambda)'$.

As $A_i \sim \sum A_i(\lambda)$, $A_i \in \widetilde{M}$, therefore $A_i(\lambda) \in N(\lambda)'$, except for a set N_1^i of $\sigma(\lambda)$-measure 0. $N_2 = \sum_{i=1}^{\infty} N_1^i$ does the same, and is independent of i. So for $\lambda \notin N_2$ all $A_i(\lambda) \in N(\lambda)'$, and therefore $M(\lambda) \subset N(\lambda)'$. As the behavior of $N(\lambda)$ on the set of $\sigma(\lambda)$-measure 0, N_2, does not matter, we may replace it there by the set of all $\alpha 1$, then $N(\lambda)'$ contains all operators. Thus we can assume $M(\lambda) \subset N(\lambda)'$ for all λ's. As $M(\lambda), N(\lambda)'$ are both $\sigma(\lambda)$-measurable, Lemma 9 applies. The set $\mathcal{M}$ of all λ with $M(\lambda) \neq N(\lambda)'$ is $\sigma(\lambda)$-measurable, and there exists a $\sigma(\lambda)$-measurable $E(\lambda)$, which is a projection for every λ, so that $\lambda \in \mathcal{M}$ implies $E(\lambda) \notin M(\lambda)$, $E(\lambda) \in N(\lambda)'$.

[13] P is the ring discussed in §11 and §12.

[14] $N \subset L$ clearly implies $L' \subset N'$; $N = N''$ by (2, p. 388).

Replacing $E(\lambda)$ by 0 for $\lambda \notin \mathcal{M}$, which does not affect its $\sigma(\lambda)$-measurability as $\mathcal{M}$ is $\sigma(\lambda)$-measurable, we obtain furthermore $E(\lambda) \in N(\lambda)'$ for all λ's.

$R(E(\lambda))$ is $\sigma(\lambda)$-measurable and so by Lemmas 7, 8, $R(M(\lambda), E(\lambda))$, $R(M(\lambda), E(\lambda))'$ are $\sigma(\lambda)$-measurable too. We have always $R(M(\lambda), E(\lambda)) \supset M(\lambda)$, and $\neq$ for $\lambda \in \mathcal{M}$ (as $E(\lambda) \notin M(\lambda)$). Consequently we have always $R(M(\lambda), E(\lambda))' \subset M(\lambda)'$, and $\neq$ for $\lambda \in \mathcal{M}$. Apply Lemma 9 again. There exists a $\sigma(\lambda)$-measurable $F(\lambda)$, which is a projection for every λ, and $F(\lambda) \in M(\lambda)'$, $F(\lambda) \notin R(M(\lambda), E(\lambda))'$ for $\lambda \in \mathcal{M}$. We can again obtain $F(\lambda) \in M(\lambda)'$ for all λ's.

By the Remark after Lemma 9 there exist two projections E, F with $E \sim \sum E(\lambda)$, $F \sim \sum F(\lambda)$. As $E(\lambda) \in N(\lambda)'$ for all λ, we have $E \in \widetilde{M}$. As $F(\lambda) \in M(\lambda)'$, $F(\lambda)$ commutes with all $A_i(\lambda), A_i(\lambda)^*$, therefore F commutes with all A_i, A_i^*. By Theorem V, $F \in P'$, so F commutes with all P too. Thus F commutes with $\widetilde{M}$, hence with E, and therefore $F(\lambda)$ with $E(\lambda)$, except for a set of $\sigma(\lambda)$-measure 0. For the non-exceptional λ, $F(\lambda)$ commutes with $E(\lambda)$, and as $F(\lambda) \in M(\lambda)'$, with all $M(\lambda)$ too, thus $F(\lambda)$ commutes with the ring generated by $M(\lambda)$ and $E(\lambda)$ too. This means $F(\lambda) \in R(M(\lambda), E(\lambda))'$, and thus implies $\lambda \notin \mathcal{M}$. Therefore $\mathcal{M}$ must have the $\sigma(\lambda)$-measure 0.

Hence $M(\lambda) = N(\lambda)'$, except for a set of $\sigma(\lambda)$-measure 0, and therefore $\widetilde{M} \approx \sum N(\lambda)'$ implies $\widetilde{M} \approx \sum M(\lambda)$. Thus the proof is completed.

We are now in the position to prove:

Theorem VI. *If the $M(\lambda)$ are rings containing 1, depending $\sigma(\lambda)$-measurably on λ, then there exists a unique ring $M \approx \sum M(\lambda)$. This M is a ring containing 1, $P \subset M \subset P'$.*

Conversely, if M is a ring containing 1, $P \subset M \subset P'$, then there exists a system of rings containing 1, $M(\lambda)$, depending $\sigma(\lambda)$-measurably on λ, with $M \approx \sum M(\lambda)$. This system is unique, except for an arbitrary λ-set of $\sigma(\lambda)$-measure 0.

Proof. The existence of M, if the $M(\lambda)$ are given, follows from Lemmas 11, 12, the uniqueness of M has been observed immediately after Definition 6.

As every $A \sim \sum A(\lambda)$ belongs to P', necessarily $M \subset P'$. If $A \in P$, then we know from §14, that $A \sim \sum \alpha(\lambda)1$, and as $1 \in M(\lambda)$, $\alpha(\lambda)1 \in M(\lambda)$, this implies $A \in M$. Thus $P \subset M$. As $1 \in P \subset M$, we have $1 \in M$ and $P \subset M \subset P'$.

Assume now that M, $P \subset M \subset P'$, is given. Choose a sequence of operators A_i, $i = 1, 2, \ldots$, which generate the ring M (cf. the corresponding construction in the proof of Lemma 12, for $\widetilde{M}'$). As $A_i \in M \subset P'$, we can write $A_i \sim \sum A_i(\lambda)$. Denote the ring generated by the $A_i(\lambda)$, $i = 1, 2, \ldots$, by $M(\lambda)$. (We may assume $1 \in M(\lambda)$, as we can include $1 \sim \sum 1$ in the system of the A_i.) The $M(\lambda)$ depend thus measurably on λ, and by Lemma 12, $R(P, A_1, A_2, \ldots) \approx \sum M(\lambda)$. Now $R(P, A_1, A_2, \ldots) \supset R(A_1, A_2, \ldots) = M$, and as $P \subset M$ and all $A_i \in M$, $R(P, A_1, A_2, \ldots) \subset M$. Thus $R(P, A_1, A_2, \ldots) = M$, and so $M \approx \sum M(\lambda)$, proving the existence of the desired $M(\lambda)$.

That the change of the $M(\lambda)$ on a set of $\sigma(\lambda)$-measure 0 does not affect $M \approx \sum M(\lambda)$, is obvious. Thus it remains to prove only this: If $M \approx \sum M(\lambda)$ and $M \approx \sum N(\lambda)$, then $M(\lambda) = N(\lambda)$, except for a $\sigma(\lambda)$-set of measure 0. Consider $M(\lambda)N(\lambda)$ and $N(\lambda)$, both are $\sigma(\lambda)$-measurable (cf. Lemma 8). As there is always $M(\lambda)N(\lambda) \subset N(\lambda)$, we can apply Lemma 9. There exists a $\sigma(\lambda)$-measurable $E(\lambda)$, which is a projection for every λ, and $E(\lambda) \notin M(\lambda)N(\lambda)$, $E(\lambda) \in N(\lambda)$ whenever $M(\lambda)N(\lambda) \neq N(\lambda)$. As in the proof of Lemma 12, we can obtain $E(\lambda) \in N(\lambda)$ for all λ's. By the Remark after Lemma 9, a projection $E \sim \sum E(\lambda)$ exists. As $E(\lambda) \in N(\lambda)$ for all λ's, therefore $M \approx \sum N(\lambda)$ necessitates $E \in M$. Now $M \approx \sum M(\lambda)$ shows, that $E(\lambda) \in M(\lambda)$, except for a set N_1 of $\sigma(\lambda)$-measure 0. Thus $\lambda \notin N_1$ implies $E(\lambda) \in M(\lambda)$, and as $E \in N(\lambda)$, therefore $E \in M(\lambda)N(\lambda)$ too. This necessitates, as we know, $M(\lambda)N(\lambda) = N(\lambda)$, and thus $M(\lambda) \supset N(\lambda)$.

Similarly there exists a set N_2 of $\sigma(\lambda)$-measure 0, so that $\lambda \notin N_2$ implies $M(\lambda) \subset N(\lambda)$. Now $N = N_1 + N_2$ has $\sigma(\lambda)$-measure 0 too, and $\lambda \notin N$ implies obviously $M(\lambda) = N(\lambda)$.

Thus all parts of our theorem are proved.

19 Algebra of the decomposition

Our next task is to discuss the algebra of the decomposition $M \approx \sum M(\lambda)$. This is done in the three lemmas which follow.

Lemma 13. *$M \approx \sum M(\lambda)$ implies $M' \approx \sum M(\lambda)'$.*

Proof. The questions of $\sigma(\lambda)$-measurability are settled by Lemma 7. Form the $A_i \sim \sum A_i(\lambda)$, $i = 1, 2, \ldots$, in the sense of Lemma 11, for the $M(\lambda)$. We know that $A_i \in M$.

If $A \in M'$ then as $M \supset P$, $M' \subset P'$, $A \in P'$ we can write by Theorem V: $A \sim \sum A(\lambda)$. As $A \in M'$, $A_i \in M$, therefore A commutes with A_i, A_i^*, so by §14, $A(\lambda)$ commutes with $A_i(\lambda), A_i(\lambda)^*$, for all $i = 1, 2, \ldots$, except for a λ-set of $\sigma(\lambda)$-measure 0. By adding the (countably many) exceptional sets for all $i = 1, 2, \ldots$, we obtained a fixed exceptional set of $\sigma(\lambda)$-measure 0. In it we may replace $A(\lambda)$ by 0, thus we can assume that $A(\lambda)$ commutes with $A_i(\lambda), A_i(\lambda)^*$ for all $i = 1, 2, \ldots$, and all λ's. Thus it commutes with $R(A_1(\lambda), A_2(\lambda), \ldots) = M(\lambda)$, that is: $A(\lambda) \in M(\lambda)'$.

Conversely: Assume $A \sim \sum A(\lambda)$, $A(\lambda) \in M(\lambda)'$. For any $B \in M$ we have $B \sim \sum B(\lambda)$, $B(\lambda) \in M(\lambda)$, therefore we see: $A(\lambda), B(\lambda)$ commute for every λ, so A, B commute. As this is true for every $B \in M$, it implies $A \in M'$. Thus we have proved $M' \approx \sum M(\lambda)'$.

Lemma 14. *$M_s \approx \sum M_s(\lambda)$, $s = 1, 2, \ldots$ (the sequence of the s may be finite or infinite) implies*

$$M_1 \cdot M_2 \cdot \ldots \approx \sum M_1(\lambda) \cdot M_2(\lambda) \cdot \ldots$$

and

$$R(M_1, M_2, \ldots) \approx \sum R(M_1(\lambda), M_2(\lambda), \ldots).$$

Proof. The questions of $\sigma(\lambda)$-measurability are settled by Lemma 8. The statement $M_1 \cdot M_2 \cdot \ldots \approx \sum M_1(\lambda) \cdot M_2(\lambda) \cdot \ldots$ is obvious, if we only remember Definition 6. Now this gives, together with Lemma 13, the statement

$$R(M_1, M_2, \ldots)' \approx \sum R(M_1(\lambda), M_2(\lambda), \ldots)',$$

because of $R(M_1, M_2, \ldots)' = M_1' \cdot M_2' \cdot \ldots$, $R(M_1(\lambda), M_2(\lambda), \ldots)' = M_1(\lambda)' \cdot M_2(\lambda)' \cdot \ldots$ (cf. footnote 12 in the proof of Lemma 8).

Lemma 15. *If $M \approx \sum M(\lambda)$, $N \approx \sum N(\lambda)$, then $M \subset N$ is equivalent to $M(\lambda) \subset N(\lambda)$ for all λ's, except perhaps for a set of $\sigma(\lambda)$-measure 0.*

Proof. By Lemma 14, we have $M \cdot N \approx \sum M(\lambda) \cdot N(\lambda)$. As $M \approx \sum M(\lambda)$, we see by Theorem VI, that $M \cdot N = M$ means $M(\lambda) \cdot N(\lambda) = M(\lambda)$ for all λ's, except perhaps for a λ-set of $\sigma(\lambda)$-measure 0. But $M \cdot N = M$ is equivalent to $M \subset N$, and $M(\lambda) \cdot N(\lambda) = M(\lambda)$ to $M(\lambda) \subset N(\lambda)$. This gives the desired statement.

The behavior of the decomposition $M \approx \sum M(\lambda)$ under equivalence is this:

If $M \approx \sum M(\lambda)$, and the representation of $\mathfrak{H}$ as a generalized direct sum is replaced by an equivalent one, with the mapping $\lambda \rightleftarrows m(\lambda)$, then the decomposition of M in this new representation will be $M \approx \sum N(\lambda)$, where $N(\lambda)$ is defined as follows: $N(m(\lambda))$ is the ring which corresponds to $M(\lambda)$ under the isomorphism of $\mathfrak{H}_1^{(\lambda)}$ with $\mathfrak{H}_2^{(m(\lambda))}$.

One verifies this statement immediately with the help of Definition 6, and of the corresponding statement for $A \sim \sum A(\lambda)$ at the end of §14.

Part IV: Decomposition of rings with respect to their center

20 Central decomposition into factors

We can now apply the fundamental operation of the theory of finite-dimensional matrices, which is indicated in the title of this part. The results of our foregone discussions, particularly those of Theorem VI and Lemmas 13–15 in §18–§19, have reduced the general problem to the same level. The meaning and the effect of this operation are best expressed by the following theorem:

Theorem VII. *Let M be a ring in $\mathfrak{H}$ which contains 1. Those (Abelian) rings P for which a decomposition $M \approx \sum M(\lambda)$ in the sense of Definition 6 is possible, are the rings (containing 1) $P \subset M \cdot M'$ and only these. (These rings are all automatically Abelian.)*

The choice $P = M \cdot M'$ is characterized by that property, that for this and only for this P all rings $M(\lambda)$ of the decomposition $M \approx \sum M(\lambda)$ (except for a λ-set of $\sigma(\lambda)$-measure 0) are factors in the sense of (10). In other words: $M(\lambda) \cdot M(\lambda)' = 0(\lambda)$ (cf. at the end of §17).

Proof. M can be decomposed with respect to P if and only if $P \subset M \subset P'$ and P is Abelian. (Of course always $1 \in P$.) Now $M \subset P'$ means that M, P commute, which is equivalent to $P \subset M'$ too. Thus $P \subset M \subset P'$ means $P \subset M$ and $P \subset M'$, that is $P \subset M \cdot M'$. This implies the Abelian character too, because $M \cdot M'$ is Abelian: $M \cdot M' \subset M$ so $(M \cdot M')' \supset M' \supset M \cdot M'$.

By Lemmas 13, 14, $M \cdot M' \approx \sum M(\lambda) \cdot M(\lambda)'$ and by the second statement of §14, $P \approx \sum 0(\lambda)$. Thus Theorem VI shows that $M(\lambda) \cdot M(\lambda)' = 0(\lambda)$ (except for a λ-set of measure 0) is equivalent to $M \cdot M' = P$. Thus all parts of our theorem are proved.—

Theorem VII makes it possible to apply the results of (10) for *factors* to all rings. The most important tools which will be used in this connection are these: The classification of all factors ((10, Theorem VIII)), the notions of relative dimensionality ((10, Theorem VII)) and of the relative trace ((10, Theorem XIII)). The first steps towards a classification and discussion of all rings on this basis will be made in the sections which follow.

21 Weight functions and relative dimensionality

We defined loc. cit. above the notion of a relative dimensionality function $d_M(E)$ for factors M, and proved, that for every M there exists one and (up to a constant factor $> 0, < +\infty$) only one such $d_M(E)$. We now wish to introduce a similar, although somewhat more general notion which covers all rings M (containing 1):

Definition 7. *Let M be a ring (containing 1). $\Delta_M(E)$ is a weight-function (with respect to M) if it has the following properties:*

(i) $\Delta_M(E)$ is defined for all projections $E \in M$ and only these. Its values are real numbers $\geq 0, \leq +\infty$. $\Delta_M(E) = 0$ if and only if $E = 0$.

(ii) $E_1, E_2, \ldots \in M$, all E_j's projections, $E_j E_k = 0$ for $j \neq k$, imply

$$\Delta_M(E_1 + E_2 + \cdots) = \Delta_M(E_1) + \Delta_M(E_2) + \cdots .$$

(iii) $E, U \in M$, E a projection, U unitary, imply $\Delta_M(UEU^{-1}) = \Delta_M(E)$.

In order to establish a first connection between the relative dimensionality functions and the weight functions, we now prove:

Lemma 16. *If M is a factor, then the notions of relative dimensionality functions and of weight functions are equivalent, with this one exception:*

$$\Delta_M^\infty(E) = \begin{cases} 0 & \text{for } E = 0, \\ \infty & \text{for } E \neq 0 \end{cases}$$

is a relative dimensionality function in the case III only (cf. (10, Theorem VIII et seq.)), but it is always a weight function.

Remark: It will appear from the proof, that this lemma remains true if we replace the condition (ii) in Definition 7, by the weaker condition (ii)′, which arises from (ii) by permitting two addends E_j ($j = 1, 2$) only. Thus it would not affect Definition 7 essentially if we replaced in it (ii) by (ii)′, provided that M is a factor. But for general rings M this replacement is essential, as we will see in (γ) in §24.

Proof. The statements concerning $\Delta_M^\infty(E)$ are obvious. Every relative dimensionality function $d_M(E)$ is at the same time a weight function: It fulfills (i) by definition, that is, by (10, Definition 8.2.1), it fulfills (ii) by (10, Lemma 8.3.2); and (iii) follows from (10, Definition 8.2.1, condition (ii)), which states $d_M(V^*V) = d_M(VV^*)$ for every partially isometric V (cf. (10, Definition 6.1.1 and Lemma 4.3.1)), and we need only put $V = UE$ (then $V^* = E^*U^* = EU^{-1}$,

and so $V^*V = E$, $VV^* = UEU^{-1}$). Thus we must only prove the converse statement. If a weight function $d_M(E)$ is not $\Delta_M^\infty(E)$, that is, if it does not assume only the values $0, +\infty$, then it is at the same time a relative dimensionality function. By (10, Lemma 8.3.5), this is proved, if we can show that

$$E \sim F \cdots \bmod M \quad \text{implies} \quad \Delta_M(E) = \Delta_M(F).$$

Assume therefore $E \sim F \cdots \bmod M$. If $1-E \sim 1-F \cdots \bmod M$, then let V, W be partially isometric operators, with the initial projections E resp. $1-E$, and the final ones F resp. $1-F$. Then $U = V+W$ is unitary and maps E on F, that is $UEU^{-1} = F$. (Cf. (10, Lemmas 4.3.1 and 4.3.3).) Now (iii) gives $d_M(E) = d_M(F)$. So we need only to prove $1-E \sim 1-F \cdots \bmod M$. This follows from $E \sim F \cdots \bmod M$, if E, and with it F, is finite, as one sees by using a relative dimensionality function $d_M(E)$ and arguing as in (10, Lemma 8.4.1, proof of (iii)). Therefore we may assume that E, and with it F, is infinite.

In this case $E = E_1 + E_2$, $E \sim E_1 \sim E_2 \cdots \bmod M$, and $F = F_1 + F_2$, $F \sim F_1 \sim F_2 \cdots \bmod M$. (Cf. (10, Definition 6.1.1).) Thus E_1, E_2 are infinite, and so $1 - E_1 \geq E_2$ is infinite too. Similarly F_1, F_2, and $1 - F_1 \geq F_2$ are infinite. Thus $E_1 \sim F_1 \cdots \bmod M$, $1 - E_1 \sim 1 - F_1 \cdots \bmod M$, which implies, as have already been seen, $\Delta_M(E_1) = \Delta_M(F_1)$. Similarly $\Delta_M(E_2) = \Delta_M(F_2)$, and so

$$\Delta_M(E) = \Delta_M(E_1) + \Delta_M(E_2) = \Delta(F_1) + \Delta_M(F_2) = \Delta_M(F).$$

This completes the proof.—

Let M be a ring containing 1, put $P = M \cdot M'$, and decompose M in the sense of Theorem VII with respect to this P: $M \approx M(\lambda)$. As each $M(\lambda)$ is a factor, we can select in each one a weight function $\Delta_{M(\lambda)}(E)$, or even a relative dimensionality function $d_{M(\lambda)}(E)$. It is clear, that if a definite $d^*_{M(\lambda)}(E)$ has been chosen (for each $\lambda > -\infty, < +\infty$), then all $\Delta_{M(\lambda)}(E)$ are obtained by forming

$$\Delta_{M(\lambda)}(E) = a(\lambda) d^*_{M(\lambda)}(E) \tag{21.1}$$

for all possible functions $a(\lambda)$ the values of which are real numbers $> 0, \leq +\infty$. (We use the convention $0 \cdot \infty = 0$. If a $\Delta_{M(\lambda)}(E)$ is $\Delta^\infty_{M(\lambda)}(E)$, then we may put $a(\lambda) = +\infty$. If we wish to obtain the $d_{M(\lambda)}(E)$'s only, we must require $a(\lambda) > 0$, $< +\infty$.)

Observe that (21.1), that is $\Delta_{M(\lambda)}(E)$, determines $a(\lambda)$ uniquely, if an $E \in M(\lambda)$ with $d^*_{M(\lambda)}(E) > 0, < +\infty$ can be found, that is if we do not have case III in the sense of (10, Theorem VIII). In case III, (21.1) holds identically for any $a(\lambda) > 0$, $< +\infty$ (remember Lemma 16). So we see: $a(\lambda)$ is uniquely determined or perfectly arbitrary accordingly to whether we do not or do have case III.

It is clear, that this $a(\lambda)$ may be a non $\sigma(\lambda)$-measurable function. Therefore we cannot expect that $\Delta_{M(\lambda)}(E(\lambda))$ will be a $\sigma(\lambda)$-measurable (numerical) function for all $\sigma(\lambda)$-measurable $E(\lambda)$'s. We now proceed to eliminate this inconvenience.

Lemma 17. *Let $d_{M(\lambda)}(E)$ be a relative dimensionality function, let the projection $E(\lambda) \in M(\lambda)$ depend $\sigma(\lambda)$-measurably on λ. Denote the sets of all λ's where*

$d_{M(\lambda)}(E) = 0$, or $= \infty$, or $> 0, < \infty$, by M_0 or M_∞ or M_r, respectively. These sets are $\sigma(\lambda)$-measurable.

Proof. Let $\varphi_m(\lambda)$ be a measurable family.

$\lambda \in M_0$ means $E(\lambda) = 0$ that is $E(\lambda)\varphi_m(\lambda) = 0$ from $m = 1, 2, \ldots$, that is $\sum_{m=1}^{\infty} \|E(\lambda)\varphi_m(\lambda)\|^2 = 0$. As the left side is a $\sigma(\lambda)$-measurable function of λ, this proves the $\sigma(\lambda)$-measurability of M_0.

Consider now M_∞: $\lambda \in M_\infty$ means that $E(\lambda)$ is infinite (with respect to $M(\lambda)$) in the sense of (10, Definition 7.1.1), in other words: the existence of an $F \sim E(\lambda) \cdots \bmod M(\lambda)$, $F < E(\lambda)$, or: the existence of a $V \in M(\lambda)$ with $V^*V = E(\lambda)$, $VV^*E(\lambda) = VV^* \neq E(\lambda)$.[15] As $M(\lambda)$ is $\sigma(\lambda)$-measurable, $M(\lambda)'$ is also $\sigma(\lambda)$-measurable, and so $\sigma(\lambda)$-measurable operators $A_q(\lambda)$, $q = 1, 2, \ldots$, exist, such that $1, A_q(\lambda)$, $q = 1, 2, \ldots$, generate the ring $M(\lambda)'$. Now $V \in M(\lambda)$ means that V commutes with all $M(\lambda)'$, that is with every $A_q(\lambda), A_q(\lambda)^*$, $q = 1, 2, \ldots$.

Proceed now as in the proof of Lemma 6. Introduce $k(\lambda)$, and then the set Δ_p of all λ with $k(\lambda) = p$, $p = 1, 2, \ldots, \infty$. Choose there a set N_p of $\sigma(\lambda)$-measure 0, such that $\Delta'_p = \Delta_p - N_p$ is a Borel-set, and that all $(E(\lambda)\varphi_m(\lambda), \varphi_n(\lambda))$ and $(A_q(\lambda)\varphi_m(\lambda), \varphi_n(\lambda))$, $q, m, n = 1, 2, \ldots$, are Baire functions in Δ'_p. Introduce a p-dimensional unitary space $\mathfrak{H}'_p$, in it a complete normalized orthogonal set $\psi'_1, \psi'_2, \ldots$, and let $J^{(\lambda)}$ be the isomorphic mapping of $\mathfrak{H}'_p$ on $\mathfrak{H}^{(\lambda)}$ ($\lambda \in \Delta_p$), by which ψ'_m corresponds to $\varphi_m(\lambda)$.

Consider now the above condition for $\lambda \in M_\infty$. Replace the operator V in $\mathfrak{H}^{(\lambda)}$ by the $J^{(\lambda)}$-isomorphic $\widetilde{V} = J^{(\lambda)-1}VJ^{(\lambda)}$ in $\mathfrak{H}'_p$. Rewriting our conditions for this case, we see: $\lambda \in M_\infty$ means, if $\lambda \in \Delta_p$, that a $\widetilde{V}$ in $\mathfrak{H}'_p$ exists, for which

$$\begin{cases} \widetilde{V}J^{(\lambda)-1}A_q(\lambda)J^{(\lambda)} = J^{(\lambda)-1}A_q(\lambda)J^{(\lambda)}\widetilde{V}, \\ \widetilde{V}J^{(\lambda)-1}A_q(\lambda)^*J^{(\lambda)} = J^{(\lambda)-1}A_q(\lambda)^*J^{(\lambda)}\widetilde{V}, \qquad q = 1, 2, \ldots, \\ \widetilde{V}^*\widetilde{V} = J^{(\lambda)-1}E(\lambda)J^{(\lambda)}, \\ \widetilde{V}\widetilde{V}^*J^{(\lambda)-1}E(\lambda)J^{(\lambda)} = \widetilde{V}\widetilde{V}^* \neq J^{(\lambda)-1}E(\lambda)J^{(\lambda)}. \end{cases} \tag{21.2}$$

Observe that a $\widetilde{V}$ fulfilling (21.2) is partially isometric, and therefore automatically $\|\widetilde{V}\| \leq 1$.

[15] V must be partially isometric, but this follows from $V^*V = E(\lambda)$, because thus V^*V is a projection. (Cf. (10, Lemma 4.3.2).)

Denote again the set $-\infty < \lambda < +\infty$ by r, and the set of all operators $\widetilde{A}$ in $\mathfrak{H}'_p$ with $\|\widetilde{A}\| \leq 1$ (in its weak topology) by S_p. Let the set of all pairs $(\lambda, \widetilde{V})$ in $r \times S_p$ with $\lambda \in \Delta'_p$ and fulfilling (21.2) be D_p. Proceed as in the proof of Lemma 6: Replace the operator equations $X = Y$ of (21.2) by the equivalent numerical equations $(X\psi'_m, \psi'_n) = (Y\psi'_m, \psi'_n)$, $m, n = 1, 2, \ldots$. Now these expressions are convergent, infinite sums, the terms of which are polynomials of the $(\widetilde{V}\psi'_m, \psi'_n)$ and of the

$$(J^{(\lambda)-1}A_q(\lambda)J^{(\lambda)}\psi'_m, \psi'_n) = (A_q(\lambda)\varphi_m(\lambda), \varphi_n(\lambda)),$$
$$(J^{(\lambda)-1}E(\lambda)J^{(\lambda)}\psi'_m, \psi'_n) = (E(\lambda)\varphi_m(\lambda), \varphi_n(\lambda)).$$

The first expression is a (weakly) continuous function of $\widetilde{V}$, the two others are (owing to $\lambda \in \Delta'_p$) Baire functions of λ. Thus we can argue as in the proof of Lemma 6 (using (15, p. 260)): D_p is a Borel-set in $r \times S_p$.

Now we argue as in the proof of Lemma 7: As D_p is a Borel-set in S_p, it is *a fortiori* an analytic set. Therefore, Lemma 5 applies to it: Define $F((\lambda, \widetilde{V})) = \lambda$, which is obviously a continuous function. $\lambda \in \Delta'_p \cdot M_\infty$ means that a $\widetilde{V}$ with $(\lambda, \widetilde{V}) \in D_p$ exists, i.e. that an $a \in D_p$ with $F(a) = \lambda$ exists. Hence $\Delta'_p \cdot M_\infty$ is $\sigma(\lambda)$-measurable. Next $(\Delta_p - \Delta'_p) \cdot M_\infty = N_p \cdot M_\infty \subset N_p$ has the $\sigma(\lambda)$-measure 0, thus $\Delta_p \cdot M_\infty$ is $\sigma(\lambda)$-measurable too, and with it $M_\infty = \sum_{p=1,2,\ldots,\infty} \Delta_p \cdot M_\infty$.

Thus M_0, M_∞ are $\sigma(\lambda)$-measurable, and with them $M_r = r - M_0 - M_\infty$ is $\sigma(\lambda)$-measurable too. This completes the proof.

Lemma 18. *Let $d_{M(\lambda)}(E)$ be a relative dimensionality function, and let the projections $E(\lambda), F(\lambda) \in M(\lambda)$ depend $\sigma(\lambda)$-measurably on λ. Then the (numerical) function*

$$\frac{d_{M(\lambda)}(E(\lambda))}{d_{M(\lambda)}(F(\lambda))}$$

is $\sigma(\lambda)$-measurable.

Proof. Form the sets M_0, M_∞, M_r in the sense of Lemma 17 for $E(\lambda)$, and similarly P_0, P_∞, P_r for $F(\lambda)$. We have

$$r = M_0 + M_r + M_\infty = P_0 + P_r + P_\infty = (M_0 + M_r + M_\infty)(P_0 + P_r + P_\infty)$$

$$= M_0P_0 + M_rP_0 + M_\infty P_0 + M_0P_r + M_rP_r + M_\infty P_r + M_0P_\infty + M_rP_\infty + M_\infty P_\infty.$$

All terms of this sum are $\sigma(\lambda)$-measurable by Lemma 17. In terms No. 1 and 9 $d_{M(\lambda)}(E(\lambda))/d_{M(\lambda)}(F(\lambda))$ has the form $0/0$ resp. ∞/∞, and is thus nowhere defined;

in all others it is everywhere defined. In terms No. 2, 3, 4, 6, 7, 8, it is constant, being respectively $\infty, \infty, 0, \infty, 0, 0$; in term No. 5 it is variable and $> 0, < +\infty$. Thus we need only to prove the $\sigma(\lambda)$-measurability of $d_{M(\lambda)}(E(\lambda))/d_{M(\lambda)}(F(\lambda))$ in term No. 5, that is in $M_r P_r$.

Choose two numbers $u, v = 1, 2, \ldots$, and define $M_{u,v}$ as the set of those λ's, for which projections $G_1, \ldots, G_u, H_1, \ldots, H_v \in M(\lambda)$ exist, which have the following properties: $G_1 \sim \cdots \sim G_u \sim H_1 \sim \cdots \sim H_v \cdots \bmod M(\lambda)$, $G_j G_k = H_j H_k = 0$ for $j \neq k$, $G_1 + \cdots + G_u \leq E(\lambda)$, $H_1 + \cdots + H_v \geq F(\lambda)$. Our first objective is to prove that $M_{u,v}$ is $\sigma(\lambda)$-measurable.

$\lambda \in M_{u,v}$ means that the above described $G_1, \ldots, G_u, H_1, \ldots, H_v$ exist. Introducing another projection $K \in M(\lambda)$, which is $\sim$ to all of them mod $M(\lambda)$, we may describe the entire situation by speaking of the partially isometric operators in $M(\lambda)$, which have all the initial projection K, and the respective final projections $G_1, \ldots, G_u, H_1, \ldots, H_v$. We denote them by $V_1, \ldots, V_{u+v}$. Now our conditions are: Each $V_s \in M(\lambda)$, that is V_s commutes with all $M(\lambda)'$, that is with all $A_q(\lambda)$, $q = 1, 2, \ldots$ (the $A_q(\lambda)$, $q = 1, 2, \ldots$, generate $M(\lambda)'$ and are $\sigma(\lambda)$-measurable, as in the proof of Lemma 17); $V_s V_s^* V_s = V_s$ (this expresses the partial isometry, cf. (10, Lemma 4.3.2)); $V_1^* V_1 = \cdots = V_{u+v}^* V_{u+v}$; $V_j V_j^* V_k V_k^* = 0$ if $j \neq k$ and both $j, k = 1, \ldots, u$ or both $= u+1, \ldots, u+v$; $(V_1 V_1^* + \cdots + V_u V_u^*) E(\lambda) = V_1 V_1^* + \cdots + V_u V_u^*$, $(V_{u+1} V_{u+1}^* + \cdots + V_{u+v} V_{u+v}^*) F(\lambda) = F(\lambda)$.

Proceed now as in the proofs of Lemmas 6, 17: Introduce $k(\lambda)$, the set Δ_p of all λ's with $k(\lambda) = p$, $p = 1, 2, \ldots, \infty$, choose N_p, $\Delta_p' = \Delta_p - N_p$ as there, with respect to the operators $E(\lambda), F(\lambda), A_q(\lambda), A_q(\lambda)^*$, $q = 1, 2, \ldots$. Introduce $\mathfrak{H}_p'$, the $\psi_1', \psi_2', \ldots$, and the isomorphism $J^{(\lambda)}$ as there. Replace now in our

above conditions the V_s in $\mathfrak{H}^{(\lambda)}$ by their $J^{(\lambda)}$-isomorphic $\widetilde{V}_s = J^{(\lambda)-1} V_s J^{(\lambda)}$ in $\mathfrak{H}_p'$. Then our conditions become:

$$
\begin{cases}
\widetilde{V}_s J^{(\lambda)-1} A_q(\lambda) J^{(\lambda)} = J^{(\lambda)-1} A_q(\lambda) J^{(\lambda)} \widetilde{V}_s, & \\
\widetilde{V}_s J^{(\lambda)-1} A_q(\lambda)^* J^{(\lambda)} = J^{(\lambda)-1} A_q(\lambda)^* J^{(\lambda)} \widetilde{V}_s, & {\scriptstyle s=1,\ldots,u+v, \atop q=1,2,\ldots,} \\
\widetilde{V}_s \widetilde{V}_s^* \widetilde{V}_s = \widetilde{V}_s, & s = 1, \ldots, u+v, \\
\widetilde{V}_1^* \widetilde{V}_1 = \cdots = \widetilde{V}_{u+v}^* \widetilde{V}_{u+v}, & \\
\widetilde{V}_j \widetilde{V}_j^* \widetilde{V}_k \widetilde{V}_k^* = 0, & {\scriptstyle j \neq k, \quad j,k=1,\ldots,u, \atop \text{or both } j,k=u+1,\ldots,u+v,} \\
(\widetilde{V}_1 \widetilde{V}_1^* + \cdots + \widetilde{V}_u \widetilde{V}_u^*) J^{(\lambda)-1} E(\lambda) J^{(\lambda)} = \widetilde{V}_1 \widetilde{V}_1^* + \cdots + \widetilde{V}_u \widetilde{V}_u^*, & \\
(\widetilde{V}_{u+1} \widetilde{V}_{u+1}^* + \cdots + \widetilde{V}_{u+v} \widetilde{V}_{u+v}^*) J^{(\lambda)-1} F(\lambda) J^{(\lambda)} = J^{(\lambda)-1} F(\lambda) J^{(\lambda)}. &
\end{cases}
\tag{21.3}
$$

Observe that (21.3) implies for each $\widetilde{V}_s$ partial isometricity, and thus $\|\widetilde{V}_s\| \leq 1$.

Consider now again r and S_p, but now form the direct product $r \times S_p \times \cdots \times S_p$ with $u+v$ factors S_p. Let the set of all $u+v+1$-uplets $(\lambda, \widetilde{V}_1, \ldots, \widetilde{V}_{u+v})$ in $r \times S_p \times \cdots \times S_p$, with $\lambda \in \Delta'_p$ and fulfilling (21.3) be C_p. We prove as in Lemma 17 that C_p is a Borel-set in $r \times S_p \times \cdots \times S_p$. $\lambda \in \Delta'_p M_{u,v}$ means that there exists a subsystem $\widetilde{V}_1, \ldots, \widetilde{V}_{u+v}$ with $(\lambda, \widetilde{V}_1, \ldots, \widetilde{V}_{u+v}) \in C_p$. From this we infer, as in Lemma 17, that $\Delta'_p M_{u,v}$ is $\sigma(\lambda)$-measurable. $(\Delta_p - \Delta'_p)M_{u,v} = N_p M_{u,v} \subset N_p$ has the $\sigma(\lambda)$-measure 0; hence $\Delta_p M_{u,v}$ and $M_{u,v} = \sum_{p=1,2,\ldots,\infty} \Delta_p M_{u,v}$ are $\sigma(\lambda)$-measurable, too.

Let us now return to our function $d_{M(\lambda)}(E(\lambda))/d_{M(\lambda)}(F(\lambda))$ in $M_r P_r$. We wish to prove that it is $\sigma(\lambda)$-measurable, that is, that the set M_α of all $\lambda \in M_r P_r$ for which

$$\frac{d_{M(\lambda)}(E(\lambda))}{d_{M(\lambda)}(F(\lambda))} > \alpha$$

is $\sigma(\lambda)$-measurable for every (real) α. For $\alpha \leq 0$ this set is $M_r P_r$ and our statement holds, thus we may assume $\alpha > 0$.

Now if $\lambda \in M_r P_r$, then the last inequality is clearly equivalent to the existence of two $u, v = 1, 2, \ldots$, such that $u/v > \alpha$, and $\lambda \in M_{u,v}$.[16] Therefore

$$Q_\alpha = M_r P_r \sum_{\substack{u,v=1,2,\ldots \\ u/v>\alpha}} M_{u,v},$$

and so it is $\sigma(\lambda)$-measurable along with M_r, P_r, and the $M_{u,v}$. Thus the proof is completed.

Lemma 19. *There exists one and (up to a λ-set of $\sigma(\lambda)$-measure 0) only one $\sigma(\lambda)$-measurable set $\overline{M}$ with the following properties.*

(i) There exists a projection $\overline{E}(\lambda) \in M(\lambda)$, depending $\sigma(\lambda)$-measurably on λ, for which $M_0 = r - \overline{M}$, $M_\infty = 0$, $M_r = \overline{M}$.

[16] The sufficiency of this condition is obvious. In order to prove its necessity, consider the classification of (10, Theorem VIII) (for this λ). As $\lambda \in M_r P_r$, therefore $d_{M(\lambda)}(E(\lambda)) > 0, < +\infty$, hence case III cannot hold. If case I holds, choose K so as to minimize $d_{M(\lambda)}(K)$ (> 0), denoting this minimum by ε_λ. Then we can put

$$u = \frac{1}{\varepsilon_\lambda} d_{M(\lambda)}(E(\lambda)), \qquad v = \frac{1}{\varepsilon_\lambda} d_{M(\lambda)}(F(\lambda))$$

as these quantities must be positive and integers, that is $= 1, 2, \ldots$. If case II holds, then u/v may be any rational number $\leq d_{M(\lambda)}(E(\lambda))/d_{M(\lambda)}(F(\lambda))$ and $> \alpha$. Then we can put

$$d_{M(\lambda)}(K) = \frac{1}{v} d_{M(\lambda)}(F(\lambda)) \quad \text{and hence} \quad \leq \frac{1}{u} d_{M(\lambda)}(E(\lambda)).$$

(ii) *For every projection $E(\lambda) \in M(\lambda)$ which depends $\sigma(\lambda)$-measurably on λ we have* $\mu_{\sigma(\lambda)}(M_r - M_r \cdot \overline{M}) = 0$.

In other words: Disregarding sets of $\sigma(\lambda)$-measure 0, $\overline{M}$ is the greatest M_r.

Proof. We proceed stepwise.

(I) Consider all $E(\lambda)$ depending $\sigma(\lambda)$-measurably on λ, such that each $E(\lambda)$ is a projection in $M(\lambda)$. Form their sets M_r and denote the set of all M_r by T.

(II) If $M \in T$, then an $E(\lambda)$ with $M_0 = r - M$, $M_\infty = 0$, $M_r = M$ exists.

(III) If $M \in T$, then every $\sigma(\lambda)$-measurable $M' \subset M$ is $M' \in T$ too.
We prove (II), (III) together, by constructing an $E(\lambda)$ with $M_0 = r - M'$, $M_\infty = 0$, $M_r = M'$. Choose an $\widetilde{E}(\lambda)$ with $\widetilde{M}_r = M$. Define now

$$E(\lambda) = \begin{cases} \widetilde{E}(\lambda) & \text{for } \lambda \in M', \\ 0 & \text{for } \lambda \notin M'. \end{cases}$$

Then $E(\lambda)$ is $\sigma(\lambda)$-measurable along with $\widetilde{E}(\lambda)$ and M', and it obviously meets our requirements.

(IV) If $M_1, M_2, \ldots \in T$, then $M_1 + M_2 + \cdots \in T$ too.
By (III) we may replace $M_1, M_2, M_3, \ldots$ by $M_1, M_2 - M_1M_2, M_3 - (M_1 + M_2)M_3, \ldots$. Thus we may assume $M_jM_k = 0$ for $j \neq k$. Now choose $E_j(\lambda)$ for M_j by (II), then we have for every fixed λ at most one $E_j(\lambda) \neq 0$. Thus $E(\lambda) = \sum_{j=1}^{\infty} E_j(\lambda)$ is a projection, and it is clear that it belongs to $M(\lambda)$, that it depends $\sigma(\lambda)$-measurably on λ, and that its $M_r = M_1 + M_2 + \cdots$.

(V) Put $\alpha = \mathrm{gr.\,l.\,b.}_{M \in T}\, \mu_{\sigma(\lambda)}(M)$. For each $j = 1, 2, \ldots$ choose an $M_j \in T$ with $\mu_{\sigma(\lambda)}(M_j) \geq \alpha - 1/j$. By (IV) $\overline{M} = \sum_{j=1}^{\infty} M_j \in T$, so $\mu_{\sigma(\lambda)}(\overline{M}) \leq \alpha$. But $\mu_{\sigma(\lambda)}(\overline{M}) \geq \mu_{\sigma(\lambda)}(M_j) \geq \alpha - 1/j$ for every $j = 1, 2, \ldots$, thus necessarily $\mu_{\sigma(\lambda)}(\overline{M}) = \alpha$. Now if $M \in T$, then by (IV) $\overline{M} + M \in T$ too; thus $\mu_{\sigma(\lambda)}(\overline{M} + M) = \alpha$. Thus $\mu_{\sigma(\lambda)}(M - \overline{M} \cdot M) = \mu_{\sigma(\lambda)}(\overline{M} + M) - \mu_{\sigma(\lambda)}(\overline{M}) = 0$. Thus $\overline{M}$ has the property (ii) of our lemma.

(VI) Applying (II) to $\overline{M} \in T$ proves that $\overline{M}$ has the property (i) of our lemma. Its unicity (up to a λ-set of $\sigma(\lambda)$-measure 0) is an obvious consequence of (i) and (ii).

Thus the proof is completed.

Observe that the sets M_0, M_∞, M_r (for a given $E(\lambda)$) and $\overline{M}$ depend only apparently on the choice of $d_{M(\lambda)}(E)$: The equations $d_{M(\lambda)}(E) = 0$ and ∞ which enter in their definitions mean the same thing for every choice of $d_{M(\lambda)}(E)$: $E = 0$ and E infinite (with respect to $M(\lambda)$), respectively.

We now proceed as follows:

Definition 8. *A weight function $\Delta_{M(\lambda)}(E)$ is $\sigma(\lambda)$-measurable (in its dependence on λ) if the (numerical) function $\Delta_{M(\lambda)}(E(\lambda))$ is $\sigma(\lambda)$-measurable whenever*

$E(\lambda)$ is $\sigma(\lambda)$-measurable. (This nomenclature applies to the relative dimensionality functions $d_{M(\lambda)}(E)$ too, as they constitute a special case of the $\Delta_{M(\lambda)}(E)$.)

Theorem VIII. *The following statements are true:*

(i) *A $\Delta_{M(\lambda)}(E)$, and even a $d_{M(\lambda)}(E)$, which is $\sigma(\lambda)$-measurable, exists.*

(ii) *If $d^*_{M(\lambda)}(E)$ is $\sigma(\lambda)$-measurable, then all $\sigma(\lambda)$-measurable $\Delta_{M(\lambda)}(E)$'s obtain by the formula (21.1):*

$$\Delta_{M(\lambda)}(E) = a(\lambda) d^*_{M(\lambda)}(E), \qquad 0 < a(\lambda) \leq +\infty, \tag{21.1}$$

$a(\lambda)$ being restricted to those functions which are $\sigma(\lambda)$-measurable in $\overline{M}$ (cf. Lemma 19).

Proof: Observe first that for every $\Delta_{M(\lambda)}(E)$ the (numerical) function $\Delta_{M(\lambda)}(E(\lambda))$ is $\sigma(\lambda)$-measurable in $r - \overline{M}$, if $E(\lambda)$ is $\sigma(\lambda)$-measurable. Choose any $d_{M(\lambda)}(E(\lambda))$. Then $\Delta_{M(\lambda)}(E) = a(\lambda) d_{M(\lambda)}(E)$ for some $a(\lambda) > 0, \leq +\infty$, and thus $\Delta_{M(\lambda)}(E(\lambda)) > 0$, $< +\infty$ implies $d_{M(\lambda)}(E(\lambda)) > 0, < +\infty$, and therefore for $\lambda \in r - \overline{M}$ it implies $\lambda \in M_r - M_r \cdot \overline{M}$. Thus we deduce from Lemma 19: $r - \overline{M}$ is $\sigma(\lambda)$-measurable, and in it $\Delta_{M(\lambda)}(E(\lambda)) > 0, < +\infty$ occurs only on a λ-set of $\sigma(\lambda)$-measure 0. Thus we must only prove that the set of all $\lambda \in r - \overline{M}$, $\Delta_{M(\lambda)}(E(\lambda)) = 0$, is $\sigma(\lambda)$-measurable.

Now $\Delta_{M(\lambda)}(E(\lambda)) = 0$ means $E(\lambda) = 0$, and this determines a $\sigma(\lambda)$-measurable set (cf. the first part of the proof of Lemma 17), and its intersection with $r - \overline{M}$ will therefore be $\sigma(\lambda)$-measurable too.

Thus we can restrict ourselves from now on to the $\lambda \in \overline{M}$.

Choose an arbitrary $d_{M(\lambda)}(E)$, an $\overline{E}(\lambda)$ by Lemma 19, and define

$$d^*_{M(\lambda)}(E) = a^*(\lambda) d_{M(\lambda)}(E), \qquad a^*(\lambda) = \begin{cases} \dfrac{1}{d_{M(\lambda)}(\overline{E}(\lambda))}, & \text{for } \lambda \in \overline{M}, \\ 1, & \text{for } \lambda \notin \overline{M}. \end{cases}$$

If $\lambda \in \overline{M}$, then

$$d^*_{M(\lambda)}(E(\lambda)) = \frac{d_{M(\lambda)}(E(\lambda))}{d_{M(\lambda)}(\overline{E}(\lambda))},$$

which is a $\sigma(\lambda)$-measurable function along with $E(\lambda)$, by Lemma 18. Thus $d^*_{M(\lambda)}(E(\lambda))$ is $\sigma(\lambda)$-measurable in $\overline{M}$, and therefore in all r, in other words $d^*_{M(\lambda)}(E)$ is $\sigma(\lambda)$-measurable, proving (i).

Any $\Delta_{M(\lambda)}(E)$ can be represented by (21.1) with some $a(\lambda) > 0, \leq +\infty$. Its $\sigma(\lambda)$-measurability means, that $\Delta_{M(\lambda)}(E(\lambda))$ is $\sigma(\lambda)$-measurable in all r for every $\sigma(\lambda)$-measurable $E(\lambda)$, but we know that it suffices to establish this in $\overline{M}$. Thus the $\sigma(\lambda)$-measurability of $a(\lambda)$ in $\overline{M}$ is sufficient. But it is necessary too: Put $E(\lambda) = \overline{E}(\lambda)$ in (21.1), for $\lambda \in \overline{M}$ we have $d^*_{M(\lambda)}(E(\lambda)) > 0, < +\infty$, and thus we can divide by it, obtaining

$$a(\lambda) = \frac{\Delta_{M(\lambda)}(\overline{E}(\lambda))}{d^*_{M(\lambda)}(\overline{E}(\lambda))},$$

which proves the $\sigma(\lambda)$-measurability of $a(\lambda)$ in $\overline{M}$. Thus (ii) too is established, and the proof is completed.—

It is worth emphasizing that Theorem VIII shows that the $a(\lambda)$ of a $\sigma(\lambda)$-measurable $\Delta_{M(\lambda)}(E)$ need only to be $\sigma(\lambda)$-measurable in $\overline{M}$, and not in all r! Now we remarked in the discussion which preceded Lemma 17 that (21.1) determines $a(\lambda)$ in $r - C_{III}$ uniquely, but that it leaves $a(\lambda)$ completely arbitrary in C_{III}. (C_{III} is the set of all λ's, for which $M(\lambda)$ belongs to the class III, cf. (10, Theorem VIII). Cf. also the notations introduced in §22, and in particular

in Definition 9.) Besides $\overline{M} \subset r - C_{III}$, because $\lambda \in \overline{M}$ implies $d_{M(\lambda)}(\overline{E}(\lambda)) > 0, < +\infty$, and therefore $\lambda \notin C_{III}$. Thus our result would be very natural, if $\overline{M}$ coincided with $r - C_{III}$ up to a set of $\sigma(\lambda)$-measure 0. But this, that is $\mu_{\sigma(\lambda)}(r - C_{III} - \overline{M}) = 0$, is unproved! In fact we cannot prove at present even the $\sigma(\lambda)$-measurability of C_{III} (which is equivalent to that one of $r - C_{III} - \overline{M}$). (Cf. Theorem IX and the observations following it.).

A natural attempt to prove $\mu_{\sigma(\lambda)}(r - C_{III} - \overline{M}) = 0$ would be this: If $\lambda \in r - C_{III}$, then a projection $E(\lambda) \in M(\lambda)$ with $d_{M(\lambda)}(E(\lambda)) > 0, < +\infty$ exists. If we could select such an $E(\lambda)$ for every $\lambda \in r - C_{III}$, and an arbitrary $E(\lambda)$ for every $\lambda \in C_{III}$, and if this could be done so as to make the resulting $E(\lambda)$ depend $\sigma(\lambda)$-measurably on λ, then we would have for this $E(\lambda)$, $M_r = r - C_{III}$. Then $M_r - M_r \cdot \overline{M} = r - C_{III} - \overline{M}$, and therefore Lemma 19 gives the desired $\mu_{\sigma(\lambda)}(r - C_{III} - \overline{M}) = 0$. But at present we are not able to construct such an $E(\lambda)$: The "axiom of choice" does not guarantee the $\sigma(\lambda)$-measurability, while the application of Lemma 5 (which would guarantee it) is impossible because the resulting set A turns out to be the complement of an analytic set, and not an analytic set. (Cf. Theorem IX and its proof. As to the importance of this difference cf. (14, pp. 194–195, 197–202, 267–273).)

22 Classification of the factor fibres

The classification of the factors will now be applied:

Definition 9. *A classification of factors was established in (10, Theorem VIII), distributing the factors M^* in a unitary space over the following classes:*

$$(I_m, I_n), \quad m, n = 1, 2, \ldots, \infty; \qquad (II_\alpha, II_\beta), \quad \alpha, \beta = 1, \infty; \qquad (III, III).$$

(This is really a classification for the "coupled factorizations" $M^*, M^{*'}$ of $\mathfrak{H}^*$, cf. (10, Definition 3.1.3), but we wish to use it for the factors M^* themselves.) Denote the set of these classes by $\mathfrak{C}$, and the general element of $\mathfrak{C}$ by c.

Let M be a ring (containing 1) in $\mathfrak{H}$. Decompose M with respect to $P = M \cdot M'$ in the sense of Theorem VII: $M \approx \sum M(\lambda)$. Each $M(\lambda)$ is a factor in its $\mathfrak{H}^{(\lambda)}$, and so it belongs to precisely one class $c \in \mathfrak{C}$. For a given $c \in \mathfrak{C}$, denote by C_c the set of all λ's for which $M(\lambda)$ is of class c.

As a decomposition $M \approx \sum M(\lambda)$ is only determined up to an arbitrary λ-set of $\sigma(\lambda)$-measure 0, therefore we have:

Each λ-set C_c ($c \in \mathfrak{C}$) is only determined up to an arbitrary λ-set of $\sigma(\lambda)$-measure 0.

The last statement of §19 implies:

If the representation of $\mathfrak{H}$ as a generalized direct sum (with the ring $P = M \cdot M'$) is replaced by an equivalent one, with the mapping $\lambda \rightleftarrows m(\lambda)$, then each set C_c is replaced by its image under this mapping. The same is true for the set $\overline{M}$ of Lemma 19.

The following abbreviations suggest themselves:

$$C_{I_m} = \sum_{n=1,2,\ldots,\infty} C_{(I_m,I_n)}, \qquad C_I = \sum_{m=1,2,\ldots,\infty} C_{I_m} = \sum_{m,n=1,2,\ldots,\infty} C_{(I_m,I_n)},$$

$$C_{II_\alpha} = \sum_{\beta=1,\infty} C_{(II_\alpha,II_\beta)}, \qquad C_{II} = \sum_{\alpha=1,\infty} C_{II_\alpha} = \sum_{\alpha,\beta=1,\infty} C_{(II_\alpha,II_\beta)},$$

$$C_{III} = C_{(III,III)}.$$

We define furthermore two sets C_f, C_i:

C_f is the set of all λ's for which 1 (in $\mathfrak{H}^{(\lambda)}$) is finite with respect to $M(\lambda)$; C_i is the set of all λ for which it is infinite.

Let $d_{M(\lambda)}(E)$ be $\sigma(\lambda)$-measurable, as the constant operator 1 (in $\mathfrak{H}^{(\lambda)}$) is $\sigma(\lambda)$-measurable, the (numerical) function $d_{M(\lambda)}(1)$ is $\sigma(\lambda)$-measurable. Thus the set of all λ's with $d_{M(\lambda)}(1) = \infty$ is $\sigma(\lambda)$-measurable, but this is C_i, and with it $C_f = r - C_i$ is $\sigma(\lambda)$-measurable too. So we have proved:

C_f and C_i are both $\sigma(\lambda)$-measurable.

The relations

$$C_f = \sum_{m=1,2,\ldots} C_{I_m} + C_{II_1}, \qquad C_i = C_{I_\infty} + C_{II_\infty} + C_{III}$$

are obvious.

Our next objective is to discuss the $\sigma(\lambda)$-measurability of the sets C_c for all $c \in \mathfrak{C}$. It is a remarkable feature of this theory that these $\sigma(\lambda)$-measurabilities are not at all obvious—in spite of the fact that each set C_c had been obtained by an effective construction. The reason for this strange phenomenon is that a direct discussion of these sets would characterize C_c, as well as $C_I + C_{II} = r - C_{III}$ as the projection of a complement of an analytic set (that is: a "projective set of the second order"), and the measurability properties of such sets are not known as yet.[17] (Cf. (14, pp. 267–273, 290–292).) We will be able, however, to prove by an indirect method the $\sigma(\lambda)$-measurability of all C_c, $c \neq (II_\infty, II_\infty), (III, III)$, and of $C_{(II_\infty,II_\infty)} + C_{(III,III)}$, but we are not able at present to separate $C_{(II_\infty,II_\infty)}$ and $C_{(III,III)}$!

The auxiliary considerations, which we must carry out for this purpose are as follows.

Lemma 20. *Let $M \approx \sum M(\lambda)$ be the decomposition of M with respect to any Abelian P. (Cf. Theorem VII.) Let M^* be a ring (containing 1) in a unitary space $\mathfrak{H}^*$. Denote by D_{M^*} the set of all those λ's for which an isomorphic mapping J^* of $\mathfrak{H}^*$ on $\mathfrak{H}^{(\lambda)}$ exists, which carries M^* into $M(\lambda)$. Then the set D_{M^*} is $\sigma(\lambda)$-measurable.*

Proof. Let p be the dimensionality of $\mathfrak{H}^*$, $p = 1, 2, \ldots, \infty$. Then $\lambda \in D_{M^*}$ necessitates $k(\lambda) = p$, $\lambda \in \Delta_p$. Proceed now as in the proofs of Lemmas 6, 17, and 18. Let $\psi_1^*, \psi_2^*, \ldots$ be a complete normalized orthogonal set in $\mathfrak{H}^*$, $\varphi_m^{(\lambda)}$ a measurable family, $J^{(\lambda)}$ an isomorphism of $\mathfrak{H}^{(\lambda)}$ on $\mathfrak{H}^*$ in which $\varphi_m^{(\lambda)}$ corresponds to ψ_m^*. Let

[17] This was written in 1938. In the meantime K. Gödel has shown that the (Lebesgue-) measurability of all "projective sets of the second order" is indemonstrable in the conventional system of mathematics. (Oral communication to the author.) The result is implicit in Gödel's work on "The consistency of the Axiom of choice and of the generalized continuum-Hypothesis with the Axioms of Set Theory", Princeton Lecture notes, 1938–1939. The non-measurability in question follows from the relation "$V = L$", cf. loc. cit., which Gödel proves to be irrefutable in his system.

the $A_q(\lambda)$, $q = 1, 2, \ldots$, be $\sigma(\lambda)$-measurable, such that $A_q(\lambda)$, $q = 1, 2, \ldots$, generate the ring $M(\lambda)$ and the $B_q(\lambda)$, $q = 1, 2, \ldots$, the same for $M(\lambda)'$. Let the $\overline{A}_1, \overline{A}_2, \ldots$ be operators in $\mathfrak{H}^*$ generating the ring M^*, and

$\overline{B}_1, \overline{B}_2, \ldots$ the ring $M^{*'}$. Choose a set $N_p \subset \Delta_p$ of $\sigma(\lambda)$-measure 0, as in the proofs quoted above, such that $\Delta_p' = \Delta_p - N_p$ is a Borel-set, and the $(A_q(\lambda)\varphi_m(\lambda), \varphi_n(\lambda))$, $(B_q(\lambda)\varphi_m(\lambda), \varphi_n(\lambda))$, $q, m, n = 1, 2, \ldots$, are Baire-functions.

Any isomorphic mapping J^* of $\mathfrak{H}^*$ on $\mathfrak{H}^{(\lambda)}$ may be put in the form $J^{(\lambda)-1}U$ where U is an isomorphic mapping of $\mathfrak{H}^*$ on itself, that is a unitary operator in $\mathfrak{H}^*$.

That J^* carries M^* into $M(\lambda)$ means that it carries M^* into a subset of $M(\lambda)$ and at the same time $M^{*'}$ into a subset of $M(\lambda)'$ (because the latter means that M^* is mapped in a set containing $M(\lambda)$), or: that the J^*-image of M^* commutes with $M(\lambda)'$ and that the J^*-image of $M^{*'}$ commutes with $M(\lambda)$. This is equivalent to this: that the J^*-image of every $\overline{A}_r$ commutes with every $B_q(\lambda)$, and similarly for every $\overline{B}_r$ and $A_q(\lambda)$, $r, q = 1, 2, \ldots$. If we now replace J^* by $J^{(\lambda)-1}U$ then this means: $U\overline{A}_rU^{-1}$ and $J^{(\lambda)}B_q(\lambda)J^{(\lambda)-1}$ commute, $U\overline{B}_rU^{-1}$ and $J^{(\lambda)}A_q(\lambda)J^{(\lambda)-1}$ commute, $r, q = 1, 2, \ldots$. Thus $\lambda \in D_{M^*}$ is equivalent to the following equations (the last line expresses the unitarity of U, and therefore it was permissible to replace U^{-1} by U^* in the previous ones):

$$\begin{cases} U\overline{A}_rU^*J^{(\lambda)}B_q(\lambda)J^{(\lambda)-1} = J^{(\lambda)}B_q(\lambda)J^{(\lambda)-1}U\overline{A}_rU^*, \\ U\overline{B}_rU^*J^{(\lambda)}A_q(\lambda)J^{(\lambda)-1} = J^{(\lambda)}A_q(\lambda)J^{(\lambda)-1}U\overline{B}_rU^*, \quad r, q = 1, 2, \ldots, \\ \qquad\qquad U^*U = UU^* = 1. \end{cases} \tag{22.1}$$

Observe that such a U is unitary, and so $\|U\| \leq 1$.

Consider now the set C_p of all pairs (λ, U) in $r \times S_p$, for which $\lambda \in \Delta_p'$ and λ, U fulfill (22.1). One proves by the same computations as in the proofs of Lemmas 17 and 18, that C_p is a Borel-set in $r \times S_p$. $\lambda \in \Delta_p' D_{M^*}$ means that there exists a U with $(\lambda, U) \in C_p$. From this we infer as in Lemmas 17 and 18 that $\Delta_p' D_{M^*}$ is $\sigma(\lambda)$-measurable.

As $D_{M^*} \subset \Delta_p$, therefore $D_{M^*} - \Delta_p' D_{M^*} = N_p D_{M^*} \subset N_p$ has the $\sigma(\lambda)$-measure 0, therefore D_{M^*} is $\sigma(\lambda)$-measurable along with $\Delta_p' D_{M^*}$. Thus the proof is completed.

We are now in the position to prove:

Theorem IX. *The following sets are $\sigma(\lambda)$-measurable: $C_{(I_m, I_n)}$, $m, n = 1, 2, \ldots, \infty$; $C_{(II_\alpha, II_\beta)}$, $\alpha, \beta = 1, \infty$, excepting however, $\alpha = \beta = \infty$; $C_{(II_\infty, II_\infty)} + C_{(III, III)}$. The question remains, however, open for $C_{(II_\infty, II_\infty)}$ and $C_{(III, III)}$ separately.*

Proof. Choose two $m, n = 1, 2, \ldots, \infty$, an mn-dimensional unitary space $\mathfrak{H}^*$, and in it a coupled factorization $M^*, M^{*'}$ of class (I_m, I_n). (Cf. (10, Definitions 3.1.1–3.1.3 and Theorem VIII).) We know from (10, Lemma 8.6.1) that $M(\lambda)$ in $\mathfrak{H}^{(\lambda)}$ is of class (I_m, I_n) if and only if an isomorphic mapping J^* of $\mathfrak{H}^*$ on $\mathfrak{H}^{(\lambda)}$ exists, which carries M^* into $M(\lambda)$. Thus we have by Lemma 20 $C_{(I_m,I_n)} = D_{M^*}$, and this proves that it is $\sigma(\lambda)$-measurable.

As we proved at the beginning of this section, C_f is $\sigma(\lambda)$-measurable too. As

$$C_f = \sum_{m,n=1,2,\ldots,\infty} C_{(I_m,I_n)} + C_{(II_1,II_1)} + C_{(II_1,II_\infty)},$$

these results imply the $\sigma(\lambda)$-measurability of $C_{(II_1,II_1)} + C_{(II_1,II_\infty)}$. If we replace M by M', then

$M(\lambda)$ and $M(\lambda)'$ are interchanged, and so $C_{(II_1,II_1)}, C_{(II_1,II_\infty)}$ become $C_{(II_1,II_1)}, C_{(II_\infty,II_1)}$. Thus $C_{(II_1,II_1)} + C_{(II_\infty,II_1)}$ is $\sigma(\lambda)$-measurable, too. Now we see that

$$\left(C_{(II_1,II_1)} + C_{(II_1,II_\infty)}\right)\left(C_{(II_1,II_1)} + C_{(II_\infty,II_1)}\right) = C_{(II_1,II_1)}$$

is $\sigma(\lambda)$-measurable, and with it

$$\left(C_{(II_1,II_1)} + C_{(II_1,II_\infty)}\right) - C_{(II_1,II_1)} = C_{(II_1,II_\infty)},$$

$$\left(C_{(II_1,II_1)} + C_{(II_\infty,II_1)}\right) - C_{(II_1,II_1)} = C_{(II_\infty,II_1)}.$$

We have established the $\sigma(\lambda)$-measurability of all C_c with $c \neq (II_\infty, II_\infty)$ and (III, III). This implies the $\sigma(\lambda)$-measurability of

$$r - \sum_{c\in\mathfrak{C},\ c\neq(II_\infty,II_\infty),(III,III)} C_c = C_{(II_\infty,II_\infty)} + C_{(III,III)}$$

too. Thus we have proved all statements of our theorem.

A direct treatment of $C_{(II_\infty,II_\infty)}$ or $C_{(III,III)}$ fails because it leads to projections of complements of analytic sets (that is: projections of complements of projections of Borel-sets), for which the problem of measurability is yet unsolved. (Cf. the remarks preceding Lemma 20.)

The difficulties in proving the $\sigma(\lambda)$-measurability of $C_{(II_\infty,II_\infty)}$ and $C_{(III,III)}$ are remarkable under several aspects. We do not know how real they are: (III, III) is the only class, which might be empty (cf. (10, Theorem XII)), and if this were the case, we would have $C_{(II_\infty,II_\infty)} = C_{(II_\infty,II_\infty)} + C_{(III,III)}$ and $C_{(III,III)} = 0$, so that both sets would be $\sigma(\lambda)$-measurable.[18] On the other hand, if we had simple normal forms for factors of class (II_∞, II_∞) or (III, III), as (10, Lemma 8.6.1) provided some for the classes (I_m, I_n), then we could use Lemma 20, to prove the $\sigma(\lambda)$-measurability of $C_{(II_\infty,II_\infty)}$ resp. $C_{(III,III)}$, and thus of both. This connects with (10, Problem 6 (following Theorem VIII)).

The connection of these sets with the set $\overline{M}$ of Lemma 19 is of interest, too. We have already proved $\overline{M} \subset r - C_{III} = r - C_{(III,III)}$. Consider now $E(\lambda) = 1$, for this $M_r = C_f$, and so by Lemma 19 $\mu_{\sigma(\lambda)}(C_f - C_f\overline{M}) = 0$. In other words: If we disregard a set of $\sigma(\lambda)$-measure 0, then $\overline{M} \supset C_f$. As $\overline{M}$ is only defined up to such a set, we may assume $\overline{M} \supset C_f$. One can prove furthermore that $\overline{M} \supset C_{(I_\infty,I_\infty)}$, and that $\overline{M}$ is unchanged, if we replace M by M'. We will not give the details here. Now we have $\overline{M} \supset C_f \supset C_{(I_m,I_\infty)}$, $m = 1, 2, \ldots$, and $\supset C_{(II_1,II_\infty)}$; and as we can replace M by M' without affecting $\overline{M}$, but thereby changing $C_{(I_m,I_\infty)}$ into $C_{(I_\infty,I_m)}$, and $C_{(II_1,II_\infty)}$ into $C_{(II_\infty,II_1)}$, this implies $\overline{M} \supset C_{(I_\infty,I_m)}$, $m = 1, 2, \ldots$, and $\supset C_{(II_\infty,II_1)}$. Thus we have

$$\begin{aligned}\overline{M} &\supset C_f + C_{(I_\infty,I_1)} + C_{(I_\infty,I_2)} + \cdots + C_{(I_\infty,I_\infty)} + C_{(II_\infty,II_1)}\\ &= r - \big(C_{(II_\infty,II_\infty)} + C_{(III,III)}\big).\end{aligned}$$

So we have

$$r - \big(C_{(II_\infty,II_\infty)} + C_{(III,III)}\big) \subset \overline{M} \subset r - C_{(III,III)}.$$

The two first sets are $\sigma(\lambda)$-measurable, while we do not know this about the third one; but we saw that there are reasons to surmise that the two last sets are identical, and they are at any rate similar under several aspects.

23 Integration of weight functions

Our information about the weight functions $\Delta_{M(\lambda)}(E)$ of the rings $M(\lambda)$ is now sufficiently complete to make it possible to discuss the weight function $\Delta_M(E)$ of M. We do, however, need one more lemma concerning the $M(\lambda)$, before we attack the main problem directly.

Lemma 21. *Let M be a ring (containing 1) in $\mathfrak{H}$. Decompose M with respect to $P = M \cdot M'$ in the sense of Theorem VII: $M \approx \sum M(\lambda)$ so that each $M(\lambda)$ is a factor in its $\mathfrak{H}^{(\lambda)}$.*

[18] This was written in 1938. In the meantime the existence of factors belonging to the class (III, III) has been established. Cf. J. v. Neumann: "On rings of operators. III", *Annals of Math.*, Vol. 41, pp. 94–161, 1940. Cf. in particular Theorem IX and §4.4, loc. cit.

Consider two projections $E, F \in M$ and their decompositions: $E \sim \sum E(\lambda)$, $F \sim \sum F(\lambda)$. Then there exist three partially isometric operators $V_j \in M$, $j = 1, 2, 3$, with the decompositions $V_j \sim \sum V_j(\lambda)$, and with the following properties:

(i) If $\lambda > -\infty, < +\infty$, and if there exists for this λ a partially isometric $V_1 \in M(\lambda)$ with the initial projection $E(\lambda)$ and with the final projection $F(\lambda)$, then $V_1(\lambda)$ is such a V_1. Otherwise $V_1(\lambda) = 0$.

(ii) If $\lambda > -\infty, < +\infty$, and if there exists for this λ a partially isometric $V_2 \in M(\lambda)$ with the initial projection $E(\lambda)$ and with a final projection $\leq F(\lambda)$, then $V_2(\lambda)$ is such a V_2. Otherwise $V_2(\lambda) = 0$.

(iii) If $\lambda > -\infty, < +\infty$, and if there exist for this λ two partially isometric $V_3, V_3' \in M(\lambda)$, both with the initial projection $E(\lambda)$, and with final projections which have the sum $E(\lambda)$, then $V_3(\lambda)$ is such a V_3. Otherwise $V_3(\lambda) = 0$.

Proof. We first construct $\sigma(\lambda)$-measurable $V_j(\lambda)$'s, $j = 1, 2, 3$, which satisfy respectively (i), (ii), (iii). It suffices, however, to do this within each set Δ_p, defined by $k(\lambda) = p$, for each $p = 1, 2, \ldots$—as we can then put together all these solutions, owing to $\sum_{p=1}^{\infty} \Delta_p = r$. Furthermore we may replace Δ_p by any $\Delta_p' = \Delta_p - N_p$ if $\mu_{\sigma(\lambda)}(N_p) = 0$: Because we can define the $V_j(\lambda)$ for $\lambda \in N_p$ by the axiom of choice, as $\mu_{\sigma(\lambda)}(N_p) = 0$, this does not affect their $\sigma(\lambda)$-measurability.

Now choose $\sigma(\lambda)$-measurable operators $A_q(\lambda)$, $q = 1, 2, \ldots$, generating $M(\lambda)'$. For each $p = 1, 2, \ldots, \infty$ proceed as in the proofs of Lemmas 17, 18, and 20: Choose N_p as a set with $\mu_{\sigma(\lambda)}(N_p) = 0$, such that $\Delta_p' = \Delta_p - N_p$ is a Borel-set, and the (numerical) functions $(E(\lambda)\varphi_m(\lambda), \varphi_n(\lambda))$, $(F(\lambda)\varphi_m(\lambda), \varphi_n(\lambda))$, $(A_q(\lambda)\varphi_m(\lambda), \varphi_n(\lambda))$, $q, m, n = 1, 2, \ldots$, are Baire-functions in Δ_p'. As there, introduce a p-dimensional unitary space $\mathfrak{H}_p$, in it a complete normalized orthogonal set $\psi_1', \psi_2', \ldots$, and (for $\lambda \in \Delta_p$) an isomorphism $J^{(\lambda)}$ of $\mathfrak{H}^{(\lambda)}$ on $\mathfrak{H}_p$ which carries $\varphi_m(\lambda)$ into ψ_m'.

Under these conditions our $V_j(\lambda)$'s are characterized by

$$\begin{gathered}(j = 1) \qquad V^*V = E(\lambda), \quad VV^* = F(\lambda), \\ VA_q(\lambda) = A_q(\lambda)V, \quad VA_q(\lambda)^* = A_q(\lambda)^*V \quad (q = 1, 2, \ldots),\end{gathered} \tag{23.1$_1$}$$

and

$$\begin{gathered}(j = 2) \qquad V^*V = E(\lambda), \quad VV^*F(\lambda) = VV^*, \\ VA_q(\lambda) = A_q(\lambda)V, \quad VA_q(\lambda)^* = A_q(\lambda)^*V \quad (q = 1, 2, \ldots),\end{gathered} \tag{23.1$_2$}$$

and

$$\begin{cases} (j=3) \qquad V^*V = V'^*V' = E(\lambda), \quad VV^* + V'V'^* = F(\lambda), \\ \qquad\qquad VA_q(\lambda) = A_q(\lambda)V, \quad VA_q(\lambda)^* = A_q(\lambda)^*V, \\ \qquad\qquad V'A_q(\lambda) = A_q(\lambda)V', \quad V'A_q(\lambda)^* = A_q(\lambda)^*V', \quad q = 1, 2, \ldots, \end{cases} \tag{23.1$_3$}$$

respectively. (We write V for $V_j(\lambda)$, $j = 1, 2, 3$, and V' for $V_3'(\lambda)$.)

Put now $\widetilde{V} = J^{(\lambda)}VJ^{(\lambda)-1}$, $\widetilde{V}' = J^{(\lambda)}V'J^{(\lambda)-1}$, then $\widetilde{V}, \widetilde{V}'$ are operators in $\mathfrak{H}_p$, and if they satisfy the equivalent of any of (23.1$_1$), (23.1$_2$), and (23.1$_3$), then they are partially isometric, implying $\|\widetilde{V}\| \le 1$, $\|\widetilde{V}'\| \le 1$. Denote the sets of all pairs $(\lambda, \widetilde{V})$ and triplets $(\lambda, \widetilde{V}, \widetilde{V}')$ fulfilling $\lambda \in \Delta_p'$ and (23.1$_1$), (23.1$_2$), and (23.1$_3$), respectively by G_p^1, G_p^2 and G_p^3. We prove as in the proofs of Lemmas 17, 18 and 20 that these are respectively Borel-sets in $r \times S_p$ and $r \times S_p \times S_p$. Now apply Lemma 5 to $S = r \times S_p$ and $r \times S_p \times S_p$, $A = G_p^1, G_p^2$ and G_p^3 and $F((\lambda, V)) = \lambda$ and $F((\lambda, V, V')) = \lambda$ respectively. K_j and $f_j(\lambda)$, $j = 1, 2, 3$, will result. For $j = 1, 2$, $f_j(\lambda) = (\lambda, \widetilde{V}_j(\lambda)) \in r \times S_p$; for $j = 3$, $f_3(\lambda) = (\lambda, \widetilde{V}_3(\lambda), \widetilde{V}_3'(\lambda)) \in r \times S_p \times S_p$, considering Lemma 5, (ii)$_2$. The set of all $\widetilde{V}$ with $\mathrm{Re}(\widetilde{V}\psi_m', \psi_n') > \alpha$ and similarly that one with $\mathrm{Im}(\widetilde{V}\psi_m', \psi_n') > \alpha$ is clearly an open set in $S = r \times S_p$ resp. $r \times S_p \times S_p$. Thus the sets of all λ's with $\mathrm{Re}(\widetilde{V}_j(\lambda)\psi_m', \psi_n') > \alpha$ or $\mathrm{Im}(\widetilde{V}_j(\lambda)\psi_m', \psi_n') > \alpha$ are $\sigma(\lambda)$-measurable, owing to Lemma 5, (ii)$_1$. Put $V_j^0(\lambda) = J^{(\lambda)-1}\widetilde{V}_j(\lambda)J^{(\lambda)}$, then the sets of all λ's with $\mathrm{Re}(V_j^0(\lambda)\varphi_m(\lambda), \varphi_n(\lambda)) > \alpha$ or with $\mathrm{Im}(V_j^0(\lambda)\varphi_m(\lambda), \varphi_n(\lambda)) > \alpha$ are $\sigma(\lambda)$-measurable, and thus $V_j^0(\lambda)$ is $\sigma(\lambda)$-measurable. Define now

$$V_j(\lambda) = \begin{cases} V_j^0(\lambda) & \text{for } \lambda \in K_j, \\ 0 & \text{for } \lambda \notin K_j, \end{cases} \qquad j = 1, 2, 3.$$

As K_j is $\sigma(\lambda)$-measurable by Lemma 5, (i), the $V_j(\lambda)$ are $\sigma(\lambda)$-measurable too.

One verifies immediately that the $V_j(\lambda)$, $j = 1, 2, 3$, fulfill the corresponding conditions (i), (ii), (iii) of our Lemma. Thus we have the desired $V_j(\lambda)$'s in each Δ_p', and therefore in r too. As $V_j(\lambda) \in M(\lambda)$ and as it depends $\sigma(\lambda)$-measurably on λ, we can form $V_j \sim \sum V_j(\lambda)$ by Definition 6. As all $V_j(\lambda)$'s are partially isometric, the V_j are also partially isometric. Thus the proof of our lemma is completed.—

Consider now a weight function $\Delta_M(E)$ with respect to M. For every $\sigma(\lambda)$-measurable set K form the function

$$1_K(\lambda) = \begin{cases} 1 & \text{for } \lambda \in K, \\ 0 & \text{for } \lambda \notin K, \end{cases}$$

and the operator 1_K (cf. §7). As $1_K(\lambda)^2 = 1_K(\lambda) = \overline{1_K(\lambda)}$, therefore 1_K is a projection, and by definition (in §11) $1_K \in P = M \cdot M'$. Thus for every $E \in M$, E and 1_K commute, and therefore $1_K E$ is a projection along with E. So we can form $\Delta_M(1_K E)$. We write provisionally

$$m(K) = \Delta_M(1_K E).$$

We enumerate these properties of $m(K)$:

(a) $m(K)$ is defined for every $\sigma(\lambda)$-measurable K, its values being ≥ 0, $\leq +\infty$.

(b) If a sequence $K_1, K_2, \ldots$ with $K_j K_k = 0$ for $j \neq k$ is given, then $1_{K_1+K_2+\cdots}(\lambda) = 1_{K_1}(\lambda) + 1_{K_2}(\lambda) + \cdots$, and so by Definition 7, (ii),

$$m(K_1 + K_2 + \cdots) = m(K_1) + m(K_2) + \cdots .$$

(c) If $\mu_{\sigma(\lambda)}(K) = 0$ then $1_K(\lambda)$ differs from 0 only on a λ-set of $\sigma(\lambda)$-measure 0, so $1_K = 0$, and thus $m(K) = 0$.

Consider those sets K for which $m(K)$ is finite, and let α be the l.u.b. of their $\mu_{\sigma(\lambda)}(K)$'s. For each $j = 1, 2, \ldots$ choose such a K_j with $\mu_{\sigma(\lambda)}(K_j) \geq \alpha - 1/j$. If $m(K)$ is finite and $K' \subset K$, then (β) gives $m(K') + m(K - K') = m(K)$, and so $m(K')$ is finite too; if $m(K')$, $m(K'')$ are finite, then $m(K' - K'K'')$ is finite, and so by (β)

$$\begin{aligned} m(K' + K'') &= m(K' + (K'' - K'K'')) \\ &= m(K') + m(K'' - K'K''). \end{aligned}$$

is finite too. Thus if we define $L_j = K_j - (K_1 + \cdots + K_{j-1})K_j$, then $m(L_j)$ is finite too; besides $L_j L_k = 0$ for $j \neq k$, and $L_1 + L_2 + \cdots = K_1 + K_2 + \cdots$. If $m(K)$ is finite, then $m(K_j + K)$ is finite too, so $\mu_{\sigma(\lambda)}(K_j + K) \leq \alpha$. Therefore

$$\begin{aligned} \mu_{\sigma(\lambda)}(K - K_j K) &= \mu_{\sigma(\lambda)}((K_j + K) - K_j) \\ &= \mu_{\sigma(\lambda)}(K_j + K) - \mu_{\sigma(\lambda)}(K_j) \\ &\leq \alpha - (\alpha - 1/j) = 1/j. \end{aligned}$$

and *a fortiori*

$$\begin{aligned} \mu_{\sigma(\lambda)}(K - (L_1 + L_2 + \cdots)K) &= \mu_{\sigma(\lambda)}(K - (K_1 + K_2 + \cdots)K) \\ &\leq \mu_{\sigma(\lambda)}(K - K_j K) \leq 1/j. \end{aligned}$$

As this holds for every $j = 1, 2, \ldots$, it gives $\mu_{\sigma(\lambda)}(K - (L_1 + L_2 + \cdots)K) = 0$. In particular: if $K \subset r - (L_1 + L_2 + \cdots) = L_\infty$ and $m(K)$ finite, then $\mu_{\sigma(\lambda)}(K) = 0$, and thus $m(K) = 0$. In other words:

(δ) For $K \subset L_\infty$ either $\mu_{\sigma(\lambda)}(K) = 0, m(K) = 0$, or $\mu_{\sigma(\lambda)}(K) > 0, m(K) = +\infty$.

Consider now $m(L_j K)$ for $j = 1, 2, \ldots$. It again fulfills (α), (β), (γ), but as $m(L_j K) \leq m(L_j K) + m(L_j - L_j K) = m(L_j) < +\infty$, more than ($\alpha$) holds:
(α') $m(L_j K)$ is defined for every $\sigma(\lambda)$-measurable K, its values being $\geq 0, \leq m(L_j) < +\infty$.

As $m(L_j K)$ fulfills (α'), (β), (γ), it is a non-negative and finite totally additive set-function, and absolutely continuous with respect to $\mu_{\sigma(\lambda)}(K)$. Therefore the theorem which holds for such functions applies to it: There exists a $\sigma(\lambda)$-measurable function $w_j(\lambda) \geq 0, < +\infty$, such that

$$m(L_j K) = \int_K w_j(\lambda)\, d\sigma(\lambda) \tag{23.2}$$

for every $\sigma(\lambda)$-measurable K. (Cf. (13, Chapter IX). The theorem is formulated there for common Lebesgue-measure only, but this restriction is unessential.) Equation (23.2) is proved for $j = 1, 2, \ldots$, but it holds for $j = \infty$ too, if we put

$$w_\infty(\lambda) = \begin{cases} +\infty & \text{for } \lambda \in L_\infty, \\ 0 & \text{for } \lambda \notin L_\infty, \end{cases}$$

considering that we know by (δ) that

$$m(L_\infty K) = \begin{cases} +\infty & \text{for } \mu_{\sigma(\lambda)}(L_\infty K) > 0, \\ 0 & \text{for } \mu_{\sigma(\lambda)}(L_\infty K) = 0. \end{cases}$$

Now put $\delta_M^{(\lambda)}(E) = \sum_{j=1}^{\infty} w_j(\lambda)$ (we

indicate in this way the dependence on λ, M, E), then summation of (23.2) over $j = 1, 2, \ldots, \infty$ gives, considering (β):

$$m(K) = \Delta_M(1_K(\lambda)E) = \int_K \delta_M^{(\lambda)}(E)\, d\sigma(\lambda). \tag{23.3}$$

For $E = 0$ we may redefine $\delta_M^{(\lambda)}(0) = 0$, without violating (23.3).

We now apply Definition 7, (ii), (iii) to (23.3). We use (ii) for two addends $E_1 = E, E_2 = F$ only (while $E_3 = E_4 = \cdots = 0$), and remember in (iii) that $1_K(\lambda) \in M \cdot M'$, $U \in M$ imply $U 1_K(\lambda) U^{-1} = 1_K(\lambda)$. Thus we obtain:

$$\int_K \delta_M^{(\lambda)}(E+F)\, d\sigma(\lambda) = \int_K \delta_M^{(\lambda)}(E)\, d\sigma(\lambda) + \int_K \delta_M^{(\lambda)}(F)\, d\sigma(\lambda) \quad \text{for } EF = 0,$$

$$\int_K \delta_M^{(\lambda)}(UEU^{-1})\,d\sigma(\lambda) = \int_K \delta_M^{(\lambda)}(E)\,d\sigma(\lambda),$$

for every $\sigma(\lambda)$-measurable set K. Therefore:

$$\begin{cases} \delta_M^{(\lambda)}(E+F) = \delta_M^{(\lambda)}(E) + \delta_M^{(\lambda)}(F), \quad EF = 0, \\ \delta_M^{(\lambda)}(UEU^{-1}) = \delta_M^{(\lambda)}(E), \end{cases} \tag{23.4}$$

except for certain λ-sets of $\sigma(\lambda)$-measure 0, which we will denote by $N_1(E,F)$ resp. $N_2(E,U)$.

We need this additional Lemma[19]:

Lemma 22: *Given two projections E, F, form*

$$G_n = F \text{ or } E \cdots EFE$$

(n factors, the first factor is F or E if n is even or odd, respectively), $n = 1, 2, \ldots$. Then

$$G = \operatorname*{strong\,lim}_{n\to\infty} G_n \tag{23.5}$$

exists. This G has the following properties:

(i) G is a projection.

(ii) For any projection H, $H \le G$ is equivalent to the conjunction of $H \le E$ and $H \le F$.

(iii) The closed linear set of G is the intersection of the closed linear sets of E and of F.

Denote this G by $E \# F$.

Proof: We note:

$$G_n \text{ is Hermitean if } n \text{ is odd.} \tag{I}$$

$$G_m^* G_n = G_p, \quad \text{where } p = m+n \text{ or } m+n-1, \text{ whichever is odd.} \tag{II}$$

$$G_{n+1} = \begin{cases} EG_n & \text{for } n \text{ even,} \\ FG_n & \text{for } n \text{ odd.} \end{cases} \tag{III}$$

From (II) with $m = n$

$$\|G_n f\|^2 = (G_{2n-1} f, f). \tag{IV}$$

(III) implies $\|G_{n+1} f\| \le \|G_n f\|$, hence (IV) gives $(G_{2n+1} f, f) \le (G_{2n-1} f, f) \ge 0$, i.e.

$$(G_1 f, f) \ge (G_3 f, f) \ge (G_5 f, f) \ge \cdots \ge 0. \tag{V}$$

[19] It is possible to avoid the use of this Lemma in the present context, but the Lemma has a certain interest of its own.

Consequently,

$$\lim_{n\to\infty}(G_{2n-1}f, f) \text{ exists and is finite.} \tag{VI}$$

Now by (II)

$$\begin{aligned}\|G_m f - G_n f\|^2 &= \|(G_m - G_n)f\|^2 \\ &= ((G_m - G_n)^*(G_m - G_n)f, f) \\ &= ((G_{2m-1} - 2G_p + G_{2n-1})f, f)\end{aligned}$$

(p according to (II)). Hence $m, n \to \infty$ and (VI) give

$$\lim_{m,n\to\infty}\|G_m f - G_n f\| = 0. \tag{VII}$$

This proves (23.5).

We now prove (i)–(iii):

$n \to \infty$ through even values and through odd values in n gives

$$G = EG = FG. \tag{VIII}$$

If $A = EA = FA$, then $A = G_n A$, hence $A = GA$:

$$A = EA = FA \quad \text{implies} \quad A = GA. \tag{IX}$$

Now $n \to \infty$ in (I) shows that G is Hermitean and (VIII) with (IX) for $A = G$ gives $G^2 = G$, hence G is a projection. This proves (i).

Now (VIII) with (IX) for projections A states precisely (ii).

Stating (ii) for the closed linear sets of the operators involved coincides with (iii).—

We infer immediately from Lemma 22:

(a) $E, F \in M$ imply $E \# F \in M$,

and replacing $\mathfrak{H}, M$ by $\mathfrak{H}^{(\lambda)}, M(\lambda)$:

(b) $E(\lambda), F(\lambda) \in M(\lambda)$ imply $E(\lambda) \# F(\lambda) \in M(\lambda)$,

(c) if $E(\lambda), F(\lambda)$ are $\sigma(\lambda)$-measurable, then $E(\lambda) \# F(\lambda)$ is also $\sigma(\lambda)$-measurable.

Now put $E \sim \sum E(\lambda)$, $F \sim \sum F(\lambda)$, and form $E_1 \sim \sum E(\lambda) \# F(\lambda)$. Applying Lemma 22 to $E(\lambda), F(\lambda)$ and to E, F then gives this: Consider a projection $H \in M$. Put $H \sim \sum H(\lambda)$. Then $H \leq E \# F$ means $H \leq E$ and $H \leq F$, hence $H(\lambda) \leq E(\lambda)$ and $H(\lambda) \leq F(\lambda)$ (except for a λ-set of $\sigma(\lambda)$-measure 0), hence $H(\lambda) \leq E(\lambda) \# F(\lambda)$ (except as above), hence $H \leq E_1$. Now $E_1 \in M$ by

its definition and $E \# F \in M$ by (a). Therefore, the equivalence of $H \leq E \# F$ and of $H \leq E_1$ for all projections $H \in M$ implies $E \# F = E_1$. We restate this

(d) $E \# F \sim \sum E(\lambda) \# F(\lambda)$.

The following further construction is necessary:

For every $A \in M$ form an operator $A_u \in M$ as follows: Form $A \sim \sum A(\lambda)$, then $A(\lambda) \in M(\lambda)$ and $\sigma(\lambda)$-measurable. Thus $1 - A(\lambda)^* A(\lambda)$ and $1 - A(\lambda) A(\lambda)^*$ are $\sigma(\lambda)$-measurable too, and therefore the set H_A where both are $= 0$ is $\sigma(\lambda)$-measurable, too. $\lambda \in H_A$ means that $A(\lambda)$ is unitary. Now define

$$A_u(\lambda) = \begin{cases} A(\lambda) & \text{for } \lambda \in H_A, \\ 1 & \text{for } \lambda \notin H_A, \end{cases}$$

then $A_u(\lambda)$ is $\sigma(\lambda)$-measurable, unitary (hence $\|A_u(\lambda)\| = 1$) and in $M(\lambda)$. Thus we can define $A_u \sim \sum A_u(\lambda)$, and we have: $A_u \in M$, A_u unitary, and $A_u(\lambda) = A(\lambda)$ for those λ's for which $A(\lambda)$ is unitary.

After these preparations we undertake the following construction: Choose an $E_0 \in M$. Take furthermore the projection $\overline{E} \in M$ from Lemma 19. Now form a sequence of operators $A_1, A_2, \ldots$ with the following properties:

(a) $1, \overline{E}, E_0$ are A_m's.

(b) $A_m^*, (A_m)_u, A_m \pm A_n, A_m A_n$ are A_p's again.

(c) If A_m, A_n are projections, then the V_1, V_2, V_3 of Lemma 21 (with $E = A_m, F = A_n$), and the $A_m \# A_n$ of Lemma 22 are A_p's again.

Choose for each A_m a fixed decomposition $A_m \sim \sum A_m(\lambda)$. Denote the projections and the unitary operators in the sequence $A_1, A_2, \ldots$ respectively by $E_1, E_2, \ldots$ and $U_1, U_2, \ldots$, and if A_m is E_n or U_n, denote $A_m(\lambda)$ by $E_n(\lambda)$ or $U_n(\lambda)$, respectively.

Let A_p be successively equal to each one of $1, A_m^*, (A_m)_u, A_m \pm A_n, A_m A_n, A_m \# A_n$, then we have correspondingly for all λ $A_p(\lambda) = 1, A_m(\lambda)^*, (A_m)_u(\lambda), A_m(\lambda) \pm A_n(\lambda), A_m(\lambda) A_n(\lambda), A_m(\lambda) \# A_n(\lambda)$, except for certain λ-sets of $\sigma(\lambda)$-measure 0, say $N_3, N_4(m), N_5(m), N_6(m,n), N_7(m,n), N_8(m,n), N_9(m,n)$. ($N_1, N_2$, or to be more specific, $N_1(E,F), N_2(E,U)$, were defined earlier, in this section, in connection with (23.4) further above.)

Compare now two projections E_m, E_n. Denote the set of all λ's with $E_m(\lambda) = E_n(\lambda)$, $\delta_M^{(\lambda)}(E_m) > \delta_M^{(\lambda)}(E_n)$, by $K_{m,n}$; it is clearly $\sigma(\lambda)$-measurable. Now $1_{K_{m,n}}(\lambda) E_m(\lambda) = 1_{K_{m,n}}(\lambda) E_n(\lambda)$ for all λ, so $1_{K_{m,n}}(\lambda) E_m = 1_{K_{m,n}}(\lambda) E_n$,

$$\Delta_M(1_{K_{m,n}}(\lambda) E_m) = \Delta_M(1_{K_{m,n}}(\lambda) E_n),$$

$$\int_{K_{m,n}} \delta_M^{(\lambda)}(E_m)\, d\sigma(\lambda) = \int_{K_{m,n}} \delta_M^{(\lambda)}(E_n)\, d\sigma(\lambda).$$

As $\delta_M^{(\lambda)}(E_m) > \delta_M^{(\lambda)}(E_n)$ for all $\lambda \in K_{m,n}$, this necessitates $\mu_{\sigma(\lambda)}(K_{m,n}) = 0$. Put $N_{10}(m,n) = K_{m,n} + K_{n,m}$, then $\mu_{\sigma(\lambda)}(N_{10}(m,n)) = 0$, and if $\lambda \notin N_{10}(m,n)$, then $E_m(\lambda) = E_n(\lambda)$ implies $\delta_M^{(\lambda)}(E_m) = \delta_M^{(\lambda)}(E_n)$.

Now put

$$\begin{aligned} N_{11} = & \sum_{m,n=1}^{\infty} N_1(E_m, E_n) + \sum_{m,n=1}^{\infty} N_2(E_m, U_n) + N_3 \\ & + \sum_{m=1}^{\infty} \big(N_4(m) + N_5(m)\big) + \sum_{m,n=1}^{\infty} \big(N_6(m,n) + \cdots + N_{10}(m,n)\big). \end{aligned}$$

Then $\mu_{\sigma(\lambda)}(N_{11}) = 0$ and $\lambda \notin N_{11}$ has the following consequences:

(a′) $\delta_M^{(\lambda)}(E_m + E_n) = \delta_M^{(\lambda)}(E_m) + \delta_M^{(\lambda)}(E_n)$ for $E_m E_n = 0$,

(b′) $\delta_M^{(\lambda)}(U_n E_m U_n^{-1}) = \delta_M^{(\lambda)}(E_m)$,

(c′) If A_p is successively equal to each one of $1, A_m^*, (A_m)_u, A_m \pm A_n, A_m A_n, A_m \# A_n$, then correspondingly for all λ $A_p(\lambda) = 1, A_m(\lambda)^*, (A_m)_u(\lambda), A_m(\lambda) \pm A_n(\lambda), A_m(\lambda)A_n(\lambda), A_m(\lambda) \# A_n(\lambda)$.

(d′) If $E_m(\lambda) = E_n(\lambda)$ (for one particular λ!), then $\delta_M^{(\lambda)}(E_m) = \delta_M^{(\lambda)}(E_n)$.

Now introduce a ($\sigma(\lambda)$-measurable) relative dimensionality function $d_{M(\lambda)}(E)$. Then we prove:

If E_m, E_n are given, then there exists a U_p with the following property. For every $\lambda \notin N_{11}$ for which $d_{M(\lambda)}(E_m(\lambda))$ is finite and $\leq d_{M(\lambda)}(E_n(\lambda))$, the projection $U_p(\lambda)E_m(\lambda)U_p(\lambda)^{-1}$ is $\leq E_n(\lambda)$.

Indeed if for a certain λ^*, $d_{M(\lambda)}(E_m(\lambda^*)) \leq d_{M(\lambda)}(E_n(\lambda^*))$, then a partially isometric $V \in M(\lambda)$ exists, which has the initial projection $E_m(\lambda)$ and a final projection $\leq E_n(\lambda)$. Thus if $A_r = V_2$ (in the sense of Lemma 21 for $E = E_m$ and $F = E_n$), then $A_r(\lambda)$ is such a V, and its final projection is $A_r(\lambda^*)A_r(\lambda^*)^*$. As A_r is partially isometric, therefore $A_r A_r^*$ is a projection. It is an A_l, and hence an E_q. Thus $A_r(\lambda^*)A_r(\lambda^*)^* = E_q(\lambda^*)$. Now $d_{M(\lambda)}(E_q(\lambda)) = d_{M(\lambda)}(E_m(\lambda))$ is finite, so $d_{M(\lambda)}(1 - E_q(\lambda)) = d_{M(\lambda)}(1 - E_m(\lambda))$. Therefore a partially isometric $W \in M(\lambda)$ exists, which has the initial and final projections $1 - E_m(\lambda)$ resp. $1 - E_q(\lambda)$. Therefore if $A_s = V_1$ (in the sense of Lemma 21 for $E = 1 - E_m$, $F = 1 - E_q$), then $A_s(\lambda)$ is such a W. Thus if $A_r + A_s = A_t$, then $A_t(\lambda) = A_r(\lambda) + A_s(\lambda)$ is a unitary operator, which maps $E_m(\lambda)$ on $E_q(\lambda) \leq E_n(\lambda)$. Now consider $(A_t)_u$, which is unitary, say U_p. As $A_t(\lambda)$ is unitary we have $U_p(\lambda) = (A_t)_u(\lambda) = A_t(\lambda)$. Thus U_p, E_q meet our requirements.

Next we prove:

$\lambda \notin N_{11}$ and $E_m(\lambda) \leq E_n(\lambda)$ imply $\delta_M^{(\lambda)}(E_m) \leq \delta_M^{(\lambda)}(E_n)$.

Indeed: Put $E_m \# E_n = E_p$, then $E_p(\lambda) = E_m(\lambda) \# E_n(\lambda) = E_m(\lambda)$, $\delta_M^{(\lambda)}(E_p) = \delta_M^{(\lambda)}(E_m)$. But $E_p \leq E_n$, so $E_n - E_p$ is a projection, say E_q. Thus $\delta_M^{(\lambda)}(E_n) = \delta_M^{(\lambda)}(E_p + E_q) = \delta_M^{(\lambda)}(E_p) + \delta_M^{(\lambda)}(E_q) \geq \delta_M^{(\lambda)}(E_p)$, proving $\delta_M^{(\lambda)}(E_m) \leq \delta_M^{(\lambda)}(E_n)$.

We now proceed to discuss the following question:

For which (real) numbers $\theta > 0$ does $\lambda \notin N_{11}$ and $d_{M(\lambda)}(E_m(\lambda)) \leq$ or $\geq \theta d_{M(\lambda)}(E_n(\lambda))$ ($d_{M(\lambda)}(E_m)$ or $d_{M(\lambda)}(E_n)$ being finite for $\leq$ or $\geq$, respectively) imply $\delta_M^{(\lambda)}(E_m) \leq$ resp. $\geq \theta\delta_M^{(\lambda)}(E_n)$?

We proceed stepwise:

(I) $\theta = 1$ is such. As $\leq$ and $\geq$ differ only by an interchange of m and n, we consider only $\leq$. Then this follows by combining our two above results.

(II) $1/\theta$ is such, along with θ. Obvious, as interchanging m and n, as well as $\leq$ and $\geq$, replaces θ by $1/\theta$.

(III) $\theta + 1$ is such along with θ. We have $d_{M(\lambda)}(E_m(\lambda)) \leq$ or $\geq (\theta + 1)d_{M(\lambda)}(E_n(\lambda))$, the finiteness restriction as above and wish to prove $\delta_M^{(\lambda)}(E_m) \leq$ resp. $\geq (\theta + 1)\delta_M^{(\lambda)}(E_n)$.

Assume first $d_{M(\lambda)}(E_m(\lambda)) < d_{M(\lambda)}(E_n(\lambda))$. This excludes the $\geq$ in the assumption, so it can only occur in the case with $\leq$, and then $\delta_M^{(\lambda)}(E_m)$ is finite.

Now we have $d_{M(\lambda)}(E_m(\lambda)) \leq d_{M(\lambda)}(E_n(\lambda))$, so by (I) $\delta_M^{(\lambda)}(E_m) \leq \delta_M^{(\lambda)}(E_n)$, and *a fortiori* $\leq (\theta + 1)\delta_M^{(\lambda)}(E_n)$, disposing of this case.

Assume next $d_{M(\lambda)}(E_m(\lambda)) \geq d_{M(\lambda)}(E_n(\lambda))$, which makes $d_{M(\lambda)}(E_n(\lambda))$ finite in any event. Applying our previous results, this implies the existence of a U_p with $U_p(\lambda)E_n(\lambda)U_p(\lambda)^{-1} \leq E_m(\lambda)$. (For this only: We have interchanged the roles of m and n.) Now form successively $E_r = U_pE_nU_p^{-1}$, $E_s = E_m \# E_r \leq E_m$, $E_t = E_m - E_s$. Then $d_{M(\lambda)}(E_m(\lambda)) = d_{M(\lambda)}(E_s(\lambda)) + d_{M(\lambda)}(E_t(\lambda))$, and therefore $\delta_M^{(\lambda)}(E_m) = \delta_M^{(\lambda)}(E_s) + \delta_M^{(\lambda)}(E_t)$. But as $E_s(\lambda) = E_r(\lambda) \# E_m(\lambda) = E_r(\lambda)$ (owing to $E_r(\lambda) = U_p(\lambda)E_n(\lambda)U_p(\lambda)^{-1} \leq E_m(\lambda)$), we may replace s by r in these equations; and as $E_r(\lambda) = U_p(\lambda)E_n(\lambda)U_p(\lambda)^{-1}$, $E_r = U_pE_nU_p^{-1}$, even by n. Thus we have

$$d_{M(\lambda)}(E_m(\lambda)) = d_{M(\lambda)}(E_t(\lambda)) + d_{M(\lambda)}(E_n(\lambda)), \quad \delta_M^{(\lambda)}(E_m) = \delta_M^{(\lambda)}(E_t) + \delta_M^{(\lambda)}(E_n).$$

As $d_{M(\lambda)}(E_n(\lambda))$ is finite, the first equation leads respectively to $d_{M(\lambda)}(E_t(\lambda)) \leq$ and $\geq \theta d_{M(\lambda)}(E_n(\lambda))$. As our question is assumed to have an affirmative answer for θ, this implies respectively $\delta_M^{(\lambda)}(E_t) \le$ and $\ge \theta\delta_M^{(\lambda)}(E_n)$ (apply to t, n instead of m, n, the finiteness of $d_{M(\lambda)}(E_m(\lambda))$ implies clearly that one of $d_{M(\lambda)}(E_t(\lambda))$ for $\le$), and now the second equation gives respectively $\delta_M^{(\lambda)}(E_m) \le$ and $\ge (\theta+1)\delta_M^{(\lambda)}(E_n)$, as desired.

(IV) All rational θ's are such. By (I), (III) every $\theta = k = 1, 2, \dots$ is such; by (II), (III) $k + 1/\theta$, $k = 0, 1, 2, \dots$, is such, along with θ. Thus all θ equal to a finite continued fraction

$$k_0 + 1/k_1 + 1/k_2 + \cdots + 1/k_\nu \quad (k_0 = 0, 1, 2, \dots, \quad k_1, \dots, k_\nu = 1, 2, \dots)$$

have the property in question. That is: All rational θ's have it.

(V) By continuity we can extend the property to all real $\theta > 0$, considering that (IV) stated it for all rational $\theta > 0$. Thus if $d_{M(\lambda)}(E_m(\lambda)), d_{M(\lambda)}(E_n(\lambda)) > 0$, $< +\infty$, we can put $\theta = d_{M(\lambda)}(E_m(\lambda))/d_{M(\lambda)}(E_n(\lambda))$ and apply $\le$ and $\ge$ simultaneously: This gives

$$\delta_M^{(\lambda)}(E_m) = \frac{d_{M(\lambda)}(E_m(\lambda))}{d_{M(\lambda)}(E_n(\lambda))}\delta_M^{(\lambda)}(E_n),$$

$$\frac{\delta_M^{(\lambda)}(E_m)}{d_{M(\lambda)}(E_m(\lambda))} = \frac{\delta_M^{(\lambda)}(E_n)}{d_{M(\lambda)}(E_n(\lambda))}.$$

In other words:

For all $\lambda \notin N_{11}$ we have

$$\frac{\delta_M^{(\lambda)}(E_m)}{d_{M(\lambda)}(E_m(\lambda))} = \gamma(\lambda) \qquad \textit{(independent of } m\textit{!)}, \tag{23.6}$$

whenever $d_{M(\lambda)}(E_m(\lambda)) > 0, < +\infty$.

$\gamma(\lambda)$ is ≥ 0, $\le +\infty$ and $\sigma(\lambda)$-measurable by its construction. Let $\overline{K}_0$ be the set of those λ, where $\gamma(\lambda) = 0$. If the projection $\overline{E}$ of Lemma 19 is substituted for E_m, then we have for $\lambda \in \overline{M}$ $d_{M(\lambda)}(\overline{E}(\lambda)) > 0, < +\infty$, so for $\lambda \in \overline{K}_0 \cdot \overline{M}$, we have $\delta_M^{(\lambda)}(\overline{E}) = 0$.

Thus

$$\Delta_M(1_{\overline{K}_0\cdot\overline{M}}(\lambda)\overline{E}) = \int_{\overline{K}_0\cdot\overline{M}} \delta_M^{(\lambda)}(\overline{E})\, d\sigma(\lambda) = 0,$$

and therefore $1_{\overline{K}_0\cdot\overline{M}}(\lambda)\overline{E} = 0$. But $1_{\overline{K}_0\cdot\overline{M}}(\lambda)\overline{E} \sim \sum 1_{\overline{K}_0\cdot\overline{M}}(\lambda)\overline{E}(\lambda)$, so $1_{\overline{K}_0\cdot\overline{M}}(\lambda)\overline{E}(\lambda) = 0$, except for a λ-set of $\sigma(\lambda)$-measure 0. Now for $\lambda \in \overline{K}_0 \cdot \overline{M}$,

$1_{\overline{K}_0 \cdot \overline{M}}(\lambda) = 1$, and as $\lambda \in \overline{M}$, $\overline{E}(\lambda) \neq 0$, therefore $1_{\overline{K}_0 \cdot \overline{M}}(\lambda)\overline{E}(\lambda) \neq 0$. All of this implies $\mu_{\sigma(\lambda)}(\overline{K}_0 \cdot \overline{M}) = 0$. Putting $N_{12} = N_{11} + \overline{K}_0 \cdot \overline{M}$, we still have $\mu_{\sigma(\lambda)}(N_{12}) = 0$, and we can state: For $\lambda \notin N_{12}$, $\lambda \in \overline{M}$ implies $\gamma(\lambda) \notin \overline{K}_0$, that is $\gamma(\lambda) > 0$. So $0 < \gamma(\lambda) \leq +\infty$, and as $0 < d_{M(\lambda)}(E_m(\lambda)) \leq +\infty$ too (if $E_m(\lambda) \neq 0$), we can write for (23.6):

$$\delta_M^{(\lambda)}(E_m) = \gamma(\lambda) d_{M(\lambda)}(E_m(\lambda)) \quad \text{if } \lambda \notin N_{12},\ \lambda \in \overline{M}, \qquad 0 < \gamma(\lambda) \leq +\infty. \tag{23.7}$$

Here $d_{M(\lambda)}(E_m(\lambda)) = 0$ or $= +\infty$ is to be excluded. But $d_{M(\lambda)}(E_m(\lambda)) = 0$ means $E_m(\lambda) = 0$, and so $\delta_M^{(\lambda)}(E_m) = \delta_M^{(\lambda)}(0) = 0$, so that (23.7) holds in this case too, if we use again the convention $0 \cdot +\infty = 0$. If $d_{M(\lambda)}(E_m(\lambda)) = +\infty$, then we have $d_{M(\lambda)}(E_m(\lambda)) \geq \theta d_{M(\lambda)}(\overline{E}(\lambda))$ for all $\theta > 0$, so

$$\delta_M^{(\lambda)}(E_m) \geq \theta \delta_M^{(\lambda)}(\overline{E}) = \theta \gamma(\lambda) d_{M(\lambda)}(\overline{E}(\lambda))$$

for all $\theta > 0$. As the second and third factors are > 0, this gives $\delta_M^{(\lambda)}(E_m) = +\infty$, and thus (23.7) holds again. Thus (23.7) holds for all E_m's.

(VI) We can use (23.7) to express $\gamma(\lambda)$ in $\overline{M}$, if we substitute the projection $\overline{E}$ of Lemma 19 for E_m. Then we see: $\gamma(\lambda)$ is a $\sigma(\lambda)$-measurable function in $\overline{M}$, and, disregarding a λ-set of $\sigma(\lambda)$-measure 0, we have:

$$\gamma(\lambda) = \frac{\delta_M^{(\lambda)}(\overline{E})}{d_{M(\lambda)}(\overline{E}(\lambda))} \qquad \text{for } \lambda \in \overline{M}. \tag{23.7$_1$}$$

(VII) We next discuss the behavior of $\delta_M^{(\lambda)}(E_m)$ for $\lambda \notin \overline{M}$. We again assume $\lambda \notin N_{12}$, too. We know from Lemma 19 that the set of all $\lambda \notin \overline{M}$ with $d_{M(\lambda)}(E_m(\lambda)) > 0, < +\infty$, say $N_{13}(m)$, has the $\sigma(\lambda)$-measure 0. So if we put $N_{14} = N_{12} + \sum_{m=1}^{\infty} N_{13}(m)$, then $\mu_{\sigma(\lambda)}(N_{14}) = 0$, and we see: If $\lambda \notin N_{14}$ and $\lambda \notin \overline{M}$, then either $E_m(\lambda) = 0$ (that is $d_{M(\lambda)}(E_m(\lambda)) = 0$) or $E_m(\lambda)$ is infinite (with respect to $M(\lambda)$, that is $d_{M(\lambda)}(E_m(\lambda)) = +\infty$). In both cases there are two partially isometric $V, V' \in M(\lambda)$, which have both the initial projection $E_m(\lambda)$, and final projections with the sum $E_m(\lambda)$: For $E_m(\lambda) = 0$ these are $0, 0$; for $E_m(\lambda)$ infinite apply (10, Lemma 7.2.3). Now form $A_p = V_3$ (in the sense of Lemma 21 for $E = E_m$). As A_p is partially isometric, $A_p A_p^*$ is a projection, say E_q. Thus $E_q(\lambda) = A_p(\lambda)A_p(\lambda)^*$ is $\sim E_m(\lambda) \cdots \bmod M(\lambda)$ and so is $E_m(\lambda) - E_q(\lambda)$. Furthermore $E_q \# E_m$ is a projection say E_r, and $\leq E_m$, thus $E_m - E_r$ too is a projection, say E_s. Now $E_q(\lambda) \leq E_m(\lambda)$, therefore $E_q(\lambda) \# E_m(\lambda) = E_q(\lambda)$, therefore $E_s(\lambda) = E_m(\lambda) - E_q(\lambda)$. Thus $E_s(\lambda)$ too is $\sim E_m(\lambda) \cdots \bmod M(\lambda)$. So we see:

$$E_m = E_q + E_s, \qquad E_m(\lambda) \sim E_q(\lambda) \sim E_s(\lambda) \cdots \bmod M(\lambda).$$

Similarly we can construct for another E_n, two E_t, E_w with

$$E_n = E_t + E_w, \qquad E_n(\lambda) \sim E_t(\lambda) \sim E_w(\lambda) \cdots \bmod M(\lambda).$$

Thus if $E_m(\lambda), E_n(\lambda) \neq 0$, then both are infinite, and with them $E_q(\lambda), E_s(\lambda), E_t(\lambda), E_w(\lambda)$ as well as $1 - E_q(\lambda) \geq E_s(\lambda)$ and $1 - E_t(\lambda) \geq E_w(\lambda)$ are infinite, too. Thus partially isometric W, W''s with the initial projections $E_q(\lambda)$,
$1 - E_q(\lambda)$ and the final projections $E_t(\lambda)$, $1 - E_t(\lambda)$ exist. If A_x, A_y are the resp. V_1 (in the sense of Lemma 21, with $E = E_q$ resp. $1 - E_q$, and $F = E_t$ resp. $1 - E_t$), then $A_x(\lambda), A_y(\lambda)$ are such W, W''s, and thus if $A_z = A_x + A_y$, then $A_z(\lambda) = A_x(\lambda) + A_y(\lambda)$ is unitary, and maps $E_q(\lambda)$ on $E_t(\lambda)$. $(A_z)_u$ is unitary, say U_a, and as $A_z(\lambda)$ is unitary, $U_a(\lambda) = (A_z)_u(\lambda) = A_z(\lambda)$. So we have $U_a(\lambda)E_q(\lambda)U_a(\lambda)^{-1} = E_t(\lambda)$. Put $U_aE_qU_a^{-1} = E_b$, then $E_b(\lambda) = E_t(\lambda)$. So we have: $\delta_M^{(\lambda)}(E_q) = \delta_M^{(\lambda)}(E_b) = \delta_M^{(\lambda)}(E_t)$. Similarly $\delta_M^{(\lambda)}(E_s) = \delta_M^{(\lambda)}(E_w)$. Thus

$$\begin{aligned}\delta_M^{(\lambda)}(E_m) &= \delta_M^{(\lambda)}(E_q + E_s) = \delta_M^{(\lambda)}(E_q) + \delta_M^{(\lambda)}(E_s)\\ &= \delta_M^{(\lambda)}(E_t) + \delta_M^{(\lambda)}(E_w) = \delta_M^{(\lambda)}(E_t + E_w) = \delta_M^{(\lambda)}(E_n).\end{aligned}$$

In other words: $\delta_M^{(\lambda)}(E_m)$ has the same value for all $E_m(\lambda) \neq 0$, say δ. Now this applies to E_m, E_q, E_s also. Then $\delta_M^{(\lambda)}(E_m) = \delta_M^{(\lambda)}(E_q) + \delta_M^{(\lambda)}(E_s)$, $\delta = 2\delta$, so that is: $\delta = 0$ or $+\infty$. As $1 \neq 0$, these alternatives are equivalent to $\delta_M^{(\lambda)}(1) = 0$ or $+\infty$.
Denote the set of all λ's where $\delta_M^{(\lambda)}(1) = 0$ by $\overline{K}_1$, it is $\sigma(\lambda)$-measurable along with $\delta_M^{(\lambda)}(1)$. Then (23.3) gives

$$\Delta_M(1_{\overline{K}_1}(\lambda)) = \int_{\overline{K}_1} \delta_M^{(\lambda)}(1)\, d\sigma(\lambda) = 0, \qquad 1_{\overline{K}_1}(\lambda) = 0,$$

and therefore $\mu_{\sigma(\lambda)}(\overline{K}_1) = 0$. Put $N_{15} = N_{14} + \overline{K}_1$, then $\mu_{\sigma(\lambda)}(N_{15}) = 0$, and we have:

$$\delta_M^{(\lambda)}(E_m) = \begin{cases} +\infty & \text{for } E_m(\lambda) \neq 0, \\ 0 & \text{for } E_m(\lambda) = 0, \end{cases} \qquad \text{if } \lambda \notin N_{15},\ \lambda \notin \overline{M}.$$

In other words: (23.7) holds again (with N_{15} instead of N_{12}), but now we have

$$\gamma(\lambda) = +\infty \qquad \text{for } \lambda \notin \overline{M}. \tag{23.7$_2$}$$

Putting (23.6) and (23.3) together we obtain

$$\Delta_M(1_K(\lambda)E_m) = \int_K \gamma(\lambda) d_{M(\lambda)}(E_m(\lambda))\, d\sigma(\lambda). \tag{A}$$

And substituting E_0 for E_m, and specializing K to r (as $1_r(\lambda) = 1$):

$$\Delta_M(E_0) = \int \gamma(\lambda) d_{M(\lambda)}(E_0(\lambda))\, d\sigma(\lambda). \tag{B}$$

The entire construction which lead to (B) depends on the projection $E_0 \in M$. But the result does not: (B) expresses the given weight-function $\Delta_M(E)$ (for $E = E_0$) by means of the arbitrarily (but $\sigma(\lambda)$-measurably) prescribed $\sigma(\lambda)$-measurable relative dimensionality function $d_{M(\lambda)}(E)$, with the sole help of the $\sigma(\lambda)$-measurable function $\gamma(\lambda) > 0, \leq +\infty$. And this $\gamma(\lambda)$ is defined by (23.7$_1$) and (23.7$_2$), in which E_0 does not enter. Thus $\gamma(\lambda)$ is a fixed function, and (B) holds for all projections $E_0 \in M$ (with the same $\gamma(\lambda)$).

We can now prove easily:

Theorem X. *If $\Delta_{M(\lambda)}(E)$ is a weight function with respect to $M(\lambda)$, depending $\sigma(\lambda)$-measurably on λ, then*

$$\Delta_M(E) = \int \Delta_{M(\lambda)}(E(\lambda))\, d\sigma(\lambda), \qquad \textit{where } E \sim \sum E(\lambda), \tag{23.8}$$

is a weight function with respect to M. Conversely, all weight functions $\Delta_M(E)$ (with respect to M) can be obtained by this process.

Proof: One verifies easily the conditions (i)–(iii) of Definition 7 for

$\Delta_M(E)$ (and M) if they do hold for all $\Delta_{M(\lambda)}(E)$ (and $M(\lambda)$). This proves the first statement.

The second statement follows from the considerations which preceded this theorem, particularly from the final formula (B) and the comments following it. We need only to put $\Delta_{M(\lambda)}(E) = \gamma(\lambda) d_{M(\lambda)}(E)$, and remember Theorem VIII. Thus the proof is complete.

24 Consequences for weights and equivalence

Theorem X permits some simple inferences which have an interest of their own, and which we are going to discuss now.
(α) Combine the results of Theorems VIII and X. Then we find: The formula

$$\Delta_M(E) = \int a(\lambda) d_{M(\lambda)}(E(\lambda))\, d\sigma(\lambda) \qquad \text{for } E \sim \sum E(\lambda) \tag{24.1}$$

expresses all weight functions $\Delta_M(E)$ (with respect to M), if $a(\lambda)$ runs over all those functions which fulfill $0 < a(\lambda) \leq +\infty$ everywhere, and which are $\sigma(\lambda)$-measurable in $\overline{M}$. It is natural to ask, how far $\Delta_M(E)$ determines $a(\lambda)$.

If $K \subset \overline{M}$ and $\sigma(\lambda)$-measurable, then

$$\Delta_M(1_K(\lambda)E) = \int_K a(\lambda) d_{M(\lambda)}(E(\lambda))\, d\sigma(\lambda),$$

thus $\int_K a(\lambda) d_{M(\lambda)}(\overline{E}(\lambda))\, d\sigma(\lambda)$ is determined for all these sets K. Thus $a(\lambda) d_{M(\lambda)}(\overline{E}(\lambda))$ is uniquely determined in $\overline{M}$, up to a set of $\sigma(\lambda)$-measure 0; and as $d_{M(\lambda)}(\overline{E}(\lambda)) > 0, < +\infty$ for all $\lambda \in \overline{M}$, therefore $a(\lambda)$ is determined to the same extent. In $r - \overline{M}$ we have by Lemma 19 always $d_{M(\lambda)}(\overline{E}(\lambda)) = 0$ or $+\infty$, except for a set of $\sigma(\lambda)$-measure 0, and therefore the numerical value of $a(\lambda)$ (which must be $> 0, \leq +\infty$) does not matter at all. So we see:

$a(\lambda)$ may be any function which fulfills $0 < a(\lambda) \leq +\infty$ and which is $\sigma(\lambda)$-measurable in $\overline{M}$. $\Delta_M(E)$ determines $a(\lambda)$ uniquely in $\overline{M}$, up to a set of $\sigma(\lambda)$-measure 0, while it leaves it entirely arbitrary in $r - \overline{M}$.

The same observation, as the one we made after Theorem VIII in §21, applies here: While it is obvious that the numerical value of $a(\lambda)$ is irrelevant if $\lambda \in C_{III} = C_{(III,III)}$ (because then $d_{M(\lambda)}(E) = 0$ or $+\infty$ for all $E \in M(\lambda)$), it is surprising, that this is the case even for $\lambda \in r - \overline{M}$, where $r - \overline{M} \subset C_{(III,III)}$. We discussed the relationship of $r - \overline{M}$ and $C_{(III,III)}$ loc. cit.
(β) We have for every partially isometric $V \in M(\lambda)$ $d_{M(\lambda)}(V^*V) = d_{M(\lambda)}(VV^*)$, by (10, Definition 8.2.1 and Theorem VIII). Formula (24.1) extends this at once to all partially isometric $V \in M$: Then $V \sim \sum V(\lambda)$, $V(\lambda) \in M(\lambda)$, $V(\lambda)$ partially isometric, thus $d_{M(\lambda)}(V(\lambda)^*V(\lambda)) = d_{M(\lambda)}(V(\lambda)V(\lambda)^*)$, and so by (24.1) $\Delta_M(V^*V) = \Delta_M(VV^*)$. Thus the condition

(iii)′ for every partially isometric $V \in M$, $\Delta_M(V^*V) = \Delta_M(VV^*)$,

follows from Definition 7, (i)–(iii). It clearly implies (iii). It suffices to put, as at the beginning of the proof of Lemma 16, $V = UE$. Then $V^* = E^*U^* = EU^{-1}$, and so $V^*V = E$, $VV^* = UEU^{-1}$, giving (iii). So we see:

Condition (iii) in Definition 7 is equivalent to (iii)′ (conserving (i), (ii)).

Another obvious way to formulate (iii)′ is this:

(iii)″ If E, F are the initial resp. final projections of a partially isometric

$V \in M$ (that is: if V maps their resp. linear, closed sets on each other; or, using the notations of (10, Definition 6.1.1): if $E \sim F \cdots \mathrm{mod}\ M$), then $\Delta_M(E) = \Delta_M(F)$.

(γ) We observed in the Remark following Lemma 16, that for factors the condition (ii) of Definition 7 can be replaced by the weaker condition (ii)′: (ii)′ arises from (ii) by permitting two addends E_j ($j = 1, 2$) only. For a general M this is not the case, that is: (ii)′ is then essentially weaker than (ii). This appears if we consider the opposite extreme to a factor M: an Abelian ring M.

The Abelian character of M implies: $M \subset M'$, $M = M \cdot M' = P \approx \sum 0(\lambda)$ (cf. the remark in the proof of Theorem VII); conversely if $M \approx \sum 0(\lambda)$, then M is Abelian as all $0(\lambda)$'s are Abelian. $0(\lambda)$ is obviously the only factor of class C_{I_1}, so we see:

M is Abelian, if and only if $C_{I_1} = r$ (up to a set of $\sigma(\lambda)$-measure 0, as usual).

Now the general element of M is (as $M = P$) $\alpha(\lambda)$, and this is a projection if $\alpha(\lambda)^2 = \alpha(\lambda) = \overline{\alpha(\lambda)}$, that is if $\alpha(\lambda) = 0$ or 1. In other words:

$$\alpha(\lambda) = 1_K(\lambda) = \begin{cases} 1 & \text{for } \lambda \in K, \\ 0 & \text{for } \lambda \notin K, \end{cases}$$

where K is any $\sigma(\lambda)$-measurable set. Thus a $\Delta_M(E)$, $E \in M$, is defined for the $E = 1_K(\lambda)$ only, and so we may write: $\Delta_M(1_K(\lambda)) = n(K) \geq 0, \leq +\infty$. Now the conditions of Definition 7 become:

(i): $n(K)$ depends on $1_K(\lambda)$ only, that is, it is unchanged, if K is changed by a set of $\sigma(\lambda)$-measure 0, and $n(K) = 0$ is equivalent to $1_K(\lambda) = 0$, that is, to $\mu_{\sigma(\lambda)}(K) = 0$.

(ii) $n(K_1 + K_2 + \cdots) = n(K_1) + n(K_2) + \cdots$ if $K_jK_k = 0$ for $j \neq k$.

(ii)′ $n(K_1 + K_2) = n(K_1) + n(K_2)$ for $K_1K_2 = 0$.

(iii) Void.

Thus the problem of finding a weight-function coincides in the case of an Abelian M with the problem of finding a Lebesgue-measure, which is absolutely continuous with respect to the $\sigma(\lambda)$-measure, and conversely. (Cf. the notion of equivalence in §2, (δ).) Now it is well known that the "countable additivity" postulated by (ii) is an essential feature of Lebesgue-measure, and cannot be replaced by the "finite additivity" which (ii)′ expresses.[20]

(δ) We can answer the following question:

If E, F are projections in M, when does a partially isometric $V \in M$ with the initial projection E and a final projection $\leq F$ exist?

In the notation of (10, Definition 6.3.1), this means: When is $E \lesssim F \cdots \bmod M$? Loc. cit. the problem was solved for factors. Then the relative dimensionality function $d_M(G)$ gives the answer, $E \lesssim F \cdots \bmod M$ being equivalent to $d_M(E) \leq d_M(F)$. We will now find the answer for a general M.

Our question is: When does a $V \in M$ with $V^*V = E$, $VV^* \leq F$ exist? Putting $E \sim \sum E(\lambda)$, $F \sim \sum F(\lambda)$, $V \sim \sum V(\lambda)$, we see that we need a partially isometric $V(\lambda) \in M(\lambda)$, depending $\sigma(\lambda)$-measurably on λ, with $V(\lambda)^*V(\lambda) = E(\lambda)$, $V(\lambda)V(\lambda)^* \leq F(\lambda)$ for each λ (except, of course, for a λ-set of $\sigma(\lambda)$-measure 0). Thus $d_{M(\lambda)}(E(\lambda)) \leq d_{M(\lambda)}(F(\lambda))$ (with the same exceptions) is necessary. But it is sufficient, too, because it involves for each λ (with the same exceptions as above) the existence of a partially isometric $V' \in M(\lambda)$ with $V'^*V' = E(\lambda)$, $V'V'^* \leq F(\lambda)$, and then the V_2 of Lemma 21 has the desired properties: $V = V_2$, $V \sim \sum V(\lambda)$.

So we have proved:

The necessary and sufficient condition is this: For all λ, except for a set of $\sigma(\lambda)$-measure 0, $d_{M(\lambda)}(E(\lambda)) \leq d_{M(\lambda)}(F(\lambda))$, where $E \sim \sum E(\lambda)$, $F \sim \sum F(\lambda)$.

We saw in (γ) above that the projections $\widetilde{E} \in P$ coincide with the $\widetilde{E} = 1_K(\lambda)$, K

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[20] This is even not possible in the totally discontinuous case, discussed in §4, where the points $\lambda_1, \lambda_2, \ldots$ (a finite or infinite sequence) have the $\sigma(\lambda)$-measures $a_1, a_2, \ldots$ (all > 0, $\sum a_\nu$ finite), and $r - (\lambda_1, \lambda_2, \ldots)$ has the measure 0: We assume that the sequence $\lambda_1, \lambda_2, \ldots$ is infinite. Now a construction of Banach permits to define a function $n'(L) \geq 0, \leq 1$ for all sets $L \subset (1, 2, \ldots)$ with the following properties:

$$n'(L) = 0 \text{ if } L \text{ is finite}; \qquad n'(1, 2, \ldots) = 1; \qquad n'(L_1 + L_2) = n'(L_1) + n'(L_2) \text{ if } L_1L_2 = 0.$$

Then denote for each $K \subset r$ the set of all $\nu = 1, 2, \ldots$ with $\lambda_\nu \in K$ by L_K, and define

$$n(K) = \sum_{\gamma \in L_K} \frac{1}{2^\gamma} + n'(L_K).$$

This $n(K)$ fulfills (i), (ii)′ ((iii) is void), but it violates (ii), because

$$\sum_{\nu=1}^{\infty} n(\lambda_\nu) = \sum_{\nu=1}^{\infty} \frac{1}{2^\nu} = 1, \qquad n(\lambda_1, \lambda_2, \ldots) = 2.$$

any $\sigma(\lambda)$-measurable set,

$$1_K(\lambda) = \begin{cases} 1 & \text{for } \lambda \in K, \\ 0 & \text{for } \lambda \notin K. \end{cases}$$

(The considerations there assumed that M is Abelian, but that part which dealt with P only made no use of this.) Hence $E \sim \sum E(\lambda)$ implies $\widetilde{E}E \sim \sum 1_K(\lambda)E(\lambda)$. Given any weight function $\Delta_M(G)$ and using the formula (24.1), we have therefore

$$\Delta_M(\widetilde{E}E) = \int_K a(\lambda) d_{M(\lambda)}(E(\lambda))\, d\sigma(\lambda). \tag{24.2}$$

Applying (24.2) to E and to F, our above condition, i.e. $d_{M(\lambda)}(E(\lambda)) \leq d_{M(\lambda)}(F(\lambda))$ for all λ excepting a set of $\sigma(\lambda)$-measure 0, is seen to imply

$$\begin{cases} \Delta_M(\widetilde{E}E) \leq \Delta_M(\widetilde{E}F) \\ \text{for all weight functions } \Delta_M(G) \text{ and all projections } \widetilde{E} \in P. \end{cases} \tag{24.3}$$

Consider the converse: (24.3) for all projections $\widetilde{E} \in P$, and we may even restrict it to one $\Delta_M(G)$, provided that the $a(\lambda)$ of (24.1) is always finite. Then (24.2) for E and F gives

$$\int_K a(\lambda) d_{M(\lambda)}(E(\lambda))\, d\sigma(\lambda) \leq \int_K a(\lambda) d_{M(\lambda)}(F(\lambda))\, d\sigma(\lambda)$$

for every $\sigma(\lambda)$-measurable set K. Now let K be the set of all λ with

$$d_{M(\lambda)}(E(\lambda)) > d_{M(\lambda)}(F(\lambda)).$$

Since $a(\lambda)$ is always finite, this implies

$$\int_K a(\lambda) d_{M(\lambda)}(E(\lambda))\, d\sigma(\lambda) > \int_K a(\lambda) d_{M(\lambda)}(F(\lambda))\, d\sigma(\lambda)$$

provided that $\mu_{\sigma(\lambda)}(K) > 0$. This, however, is a contradiction, hence $\mu_{\sigma(\lambda)}(K) = 0$. That is, $d_{M(\lambda)}(E(\lambda)) \leq d_{M(\lambda)}(F(\lambda))$, except for a set of $\sigma(\lambda)$-measure 0. This is our above condition.

We assumed that $a(\lambda)$ is always finite. We know, however, that changing $a(\lambda)$ on a set of $\sigma(\lambda)$-measure 0 or outside $\overline{M}$ does not matter. (Concerning the latter, cf. (α) above.)

So we have obtained:

Another necessary and sufficient condition is this (24.3) *above. It is even sufficient to postulate* (24.3) *for one given weight function* $\Delta_M(G)$ *only, provided that the* $a(\lambda)$ *of* (24.2) *is finite for all* $\lambda \in \overline{M}$*, except for a* λ*-set of* $\sigma(\lambda)$*-measure* 0*.*

(ϵ) A similar question is this:

When does a partially isometric operator $V \in M$ *with the initial and final projections* E *resp.* F *exist?*

In the notation of (10, Definition 6.1.1): When is $E \sim F \cdots \mathrm{mod}\ M$? The same procedure as in (δ) (only using the V_1 of Lemma 21 instead of the V_2) gives two necessary and sufficient conditions, which obtain from those of (δ) by replacing the relation $\leq$ in each case by the relation $=$. Another way to derive this result obtains by observing that $E \sim F \cdots \mathrm{mod}\ M$ is equivalent to the conjunction of $E \lesssim F \cdots \mathrm{mod}\ M$ and of $F \lesssim E \cdots \mathrm{mod}\ M$ and then by applying (δ) to E, F and F, E.